

**THE MAHARAJA SAYAJIRAO UNIVERSITY OF
BARODA**

FACULTY OF SCIENCE



A PROJECT REPORT

on

Coaching Institute Management System

for

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Submitted by

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In fulfilment for the award of the degree

Of

BACHELOR OF COMPUTER APPLICATIONS

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CERTIFICATE

This is to certify that

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Date :

Name & Signature of the Guide

Signature of Head, Department of
Computer Applications

CERTIFICATE FROM CLIENT

CERTIFICATE OF COMPLETION

THIS CERTIFICATE IS PROUDLY PRESENTED TO

Prachi Sanyal

This certificate is proudly presented for successfully developing the Shreeji Classes Web Application as part of the 6th Semester Project, ensuring seamless operational management and workflow efficiency.

S. K. Doshi

Samarth Doshi



SHREEJI CLASSES

ACKNOWLEDGMENT

In completing this project, I have been fortunate to receive support, guidance, and encouragement from many individuals. I sincerely acknowledge their invaluable contributions.

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ABSTRACT

The **Shreeji Classes Web Application** is a comprehensive coaching management system designed to enhance administrative efficiency, streamline student learning, and optimize teacher workflows. It provides a centralized digital platform where:

- **Admins** can manage inquiries, students, teachers, fees, notifications, and timetables.
- **Teachers** can mark attendance, upload test scores, send notifications, share study materials and quizzes.
- **Students** can track their attendance, view performance reports, access study resources, and pay fees online.

The system eliminates manual record-keeping by automating core functions such as inquiry handling, attendance tracking, fee management, performance evaluation, and SMS notifications. This automation reduces administrative workload, minimizes human errors, and enhances communication between stakeholders.

Some of the common challenges faced in traditional coaching management include delayed response to inquiries, difficulty in tracking student progress, manual fee collection issues, and lack of centralized study resources. The Shreeji Classes Web Application addresses these issues by offering a secure, scalable, and efficient digital solution tailored for educational institutes.

Built using the **MERN Stack**, with integrations of **Twilio**, **Razorpay**, and **Cloudinary**, the application ensures seamless communication, secure transactions, and an enriched learning experience.

By digitizing coaching operations, this project significantly improves efficiency, reduces paperwork, enhances student engagement, and fosters a structured learning environment, making it a valuable tool for modern educational institutions.

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1. NOTATION, NAMING AND CONVENTIONS

In the development of Shreeji Classes Web App, I have adhered to standardized naming, notations, and conventions to maintain a structured, readable, and maintainable codebase. Following a consistent approach helps in better collaboration, reduces errors, and ensures scalability. Below are the key guidelines followed across different parts of the project:

Naming Conventions:

- **File & Folder Naming:** All folder names use lowercase to maintain consistency across the project structure, ensuring easy identification. Files are named using camelCase.js, except for React components, which follow PascalCase.js to distinguish them clearly from other files.
- **Variable Naming:** To maintain clarity and avoid confusion, variables and function names strictly adhere to camelCase, promoting better readability. Constants are always written in UPPER_CASE to differentiate them from regular variables.
- **React Component Naming:** All React components are named using PascalCase to ensure a uniform and recognizable structure across the frontend application.
- **MongoDB Collection Naming:** Database collections are named in singular and lowercase to maintain consistency and avoid ambiguity while interacting with the database.

Notations & Code Style:

- **JavaScript & Node.js:** The project follows modern ES6+ syntax, making use of arrow functions, async/await, and destructuring to enhance efficiency and maintain clean, modular code. Code indentation follows a consistent 2-space format, ensuring a uniform layout throughout the codebase. Semi-colons (;) are included at the end of statements to improve clarity and avoid potential parsing issues.
- **React JS:** Proper indentation and spacing have been maintained to improve readability. Self-closing tags are used wherever applicable to reduce unnecessary closing tags, keeping the component structure clean and concise.

- **Tailwind CSS Notation:** Class names follow a structured order, ensuring a logical arrangement that prioritizes layout, positioning, size, color, and typography. This helps in maintaining clarity and consistency in the styling of UI components, making modifications easier.

API & Routing Conventions:

- **REST API Endpoints:** The API design strictly follows RESTful principles, ensuring a structured and predictable format for API requests. Endpoints are named using kebab-case and use appropriate HTTP methods to maintain clarity and consistency across the backend services.
- **Express.js Route Handlers:** To keep the backend well-organized, API routes are defined in separate files within the routes/ directory. The project follows a modular structure, making it easier to scale, maintain, and debug different API functionalities without affecting other parts of the system.

Database Schema Conventions:

- **Field Naming:** All database field names follow camelCase to ensure uniformity across collections and prevent inconsistencies when accessing data. This approach enhances readability and aligns with JavaScript's object notation.
- **Timestamps:** To facilitate tracking of records, timestamps are incorporated into database schemas. This helps maintain accurate logs of data creation and modifications, aiding in auditing, reporting, and debugging. These timestamps provide valuable insights into data history and lifecycle management. By following these well-defined conventions, the Shreeji Classes Web App ensures code clarity, ease of collaboration, and long-term maintainability, allowing for seamless development and future scalability.

2. BUSINESS PROBLEM

2.1 Problem Statement

Coaching institutes face significant challenges due to manual record-keeping, inefficient communication, and a lack of automation in their daily operations. Essential tasks such as handling student inquiries, tracking attendance, managing fee payments, and distributing study materials require substantial time and effort, often leading to inefficiencies, errors, and delays. Without a structured digital system, administrative processes become cumbersome, affecting the overall learning experience and operational efficiency.

Key Challenges Faced by Coaching Institutes

1. **Inquiry Management Issues** – Coaching institutes often rely on manual processes to handle student inquiries, responding via phone calls and emails. This approach is time-consuming, inefficient, and prone to delays, resulting in potential students losing interest or seeking alternative options.
2. **Inefficient Attendance Tracking** – Many coaching institutes lack a digital attendance system, leading to inaccurate records and ineffective communication of attendance updates to students and parents. Manual attendance marking is error-prone and requires additional administrative effort to maintain records.
3. **Fee Payment Hassles** – Traditional fee collection methods involve manual tracking, cash transactions, and delayed payments. The absence of an automated fee management system with reminders increases the chances of missed payments and administrative workload.
4. **Limited Parent Communication** – Parents often struggle to stay informed about their child's academic progress, attendance, and overall performance. Without real-time updates, they are unable to actively monitor or support their child's education effectively.
5. **Study Material & Test Management** – Teachers find it challenging to distribute study materials and practice quizzes efficiently due to the lack of a structured online platform. Managing assignments, quizzes, and test evaluations manually requires additional effort, leading to inconsistencies in academic tracking.

2.2 Proposed Solution

To address these challenges, the Shreeji Classes Web Application offers a centralized digital platform that streamlines operations and enhances overall efficiency. The system provides:

- **Automated Inquiry Handling** – Student inquiries are managed through a structured email and SMS response system, reducing delays and improving engagement.
- **Digital Attendance Management** – Teachers can mark attendance digitally, which is immediately accessible to both students and parents, ensuring transparency and accuracy.
- **Integrated Online Fee Payment System** – The application incorporates Razorpay for secure online transactions, along with automated reminders for timely payments.
- **Real-Time Notifications & Reports** – Parents and students receive performance updates, attendance reports, and notifications via SMS and email, fostering better communication.
- **Comprehensive Study Material & Quiz Portal** – The platform enables teachers to upload study materials, practice quiz, and manage assessments online, enhancing the learning experience.

By automating administrative processes, improving communication, and providing digital solutions for fee management and academic tracking, the Shreeji Classes Web Application enhances efficiency, accuracy, and user experience, making coaching class management more effective and technology-driven.

3. PROJECT CHARTER

3.1 Project Definition

The Shreeji Classes Web Application is a comprehensive coaching management system designed to automate and streamline the core operations of coaching institutes. The system provides a centralized platform for managing student inquiries, attendance tracking, fee payments, academic performance monitoring, study materials, and communication between students, teachers, and administrators.

This project aims to replace traditional manual processes with an intuitive, technology-driven solution, ensuring efficient administration, better student engagement, and enhanced communication.

3.2 Purpose

The primary purpose of the Shreeji Classes Web Application is to digitize and optimize the functioning of coaching institutes by providing an integrated, feature-rich solution that enhances operational efficiency.

The key objectives of this project are:

- To eliminate manual record-keeping by automating inquiries, attendance tracking, and fee management.
- To improve student engagement by providing an interactive digital platform with real-time access to performance reports, attendance records and study materials.
- To facilitate better communication between students, parents, teachers, and administrators through automated notifications and messaging services.
- To offer a secure and seamless payment system integrated with Razorpay, reducing delays and inefficiencies in fee collection.
- To enhance the overall learning experience by providing a structured digital platform for study materials and practice quizzes.

By achieving these objectives, the project will significantly improve operational effectiveness, minimize human errors, and enhance the overall learning experience for students.

3.3 Scope of Project

The Shreeji Classes Web Application encompasses the development of a full-fledged coaching management system that covers the following major functionalities:

In-Scope Features:

- **Student Inquiry Management:** Automates inquiry responses via SMS and email.
- **Attendance Tracking:** Teachers can digitally mark attendance; students and parents can view real-time updates.
- **Fee Management System:** Integrated with Razorpay for online payments, automated reminders for due payments.
- **Study Material Repository:** Teachers can upload notes, assignments, and resources, and students can access them anytime.
- **Marks Upload & Progress Monitoring:** Teachers can upload marks of the tests they conduct, and students can track their performance.
- **Performance Tracking:** Students and parents can access performance reports and test results.
- **Notification & Communication System:** Real-time updates through SMS and in-app notifications for announcements, schedule changes, and reminders.

Out of Scope:

- Multi-institute support (this version is designed for a single coaching institute).
- Mobile application (the current scope is for a web-based platform).
- AI-based student performance analysis (future enhancement).

3.4 Team Size

- Project Developer: 1 (Solo Developer – Prachi Sanyal)
- As a solo developer, I am responsible for the entire development lifecycle, including:
 - Requirement Analysis & System Design
 - Frontend & Backend Development (MERN Stack)
 - Database Management (MongoDB)
 - Third-Party Integrations (Twilio for SMS, Razorpay for payments, Cloudinary for file storage)
 - Testing & Deployment
 - Documentation & Maintenance

4. PROJECT PLAN

4.1 Objective

The Shreeji Classes Web Application is developed to serve as a comprehensive, digitalized coaching management system aimed at automating and streamlining the daily operations of educational institutes. Traditional coaching institutes often rely on manual processes for handling inquiries, student attendance, fee collection, performance tracking, and study material distribution, leading to inefficiencies, errors, and communication gaps.

The primary objective of this project is to develop an integrated web-based solution that:

- **Automates core administrative functions** such as student inquiries, attendance tracking, fee payment, and notifications.
- **Enhances communication and transparency** between students, teachers, and parents by providing real-time updates.
- **Improves student learning experiences** through easy access to study materials, quizzes, and performance reports.
- **Eliminates manual dependencies** by integrating modern technologies for digital record-keeping, notifications, and secure online transactions.

By implementing this system, the project aims to enhance efficiency, minimize errors, improve engagement, and ensure smooth management of coaching institute operations in a structured and professional manner.

4.2 Software Model - Traditional

For the development of the **Shreeji Classes Web Application**, the **Traditional Software Development Model** was adopted. This model was selected due to its structured, sequential approach, ensuring well-documented phases, systematic execution, and minimal ambiguity in project scope and requirements.

Justification for Choosing the Traditional Model

The **Traditional Model** was chosen for its well-defined phases, making it ideal for projects with clear requirements and limited scope changes. The key advantages of this model for the project include:

- **Sequential Development Approach** – The system was developed in distinct, well-structured phases (Requirement Analysis, Design, Development, Testing, Deployment, and Maintenance), ensuring clarity in execution.
- **Comprehensive Planning** – Every phase was thoroughly documented and reviewed before proceeding to the next, reducing the chances of misinterpretation.
- **Minimal Overlapping** – Each phase was completed before moving forward, ensuring a structured and methodical development process.
- **Efficient Resource Management** – With a clear plan in place, time and effort were allocated effectively, leading to better productivity even as a solo developer.
- **Predictable Outcomes** – Due to its step-by-step nature, the final system met all predefined requirements without major unexpected changes.

4.3 Strategies

To ensure systematic execution and timely completion of the project, the following strategies were implemented:

4.3.1 Development Strategy

- **Phase-Wise Execution** – The project followed a structured approach with distinct phases:
 1. **Requirement Analysis** – Understanding client needs and documenting key functionalities.
 2. **System Design** – Creating blueprints for database structures, UI designs, and system architecture.
 3. **Implementation** – Coding and development of core functionalities (e.g., user management, attendance tracking, fee management).
 4. **Testing** – Performing rigorous testing before deployment.
 5. **Deployment & Maintenance** – Launching the system and handling future maintenance.
- **Standardized Coding Practices** – Industry-standard coding practices were followed to ensure maintainability and scalability.
- **Well-Defined Documentation** – Each phase had detailed documentation to facilitate smooth transitions and reference for future enhancements.

4.3.2 Testing Strategy

- **Phase-Wise Testing** – Testing was conducted at the end of each phase to ensure correctness before moving forward.
- **Validation and Verification** – Ensured that each phase met its requirements and objectives.
- **Final System Testing** – A full system test was conducted to check overall performance, usability, and security before deployment.

4.4 Work division (w.r.t time)

Time Period	Tasks Completed
6th Jan – 10th Jan	Requirement analysis, planning, and initial documentation.
11th Jan – 15th Jan	Frontend setup, initial UI development, and system design including database schema, architecture, and wireframing.
16th Jan – 20th Jan	Backend implementation (Node.js, Express.js, MongoDB), core module structuring, and authentication module development.
21st Jan – 25th Jan	User management and authentication module completed, including admin panel configuration and route protection.
26th Jan – 27th Jan	Backend route integration and testing.
28th Jan – 1st Feb	Course management module development and integration.
2nd Feb – 5th Feb	Inquiry management module implementation.
6th Feb – 7th Feb	Timetable management functionality added.
8th Feb – 11th Feb	Notification module setup with Twilio integration.
12th Feb – 16th Feb	Attendance module development and cross-module testing.
17th Feb – 20th Feb	No significant development activity during this period.
21st Feb – 25th Feb	Implementation of study material and quiz module.
26th Feb – 29th Feb	Test marks upload feature integration.
1st Mar – 5th Mar	Performance report generation and SMS integration.

6th Mar – 10th Mar	Fee management module and Razorpay payment integration.
11th Mar – 20th Mar	UI refinements, bug resolution, and overall module polishing.
21st Mar – 6th Apr	Final testing, client-side improvements, and deployment completion.
7th Apr – 26th Apr	Post-deployment checks, performance optimization, and final presentation preparation.

Note: While the core functionalities were completed by **6th April 2024**, the remaining time until **26th April 2024** was reserved for final refinements, maintenance, and project presentation.

4.5 Work Division (w.r.t Team Members)

As this project was developed by a **single individual**, all roles and responsibilities were managed by **Prachi Sanyal**. The following table outlines the division of work:

Role	Responsibilities
Project Developer (Prachi Sanyal)	- Requirement gathering, system analysis, and documentation. - UI/UX design and wireframing. - Frontend development (React.js, Tailwind CSS). - Backend development (Node.js, Express.js, MongoDB). - API development and integration. - Razorpay payment gateway implementation. - Development of attendance tracking and student inquiry management modules. - Study material management and quiz system implementation. - Testing (Unit testing, integration testing, and UAT). - Deployment, system maintenance, and final presentation preparation.

5. HARDWARE AND SOFTWARE REQUIREMENTS

The Shreeji Classes Web Application is a system that requires minimal hardware specifications for users. The system is designed to be accessible from a wide range of devices, ensuring smooth performance for both students and administrators.

5.1 Hardware Requirements

The application is web-based and does not require high-end hardware. The following specifications are recommended for an optimal experience:

Component	Minimum Requirement	Recommended Requirement
Device	Laptop, Desktop, Tablet, or Smartphone	Laptop/Desktop with modern specifications
Processor	Intel Core i3 (or equivalent)	Intel Core i5/i7 (or equivalent)
RAM	4GB	8GB or higher
Storage	20GB free disk space	SSD for faster performance
Display	1280x720 resolution	Full HD resolution or higher
Internet	2 Mbps connection	10 Mbps or higher for seamless performance
Browser	Chrome, Firefox, Edge (latest versions)	Latest Chrome or Firefox for best compatibility

5.2 Software Requirements

The following software and platforms are required for users and administrators to access and manage the system:

- **Operating System:** Windows / macOS / Linux / Android / iOS
- **Web Browser:** Google Chrome, Mozilla Firefox, Microsoft Edge (latest versions)
- **Internet Connection:** Required for accessing the web application

6. TOOLS AND TECHNOLOGIES USED

To ensure an efficient and scalable development process, the project leveraged modern technologies and development tools.

Programming Languages & Frameworks

Technology	Purpose
React.js	Frontend development, UI components, and state management
Tailwind CSS	Responsive and optimized UI styling
Node.js	Backend development for API handling
Express.js	Framework for building RESTful APIs
MongoDB	NoSQL database for storing user and system data
Razorpay API	Secure online payment processing

Development & Collaboration Tools

Tool	Purpose
Git & GitHub	Version control and project collaboration
Postman	API testing and debugging
VS Code	Code editing and development
MongoDB Atlas	Cloud-based database storage
Google Drive	Storing monthly performance reports to share links via SMS
Twilio API	SMS & email notifications
Cloudinary	Uploading and managing study material files

7. REQUIREMENT ANALYSIS

Requirement analysis is a crucial phase in the software development lifecycle, where the functional and non-functional requirements of the Shreeji Classes Web Application were identified. This phase ensured that the system aligns with user needs and business objectives.

7.1 System Requirement Specification (SRS)

The **System Requirement Specification (SRS)** document outlines the functional and technical aspects of the project. It provides a structured approach to ensure that all system components work efficiently.

7.1.1 Introduction

The Shreeji Classes Web Application is designed to manage student inquiries, attendance tracking, fee payments, study material distribution, and notifications in a seamless digital environment. The system automates key operations, reducing manual work and improving efficiency.

7.1.2 Objectives of the System

- To provide an efficient student inquiry management system with automated responses.
- To enable attendance tracking for teachers and students in real-time.
- To facilitate online fee payments using Razorpay with automated reminders.
- To allow study material uploads and distribution for students.
- To send SMS notifications for fee reminders, attendance, performance reports, and announcements.

7.1.3 Functional Requirements

The functional requirements define the key operations and services provided by the system.

User Management

- Students, parents, and administrators can log in securely.
- Role-based access control to restrict functionalities.
- Profile management for students and teachers.

Inquiry Management

- Students and parents can submit course inquiries online.
- Automated responses to inquiries via email and SMS.
- Admins can track and manage inquiries.

Attendance Tracking

- Teachers can mark student attendance in real-time.
- Students and parents can view attendance records.
- Automated notifications for low attendance.

Fee Management

- Razorpay integration for online fee payments.
- Automated fee reminders via SMS.
- Payment history and receipts generation.

Study Material & Notifications

- Teachers can upload study materials in various formats (PDF, Word, PPTs).
- Students can access materials anytime.

Performance & Evaluation Module

- Teachers can upload test marks, visible to students on their dashboards.
- Students can view a monthly performance graph across different subjects.
- Monthly performance reports are sent to parents via SMS.

7.1.4 Non-Functional Requirements

The non-functional requirements define system performance, security, and usability standards.

Performance Requirements

- The system should support 100+ concurrent users without lag.
- Response time for major operations should be under 3 seconds.

Security Requirements

- JWT-based authentication for secure logins.
- Data encryption for sensitive information (personal data).
- Role-based access to prevent unauthorized actions.

Scalability Requirements

- Cloud-based architecture to handle increasing users.
- Optimized database structure for faster query execution.

Usability Requirements

- Mobile-friendly UI with responsive design.
- Intuitive navigation for non-technical users.

7.1.5 User Roles and Access Levels

User Role	Permissions
Admin	Full access to manage students, teachers, inquiries, payments, and notifications.
Teacher	Can mark attendance and upload study materials.
Student	Can view attendance, download study materials, and make fee payments.
Parent	Can monitor student attendance and performance report.

7.1.6 Constraints & Assumptions

Constraints

- Users must have an active internet connection to access the platform.
- Payments can only be processed via Razorpay.
- SMS/email notifications depend on third-party service availability.

Assumptions

- Users are familiar with basic web navigation.
- Students and parents will use valid credentials to access the system.
- Teachers will actively update attendance and study materials.

8. SYSTEM ANALYSIS

The Shreeji Classes Web Application was designed with a structured approach to ensure efficient management of students, courses, fees, and academic resources. This section presents the module specifications and system flow, outlining the key functionalities and workflow of the system.

8.1 Module Specifications

The system consists of eight core modules, each addressing specific functionalities essential for the smooth operation of Shreeji Classes.

8.1.1 User Authentication & Management

- Secure JWT-based authentication for students, teachers, and admins.
- Role-based access control (Students, Teachers, Admins).
- Profile management for users to update personal details.
- Password recovery and account security measures.

8.1.2 Course Management

- Admins can add, update, and remove courses.
- Course details include syllabus, duration, fees, and faculty assignments.

8.1.3 Consultation and Inquiry

- Students and parents can submit inquiries regarding courses, fees, or general queries.
- Admins can track, manage, and reply to inquiries in real-time.

8.1.4 Announcements and Notifications

- Admins can send important announcements via SMS, and in-app notifications.
- Students receive alerts for fee deadlines, exam schedules, and class changes.

8.1.5 Fee Management

- Secure online payment integration using Razorpay.
- Automated fee reminders and due date notifications.
- Students can view fee payment history and download receipts.
- Admins can generate and manage payment reports.

8.1.6 Uploading Test Marks & Attendance

- Teachers can upload test marks for each student.
- Students and parents can view marks and performance analytics.
- Attendance tracking system where teachers can mark student attendance.

8.1.7 Timetable Management

- Admins can create and update class schedules and timetables.
- Students and teachers can view their respective schedules in real time.

8.1.8 Uploading Study Material and Resources

- Teachers can upload study materials in various formats (PDFs, videos, links).
- Students can download materials and access them anytime.
- Organized course-wise content management for better usability.

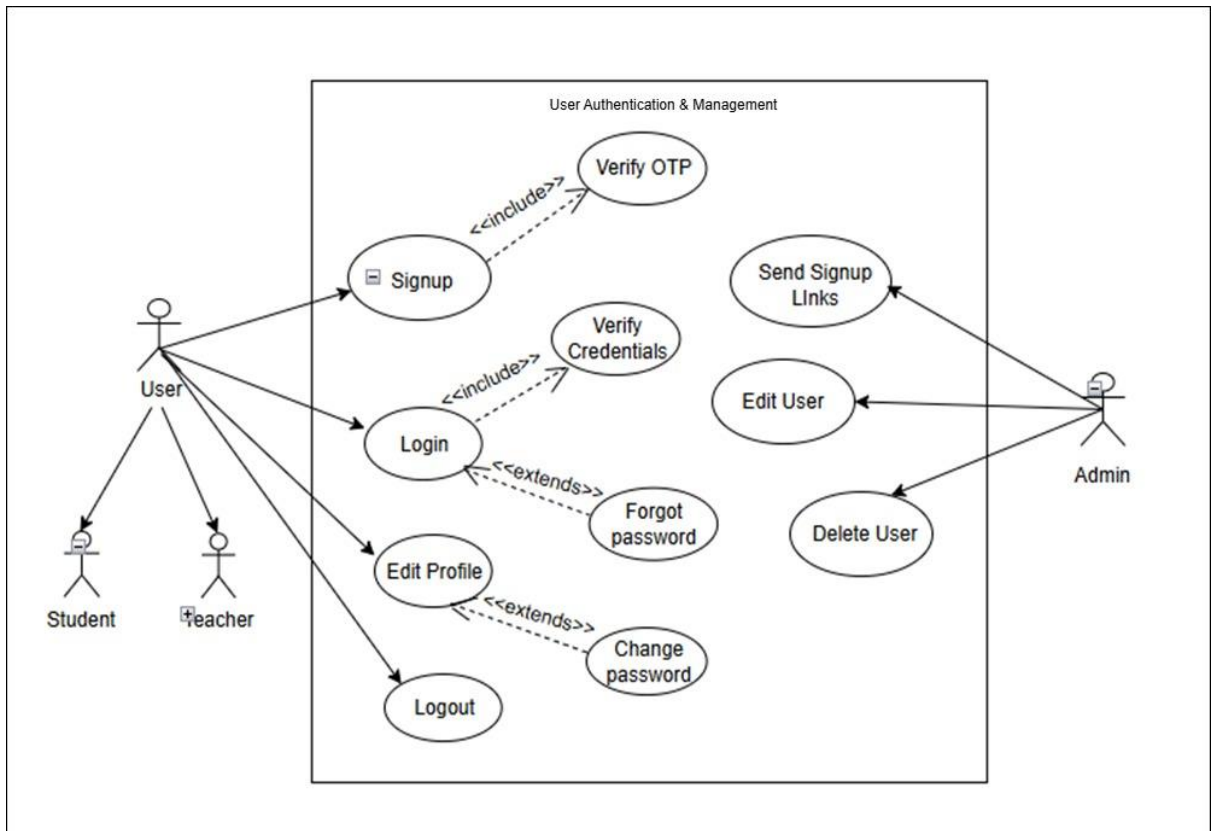
8.2 System Flow

The system workflow outlines how different modules interact and ensure seamless user experience.

The system begins with users submitting inquiries, which admins review. If confirmed, users proceed with signup, verifying their OTP via email before successful registration and login. Admins manage users, courses, inquiries, and timetables, while teachers handle attendance, test marks, study materials, and quizzes. Both admins and teachers can send notifications to students and parents. Students can view timetables, track attendance, check performance reports, access and download study materials, play quizzes anytime for practice, and make secure fee payments.

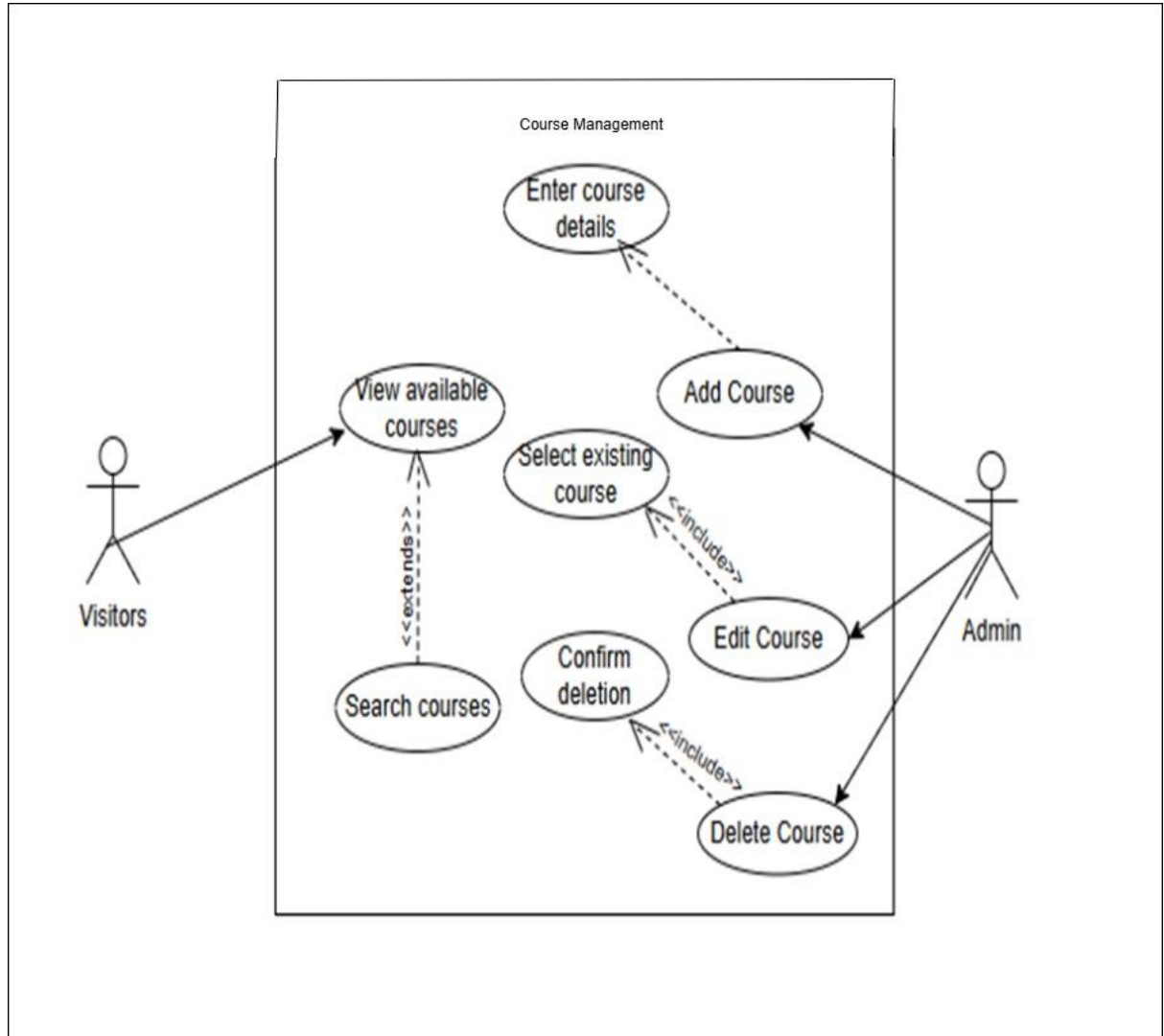
8.2.1 Use Case Diagrams

■ Module – 1 User Authentication & Management



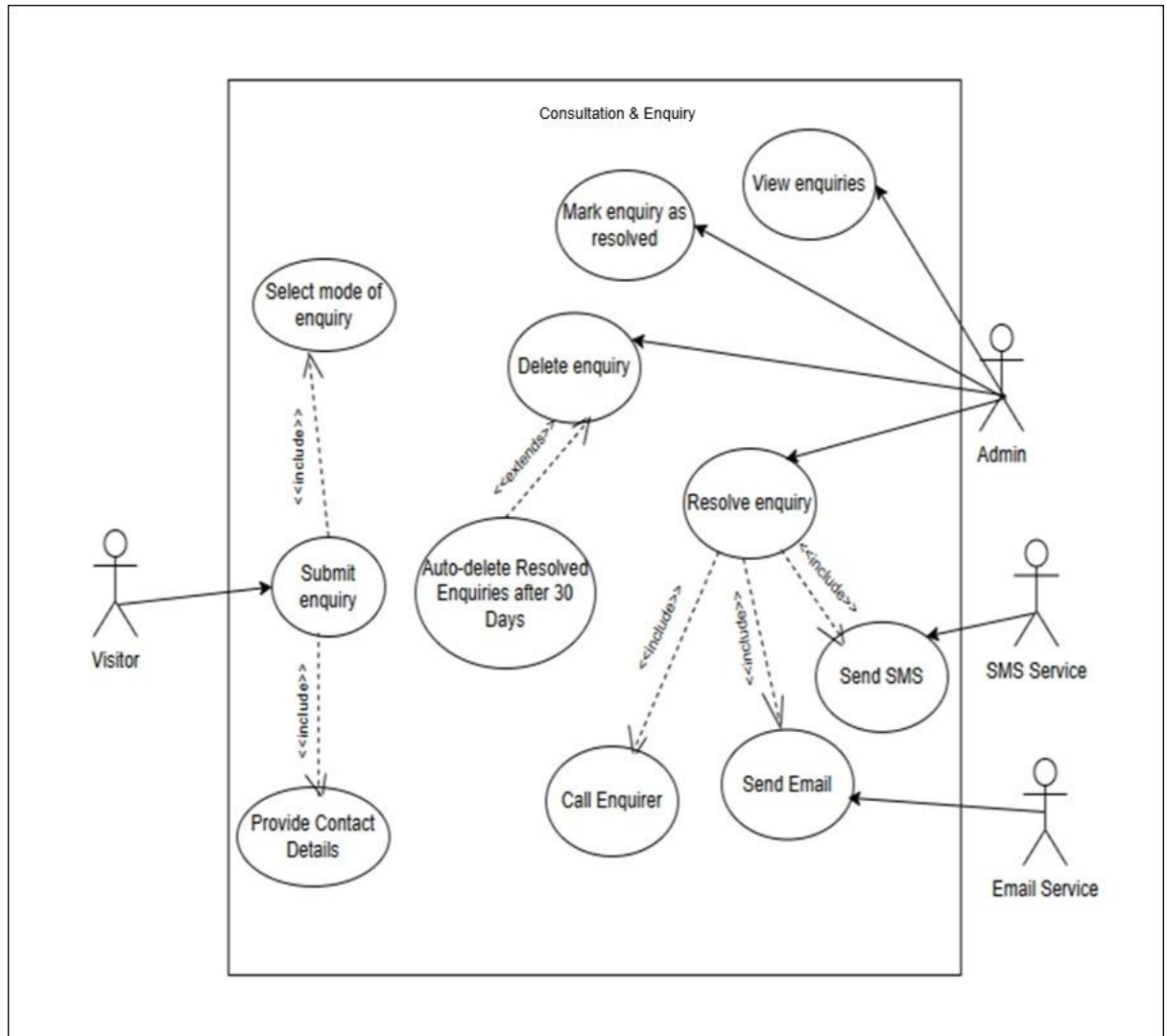
This Use Case Diagram for the Shreeji Classes Web App illustrates interactions between the primary actors: User (Student/Teacher) and Admin. Users can sign up, log in (with OTP verification and credential check), edit their profile (including password changes), and log out. They can also recover accounts via the forgot password feature. The Admin has control over sending signup links, editing users, and deleting users. This ensures role-based access and secure authentication within the system.

■ Module – 2 Course Management



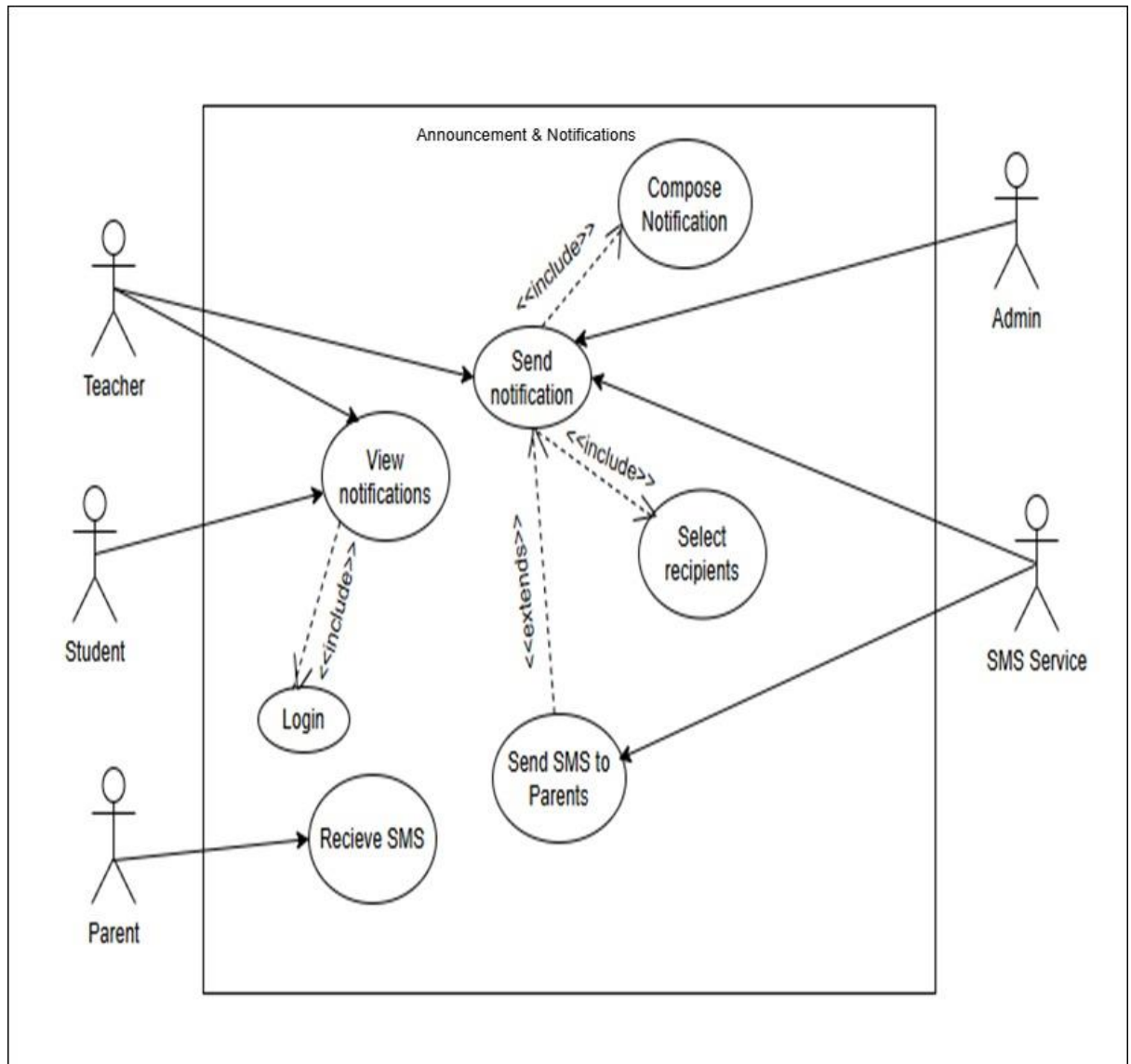
The Use Case Diagram for the Course Management system defines interactions between Visitors and Admins. Visitors can view and search available courses. Admins have comprehensive control, including adding new courses by entering details, editing existing courses after selection, and deleting courses with confirmation. The "Select existing course" and "Confirm deletion" use cases are included within "Edit Course" and "Delete Course" respectively, ensuring proper workflow. The "Search courses" use case extends the "View available courses" functionality.

■ Module – 3 Consultation & Enquiry



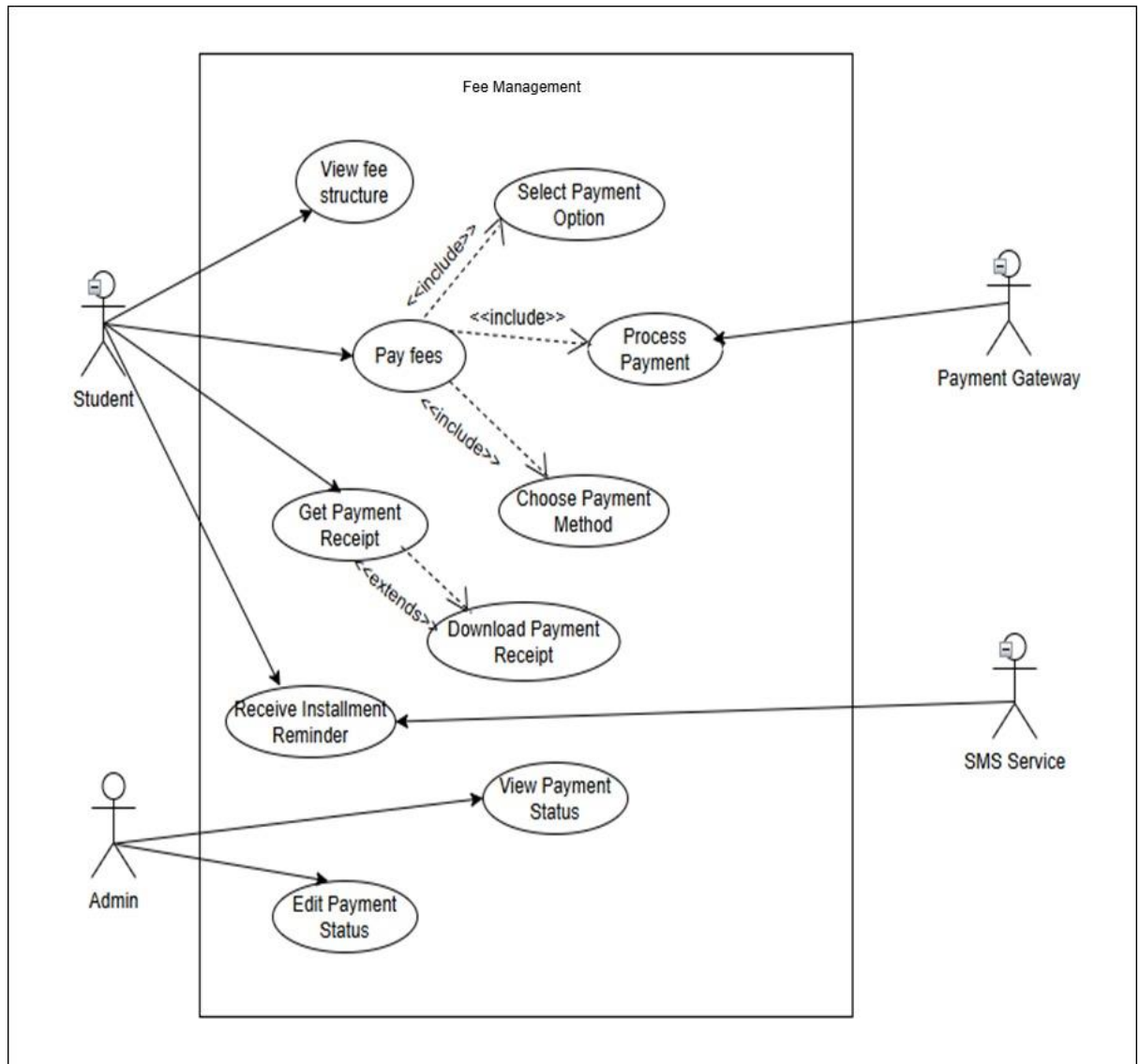
The Use Case Diagram for the "Consultation & Enquiry" system illustrates interactions between Visitors, Admins, and external services (SMS and Email). Visitors can submit enquiries by selecting a mode and providing contact details. Admins manage enquiries by viewing, marking them as resolved, and deleting them. The "Resolve enquiry" use case includes sending SMS, email, or calling the enquirer. "Auto-delete Resolved Enquiries after 30 Days" extends the "Delete enquiry" use case, automating cleanup.

■ Module – 4 Announcement & Notifications



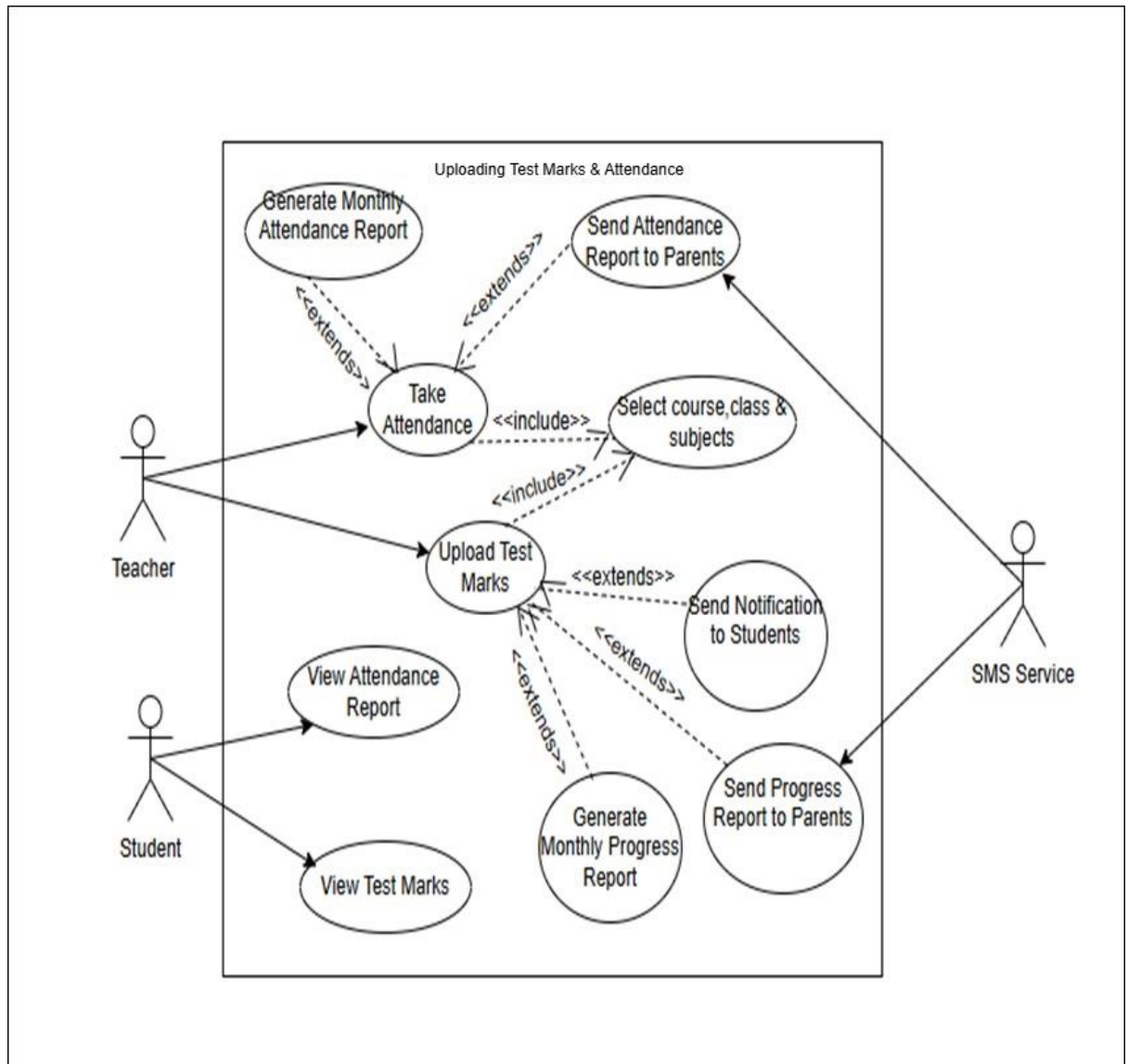
The Use Case Diagram for "Announcement & Notifications" depicts interactions between Teachers, Students, Parents, Admins, and an SMS Service. Admins compose and send notifications, which requires recipient selection. Teachers, Students, and Parents can view notifications after logging in. Parents also receive SMS notifications. The "Login" use case is included within "View notifications," ensuring authenticated access. "Send SMS to Parents" extends the "Send notification" use case.

■ **Module – 5 Fee Management**



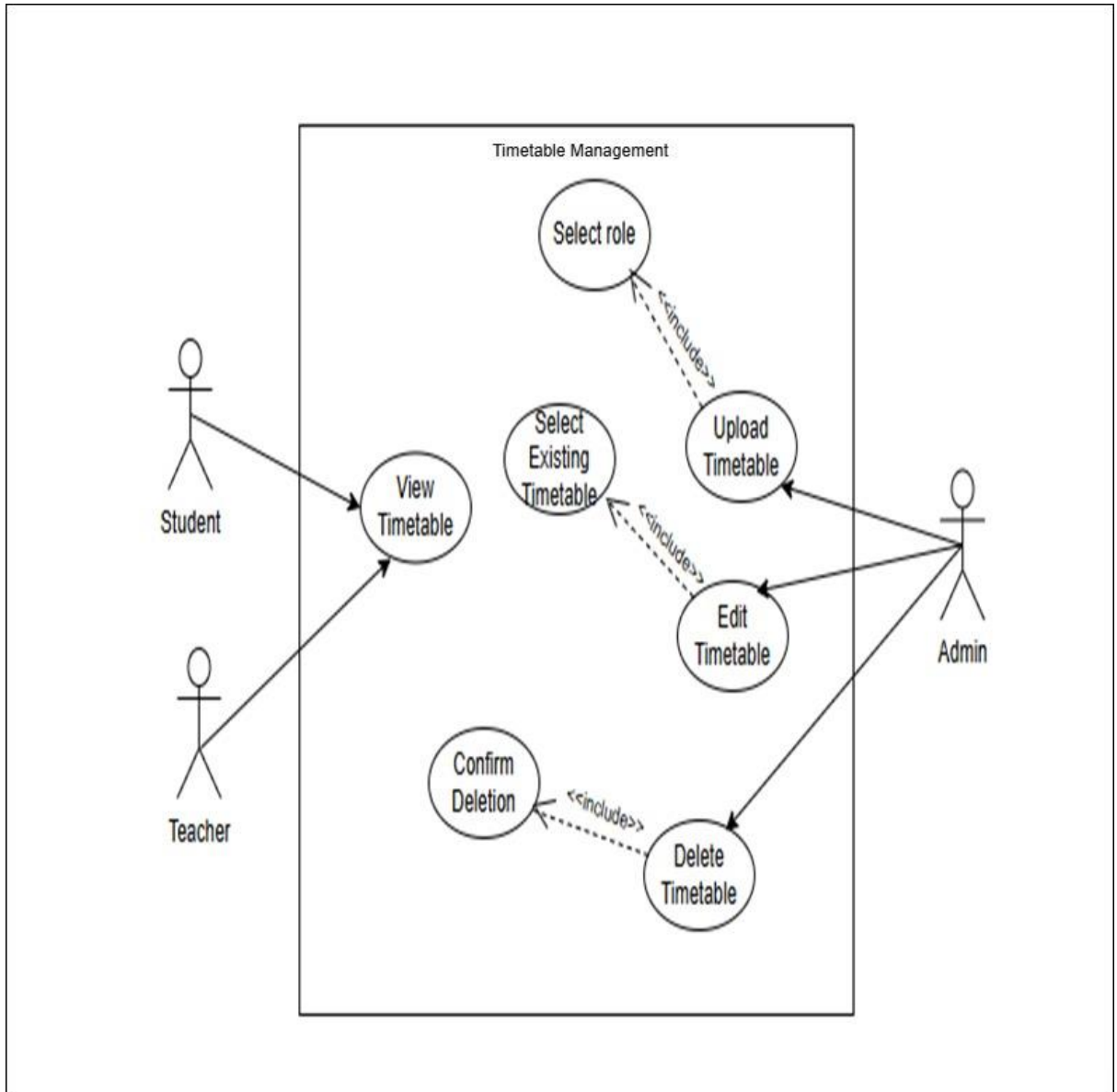
The Use Case Diagram for "Fee Management" outlines interactions between Students, Admins, a Payment Gateway, and an SMS Service. Students can view the fee structure, pay fees and get payment receipts. They can also receive installment reminders via SMS and view their payment status. Downloading a payment receipt extends the "Get Payment Receipt" use case. Admins can view and edit payment statuses.

▪ **Module – 6 Uploading Test Marks & Attendance**



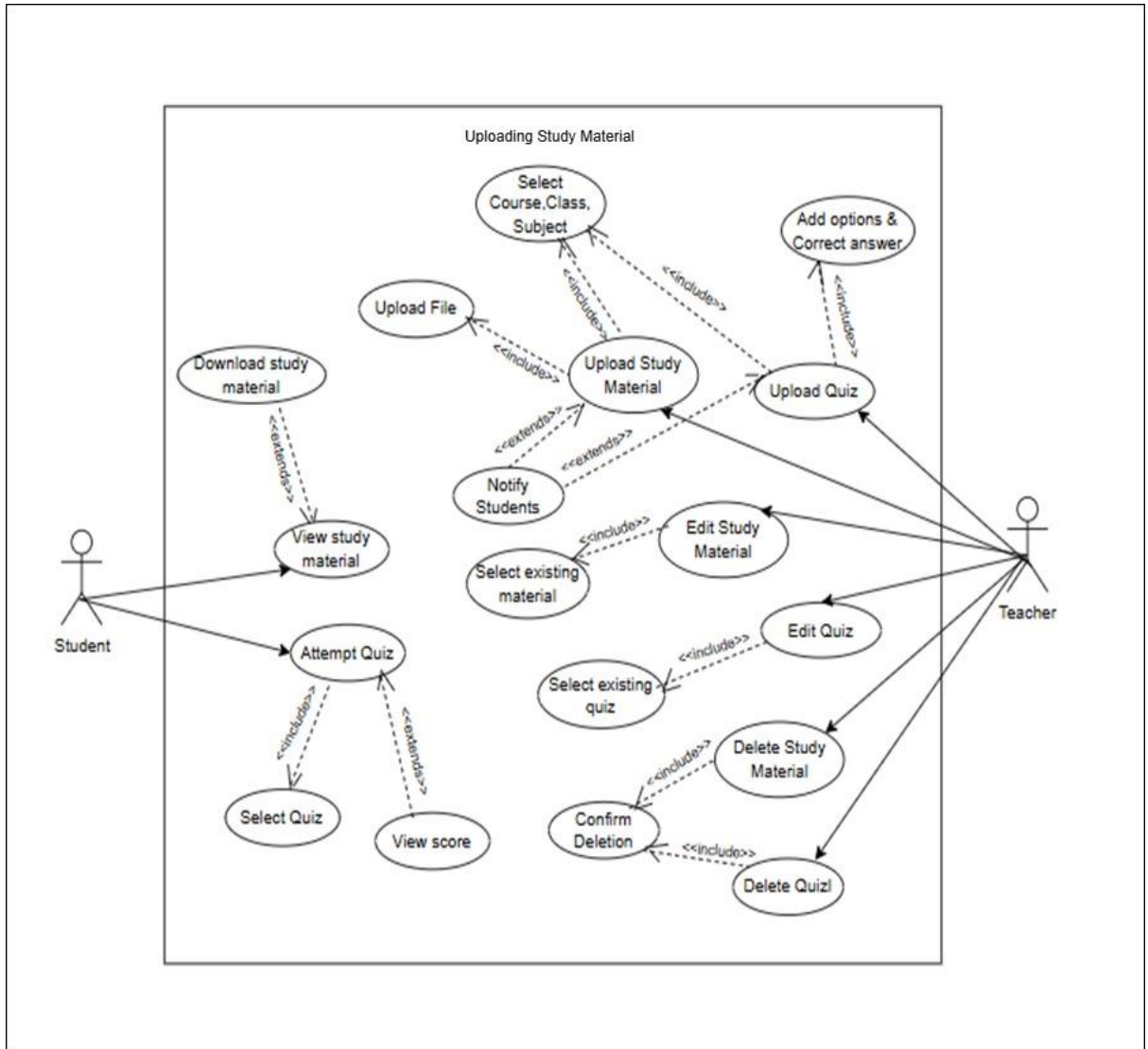
The Use Case Diagram for "Uploading Test Marks & Attendance" shows how Teachers, Students, and an SMS Service interact. Teachers can take attendance, upload marks, and generate monthly reports. Sending attendance and progress reports to parents, and notifications to students, are extended use cases. Students can view their attendance and marks. The system uses the SMS Service to send updates, ensuring efficient academic data management and communication.

■ Module – 7 Timetable Management



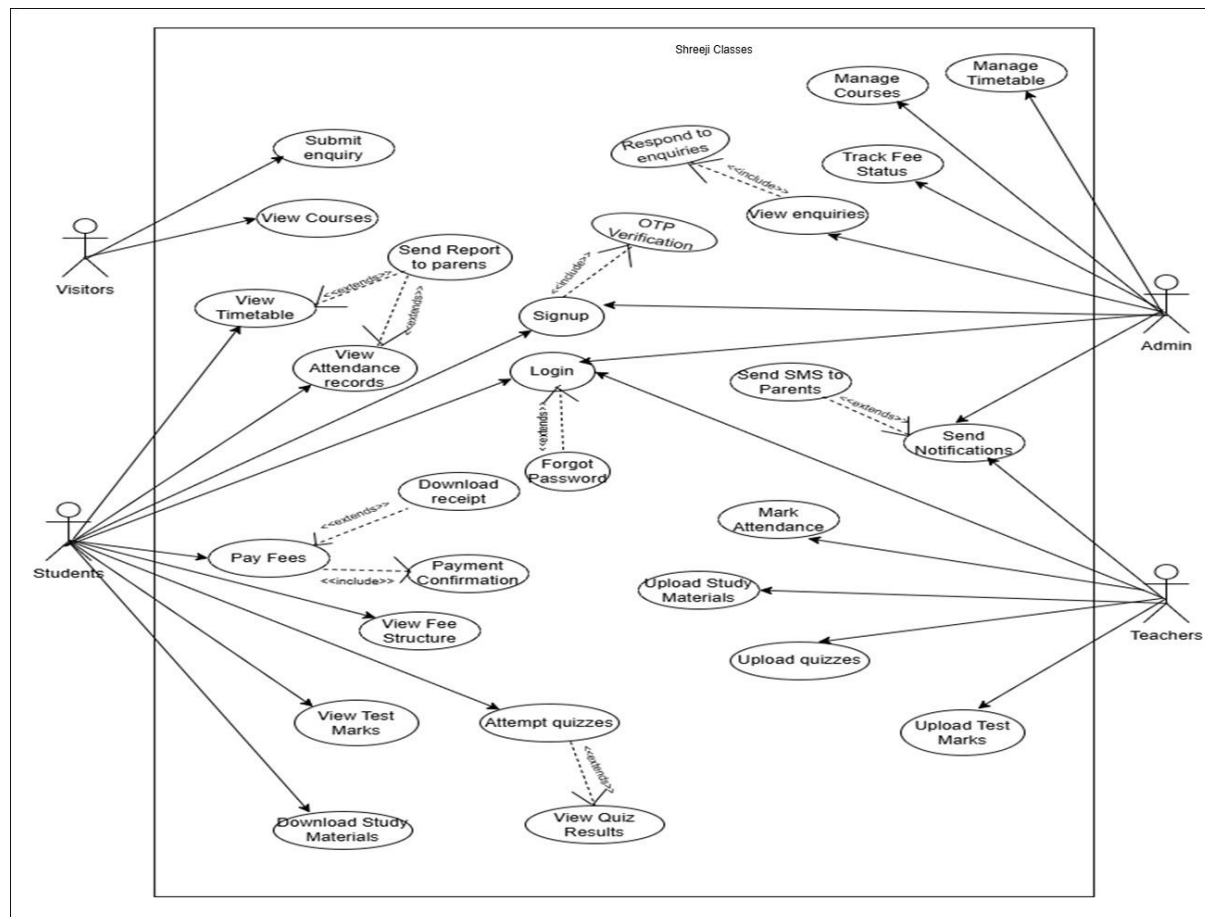
The Use Case Diagram for "Timetable Management" outlines interactions between Students, Teachers, and Admins. Students and Teachers can view the timetable. Admins manage timetables by uploading new ones, editing existing ones, and deleting them. "Select role" is included in "Upload Timetable" and "Edit Timetable" to specify the target user group. "Select Existing Timetable" is included in "Edit Timetable," and "Confirm Deletion" is included in "Delete Timetable" to ensure proper workflow.

■ **Module – 8 Uploading Study Material**



The Use Case Diagram for "Uploading Study Material" outlines interactions between Students and Teachers. Teachers can upload study materials (including files, quizzes with options and answers), edit and delete them, and notify students. "Select Course, Class, Subject" is included in "Upload Study Material," and "Add options & Correct answer" is included in "Upload Quiz." Students can view and download study materials, attempt quizzes (selecting them first), and view scores. "Notify Students" extends "Upload Study Material," and "View score" extends "Attempt Quiz."

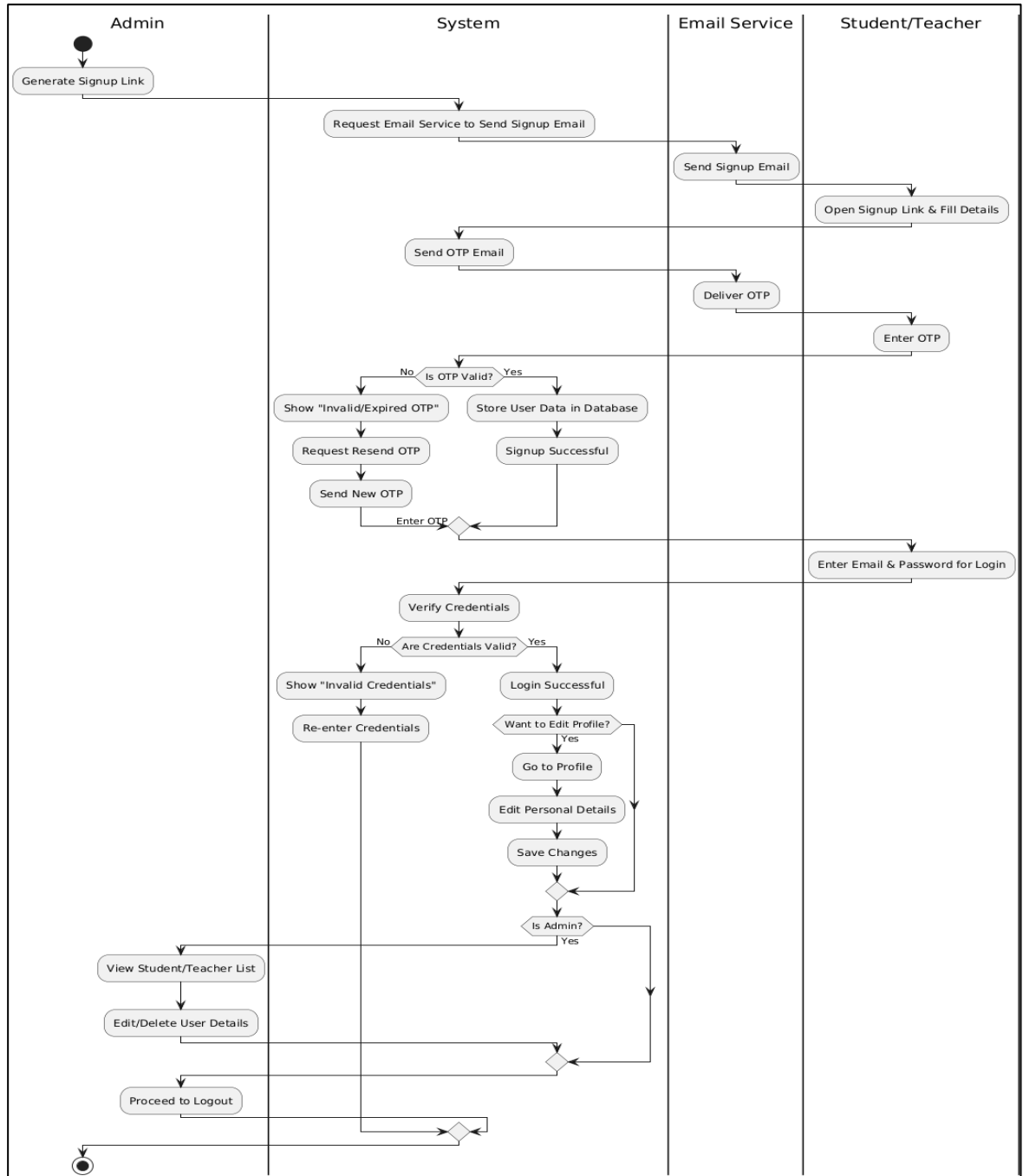
Overall Use Case Diagram



This Use Case Diagram for Shreeji Classes outlines interactions between Visitors, Students, Teachers, and Admins. Visitors can view courses and submit enquiries. Students can pay fees, view fee structures, download materials, take quizzes, and view results and attendance. Teachers send notifications, upload materials, quizzes, and marks, and mark attendance. Admins manage courses, timetables, fees, enquiries, notifications. Login (with OTP and password recovery) is central for authenticated access. This design ensures comprehensive functionality and communication within the Shreeji Classes system.

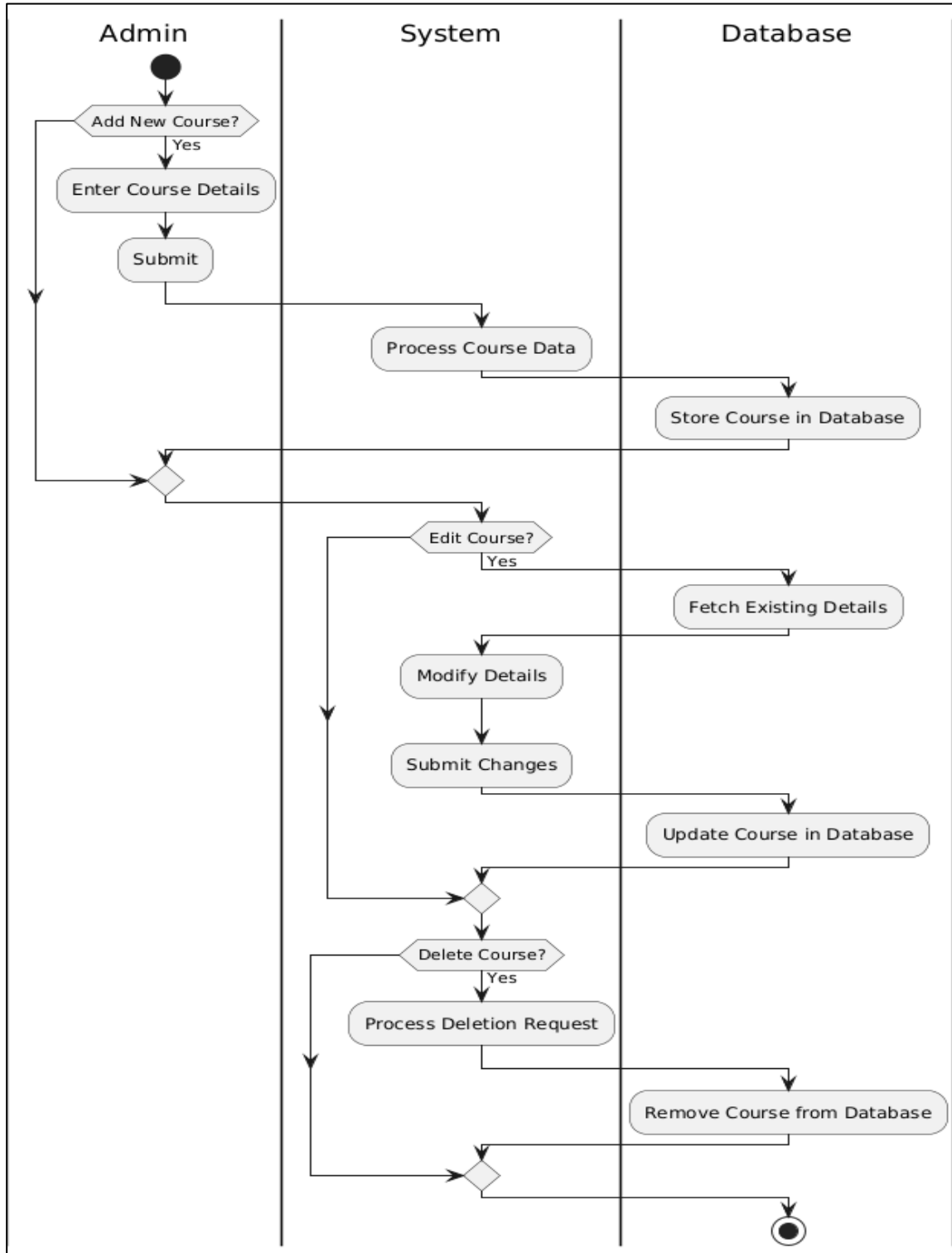
8.2.2 Activity Diagram

■ Module – 1 User Authentication & Management



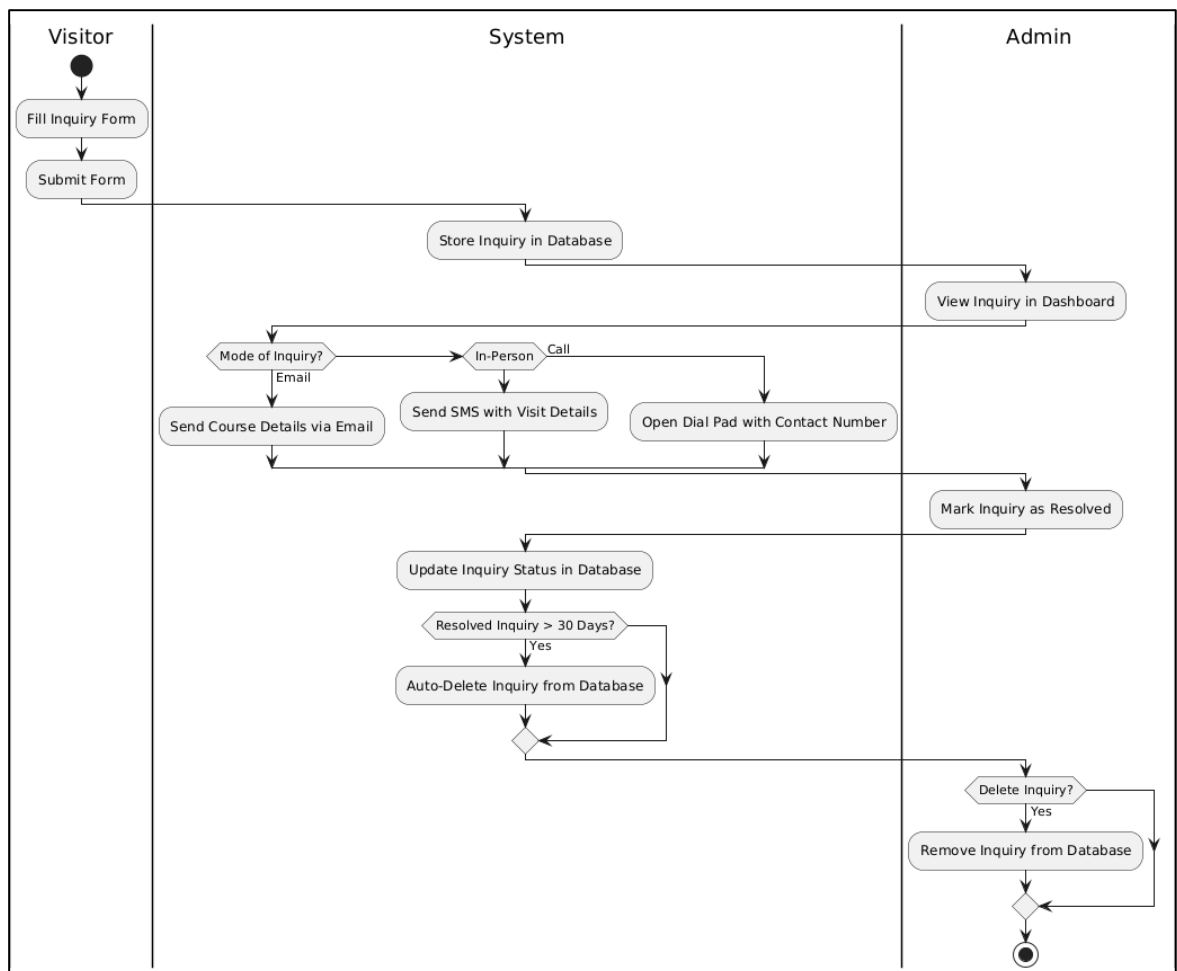
This Activity Diagram illustrates the user authentication and management module, depicting interactions between Admin, System, Email Service, and Student/Teacher. The Admin generates signup links, triggering the System to request signup emails from the Email Service. Students/Teachers open the link, fill details, and receive OTPs via email. The System validates OTPs and stores user data upon successful signup. Login involves credential verification by the System. Users can edit profile details. Admins have additional privileges to view, edit, and delete user details. The process concludes with logout, ensuring secure access and management of user accounts.

▪ Module – 2 Course Management



This activity diagram depicts the workflow for managing courses within a system. The process begins with an **Admin** who can choose to add a new course. If "Yes," the admin enters course details and submits them. The **System** then processes this data and stores it in the **Database**. Alternatively, the admin can edit an existing course. If "Yes," the system fetches the current details, allows the admin to modify them, and updates the database with the changes. Lastly, the admin can delete a course. If "Yes," the system processes the deletion request and removes the course from the database. The diagram effectively illustrates the interactions between the Admin, System, and Database for course management operations.

■ Module – 3 Consultation & Enquiry

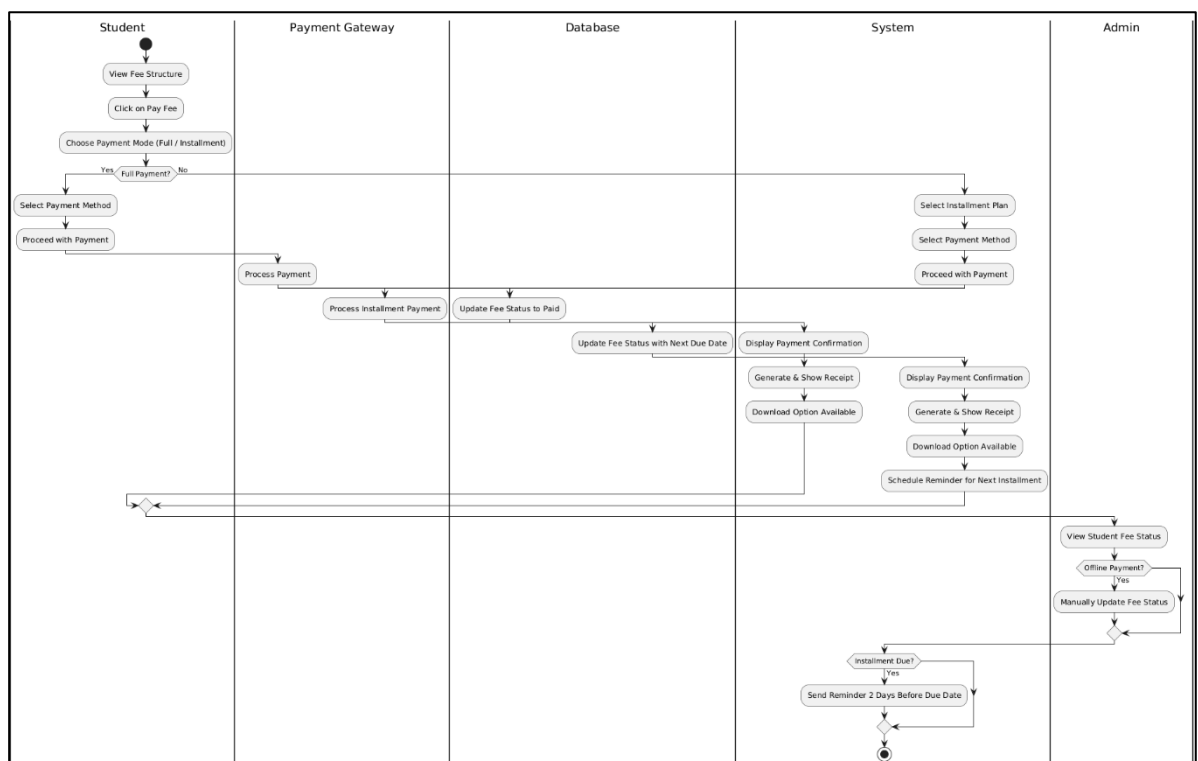


This diagram illustrates the process of handling inquiries in a consultation module. A **Visitor** fills and submits an inquiry form. The **System** stores this in the database. An **Admin** views the inquiry and selects the mode of inquiry (email, in-person, call). Based on the mode, the system sends course details via email, SMS with visit details, or opens a dial pad. The admin marks the inquiry as resolved, updating its status in the database. If a resolved inquiry is older than 30 days, the system auto-deletes it. The admin can also manually delete inquiries.

- **Module – 4 Announcement & Notifications**

This diagram outlines the notification process. An **Admin** initiates by entering a title and message, then selects the recipient category (teachers or students). For teachers, the **System** delivers the notification. For students, the admin chooses a student group, the system delivers the notification, and optionally sends SMS to parents. A **Teacher** can also send notifications, entering a title, message, and selecting a student group. The system delivers the notification to students and again, optionally sends SMS to parents.

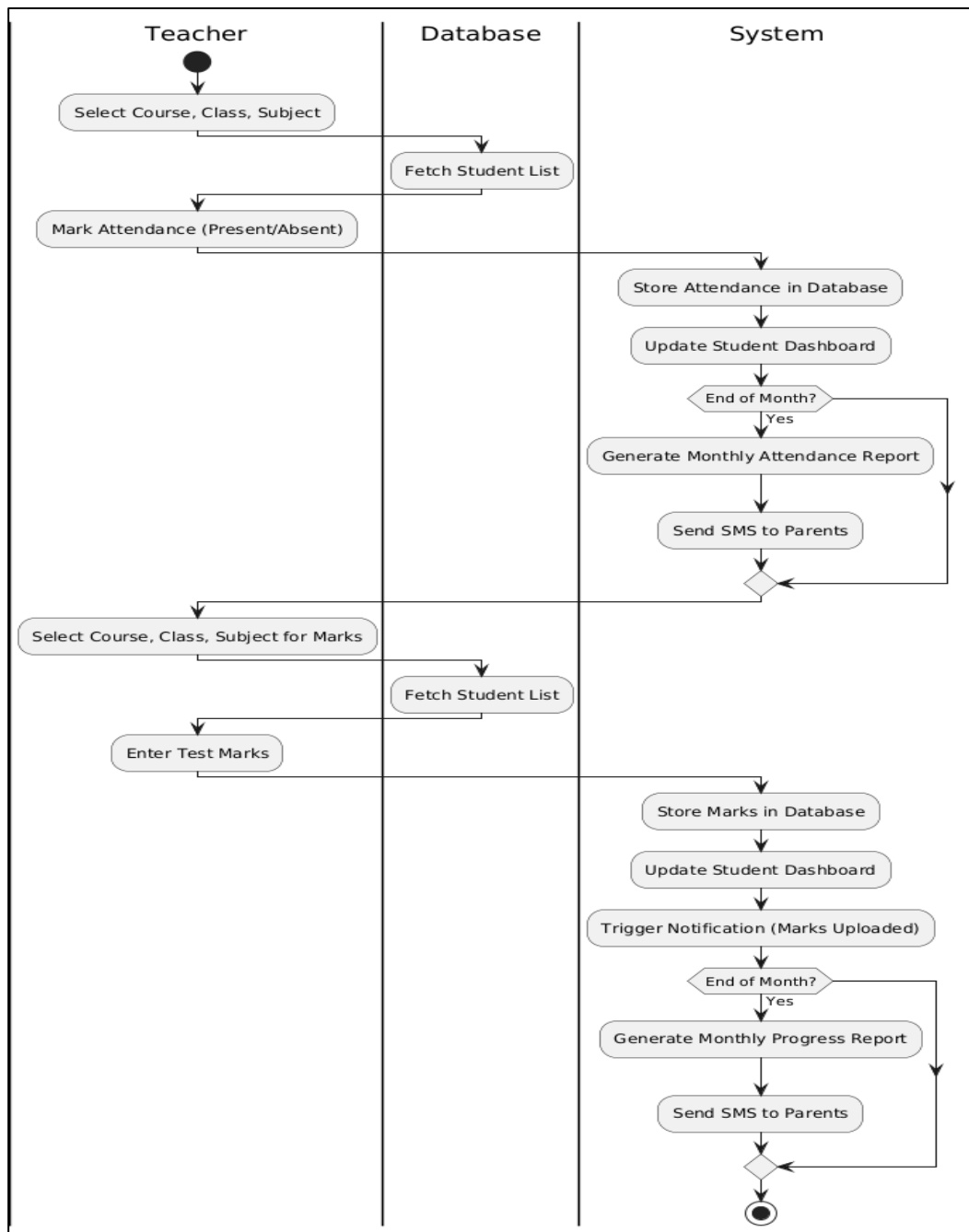
■ Module – 5 Fee Management



This diagram illustrates the fee payment process. A **Student** views the fee structure and chooses to pay, selecting either full or installment payment. For full payment, they choose a payment method, proceed, the **Payment Gateway** processes it, the **Database** updates the fee status, and the **System** displays confirmation with a receipt download option. For installments, the student selects a plan, payment method, proceeds, the gateway processes the installment, the database updates the fee status with the next due date, and the system shows confirmation with receipt download. The system also schedules reminders for the next installment. An **Admin** can view

student fee status, manually update it for offline payments, and the system sends reminders two days before installment due dates. This diagram outlines the payment workflow and administrative management of student fees.

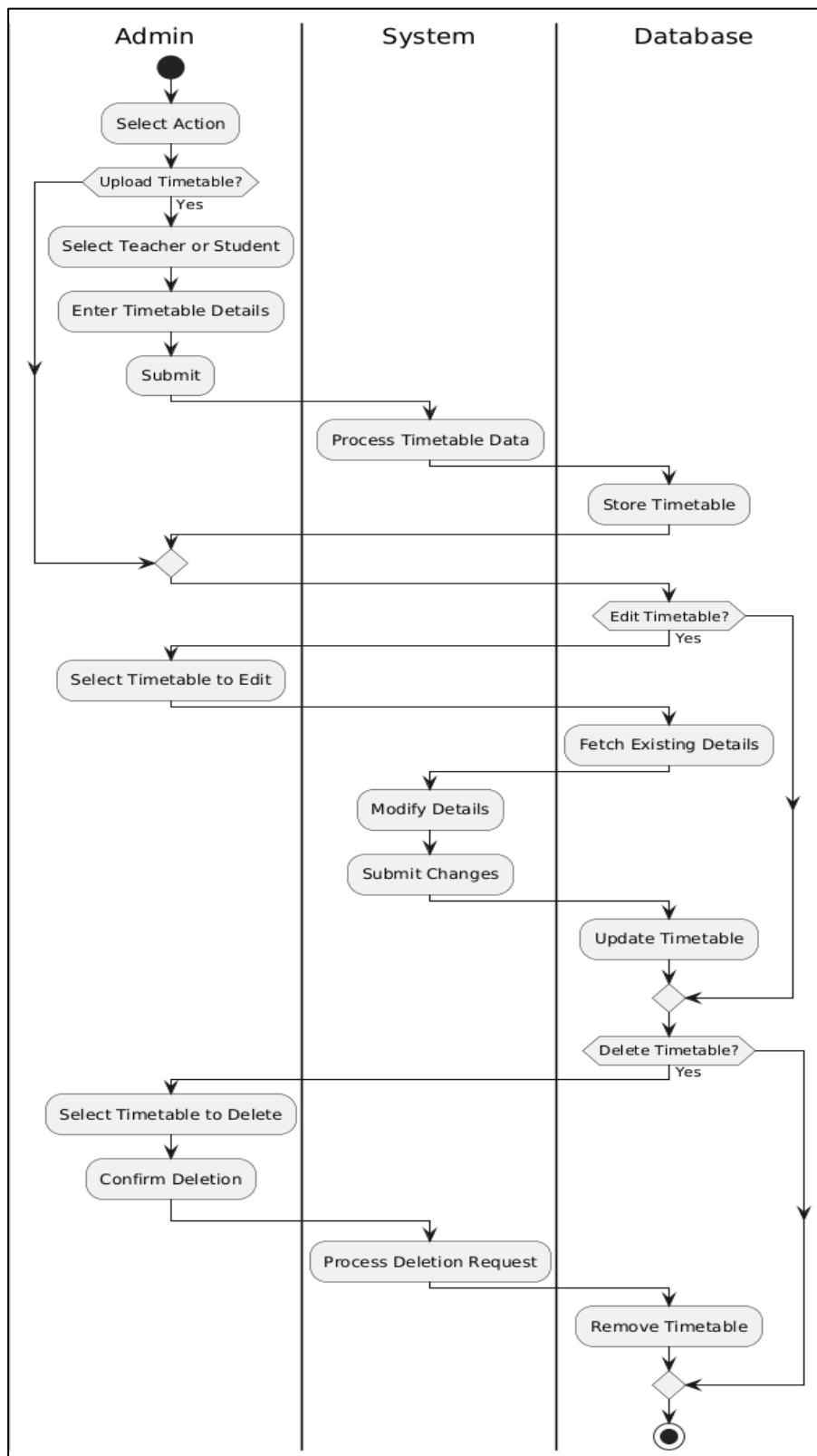
▪ **Module – 6 Uploading Test Marks & Attendance**



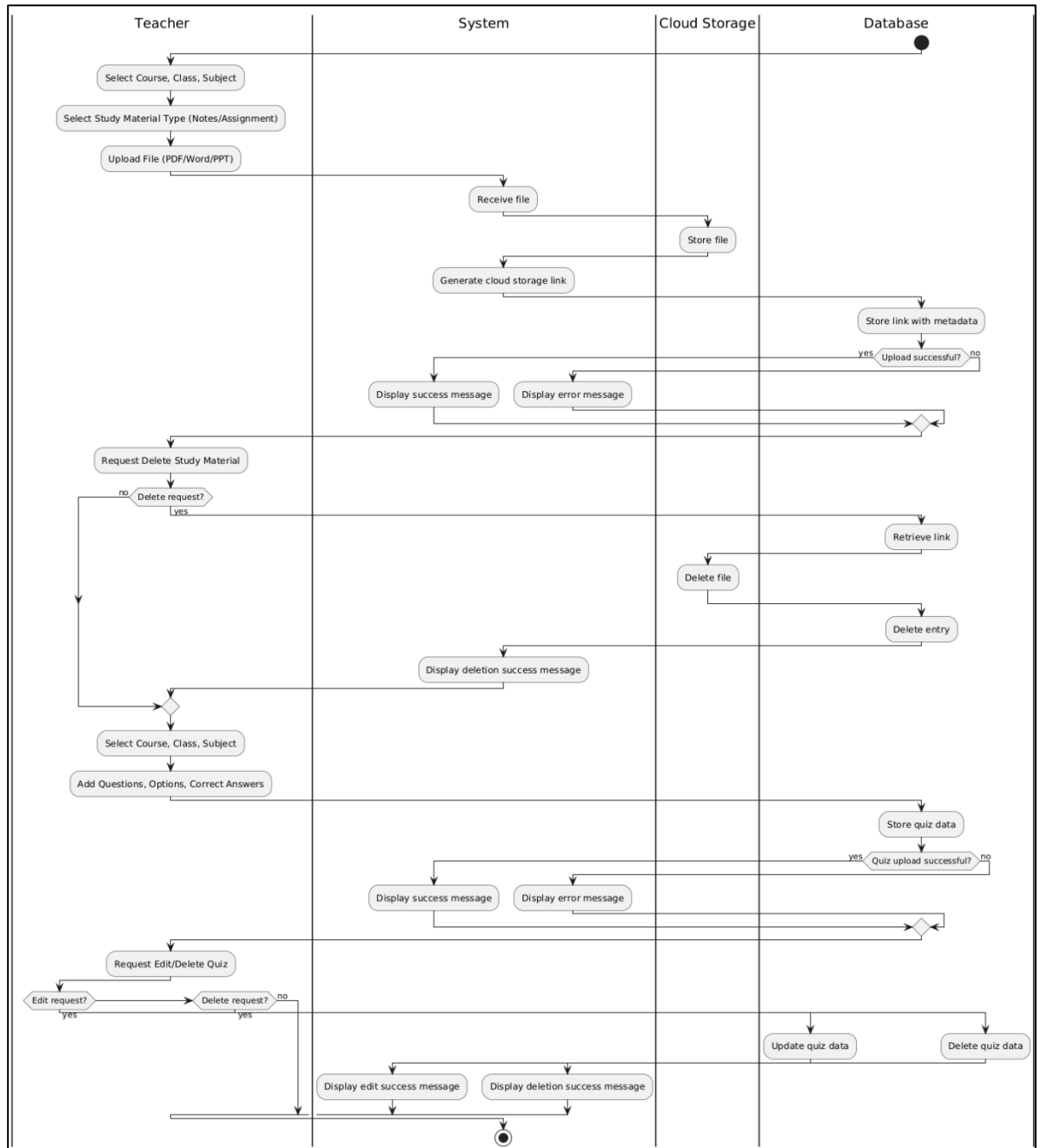
This diagram illustrates the process of uploading attendance and test marks. A **Teacher** first selects a course, class, and subject. The **System** fetches the student list from the **Database**. The teacher marks attendance, which the system stores in the database and updates student dashboards. At month's end, the system generates an attendance report and sends SMS to parents. The teacher then selects the same course, class, and subject to enter test marks. The system fetches the student list again. The teacher enters marks, which are stored in the database, student dashboards are updated, and a notification is sent. At month's end, the system generates a progress report and sends SMS to parents.

▪ **Module – 7 Timetable Management**

This diagram depicts the timetable management process. An **Admin** selects an action, choosing to upload a timetable. If "Yes," they specify whether it's for teachers or students, enter timetable details, and submit. The **System** processes this data and stores it in the **Database**. The admin can also edit a timetable. If "Yes," they select the timetable, the system fetches its details, the admin modifies them, and the system updates the database. Finally, the admin can delete a timetable. If "Yes," they select it, confirm deletion, the system processes the request, and removes the timetable from the database. This outlines the workflow for creating, editing, and deleting timetables.



■ Module – 8 Uploading Study Material



This diagram illustrates the process of uploading study materials and quizzes. A **Teacher** selects a course, class, and subject. They choose the material type (notes/assignment), upload a file (PDF/Word/PPT), and the **System** receives it. The system stores the file in **Cloud Storage**, generates a link, and stores the link with

metadata in the **Database**. If the upload is successful, a success message is displayed; otherwise, an error message appears. The teacher can request to delete study material. If "Yes," the system retrieves the link, deletes the file from Cloud Storage, and removes the entry from the database, displaying a success message. The teacher can also add quiz questions, options, and answers. This data is stored in the database. A success or error message is shown based on the upload. The teacher can request to edit or delete a quiz. If "Edit," the quiz data is updated; if "Delete," it's removed from the database. A success message is displayed accordingly. This outlines the flow for teachers to manage study materials and quizzes.

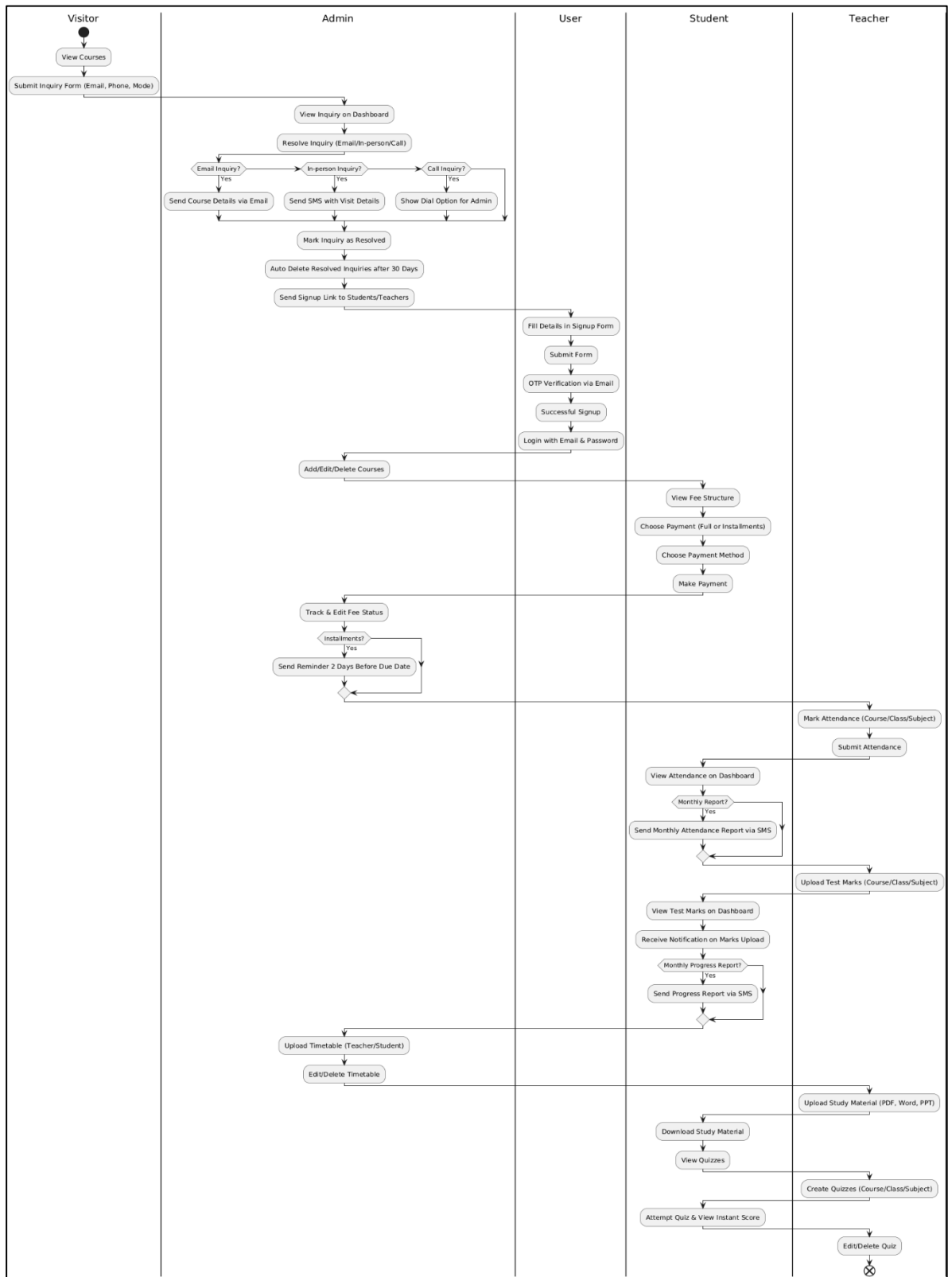
Overall Activity Diagram

This diagram presents a comprehensive overview of user interactions across various roles within a learning management system. A **Visitor** can submit inquiry forms, choosing different communication modes (email, in-person, call). The **Admin** views and resolves these inquiries, sending relevant information and managing follow-ups. The admin also handles user sign-ups, approving or rejecting them.

Users (including students and teachers) receive OTPs for verification, log in, and view available courses. The **Student** role encompasses fee payment, attendance tracking, viewing study materials, attempting quizzes, and receiving notifications and progress reports. They can also view the fee structure and select payment modes.

The **Teacher** role focuses on marking attendance, uploading study materials and test marks, and creating quizzes.

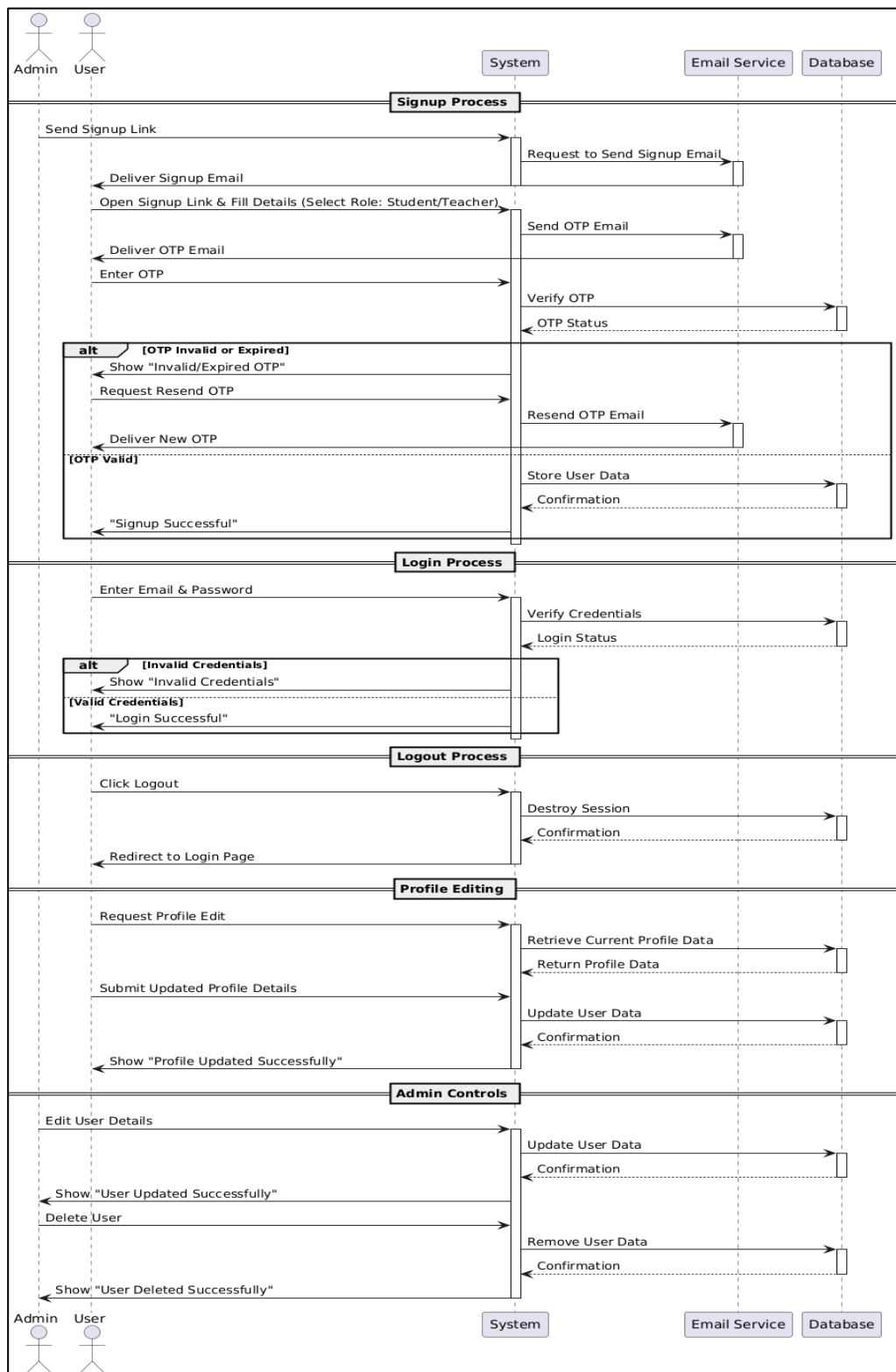
The **System** automates processes like sending reminders, generating reports, managing file storage, and handling database interactions. It ensures seamless communication and data management across all user roles. This diagram effectively encapsulates the end-to-end workflow of the learning platform.



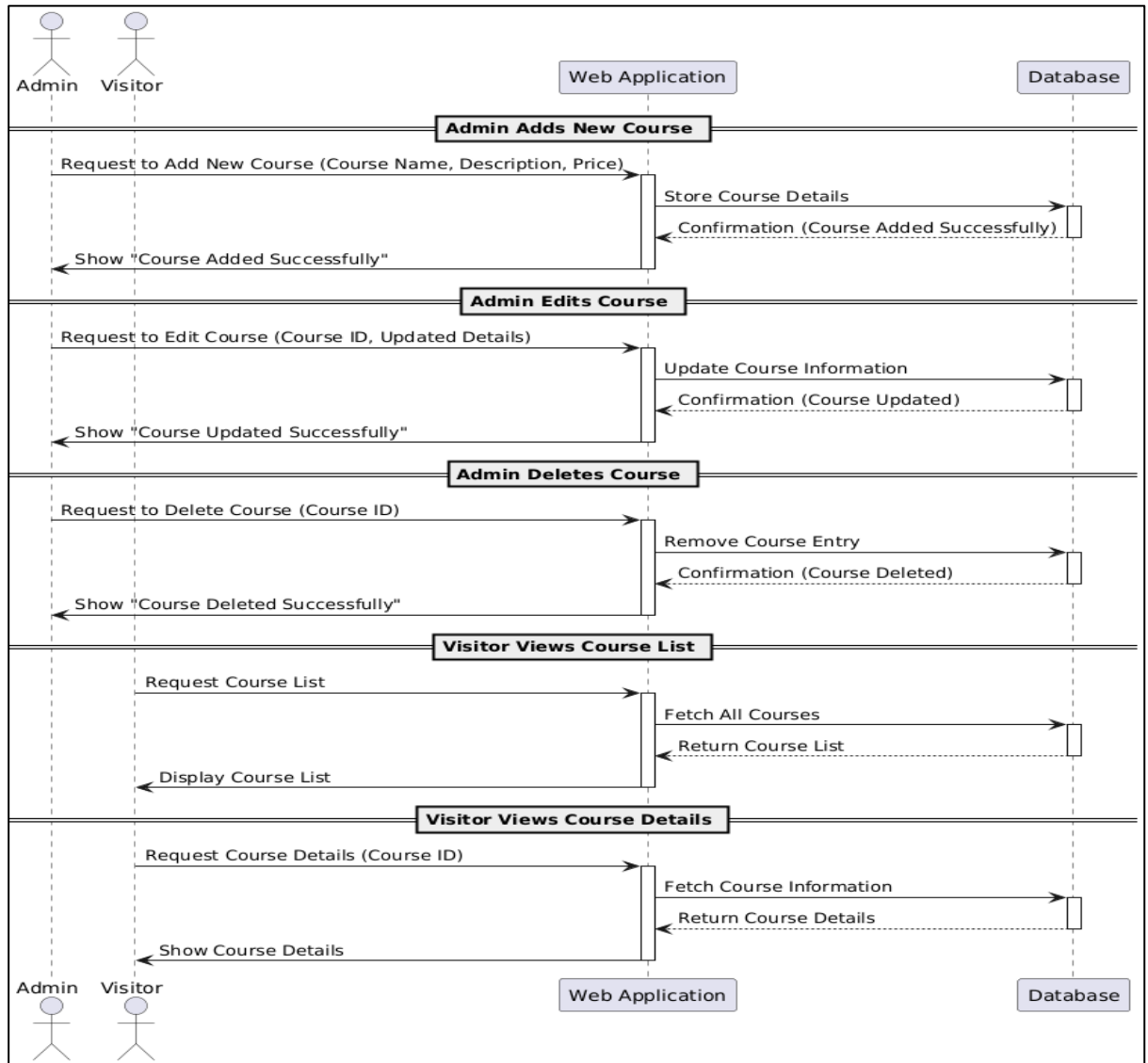
8.2.3 Sequence Diagram

▪ Module – 1 User Authentication & Management

This sequence diagram outlines the user authentication and management processes. An **Admin** or **User** initiates the signup process by requesting a signup link. The **System** sends an email with the link. The user fills out the form and receives an OTP via email. The system verifies the OTP. If valid, the user's data is stored, and a confirmation is displayed. For login, the user enters credentials, which the system verifies. If valid, the user logs in; otherwise, an error message is shown. Logout involves the user clicking logout, the system destroying the session, and redirecting to the login page. Users can request to edit their profiles, with the system retrieving, updating, and confirming the changes. **Admin** users have additional controls to edit or delete other user accounts, with corresponding system updates and confirmations. This diagram effectively illustrates the flow of user authentication and account management within the system.



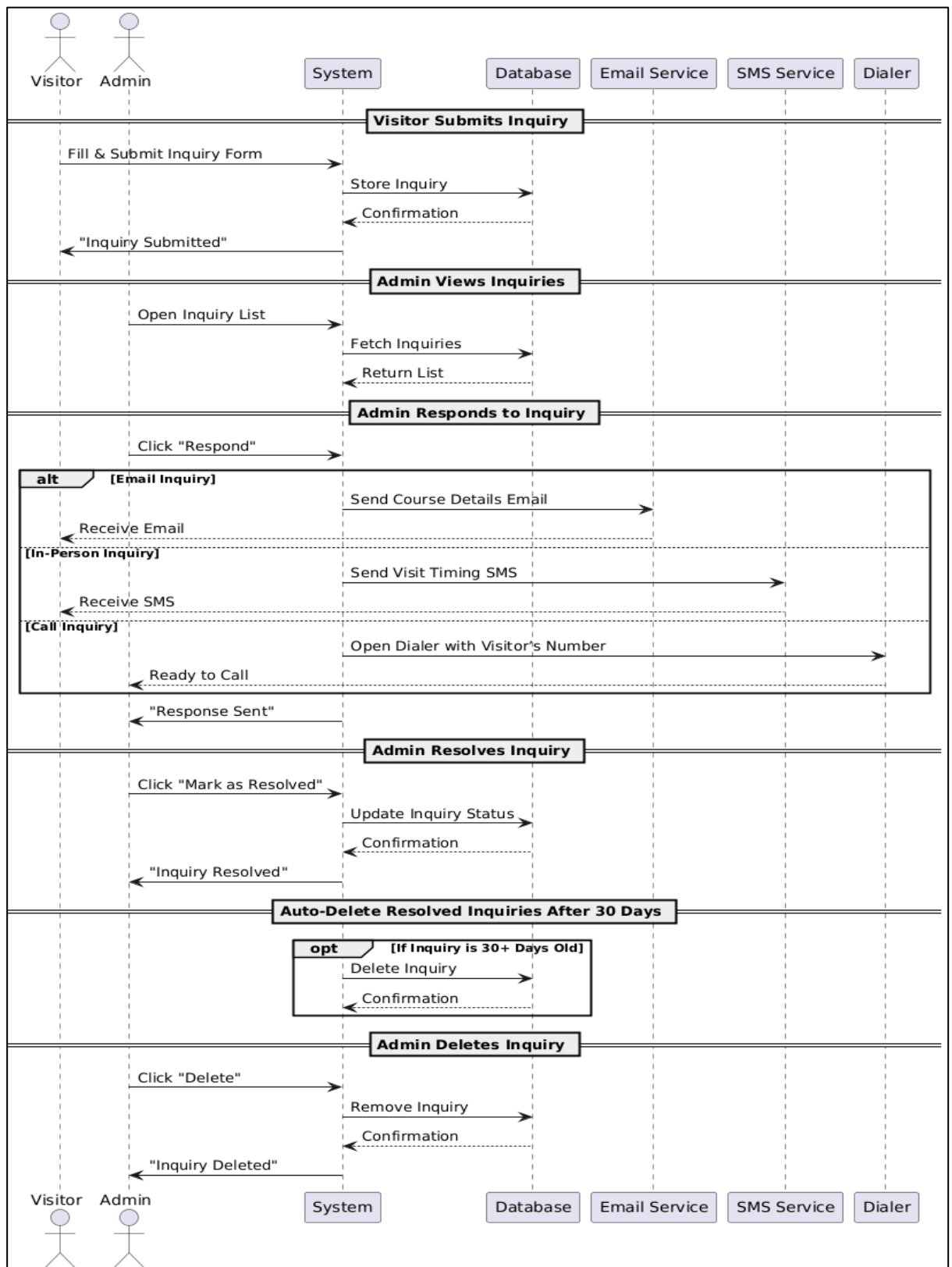
■ Module – 2 Course Management



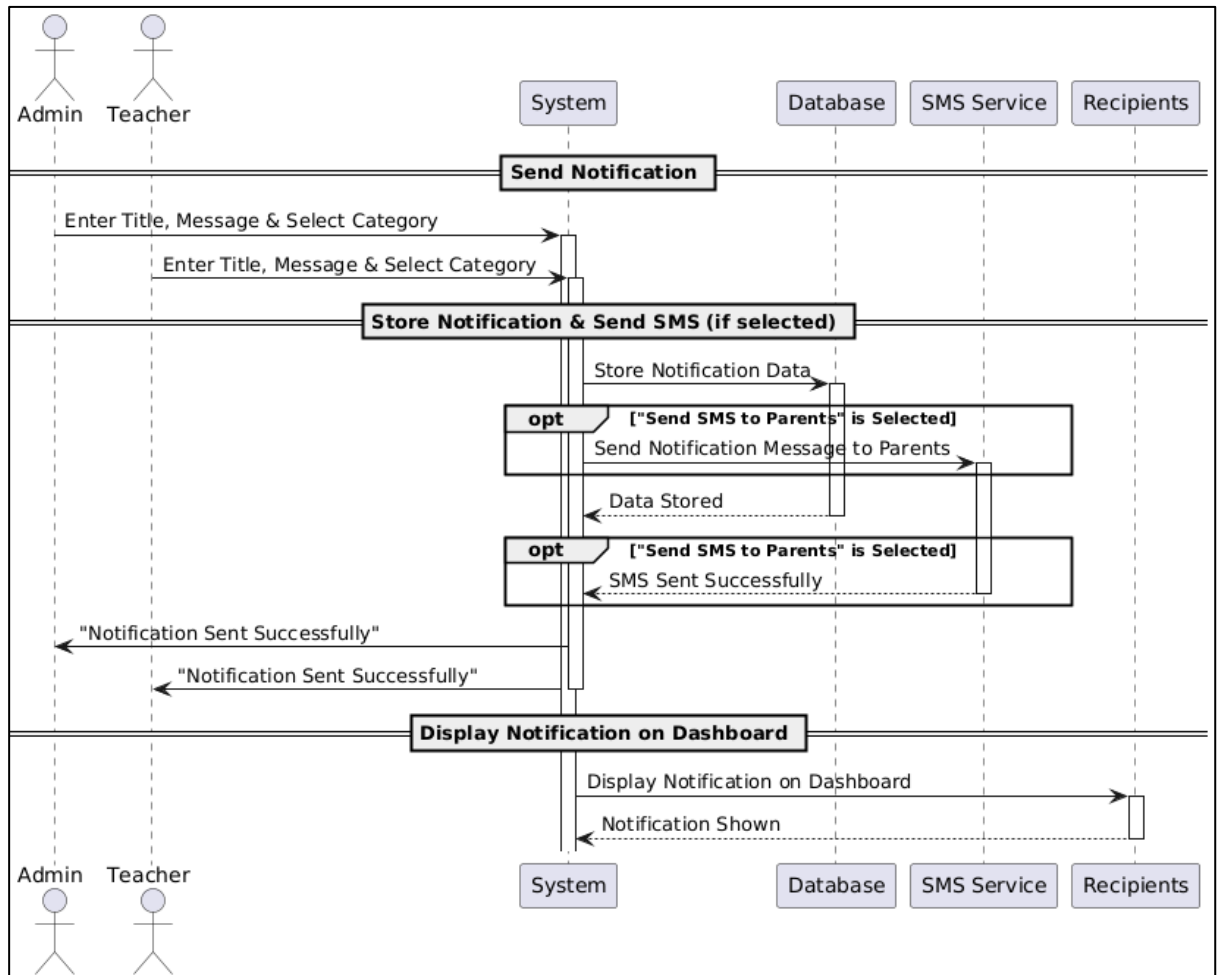
This sequence diagram illustrates the interactions for managing courses. An **Admin** can add a new course by sending a request with course details. The application stores this data in the **Database** and returns a confirmation message. The admin can also edit an existing course by providing the course ID and updated details. The application updates the course in the database and confirms the change. To delete a course, the admin sends the course ID, the application removes the entry from the database, and confirms the deletion. A **Visitor** can view the list of courses by requesting it from the web application. The application fetches all courses from the database and displays them to the visitor.

▪ **Module – 3 Consultation & Enquiry**

This sequence diagram illustrates the flow of a consultation and inquiry process. A **Visitor** initiates by filling and submitting an inquiry form. The **System** stores this inquiry in the **Database** and confirms the submission to the visitor. An **Admin** then views the inquiries by opening the inquiry list, which the system fetches from the database and returns. The admin selects an inquiry to respond to. Depending on the inquiry mode (email, in-person, or call), the system sends course details via **Email Service**, sends visit timing via **SMS Service**, or opens the **Dialer** with the visitor's number. After responding, the admin marks the inquiry as resolved, updating its status in the database. The system automatically deletes resolved inquiries older than 30 days from the database. Finally, the admin can manually delete an inquiry, which the system removes from the database. This diagram effectively outlines the interactions between the visitor, admin, system, and external services for managing consultation and inquiries.



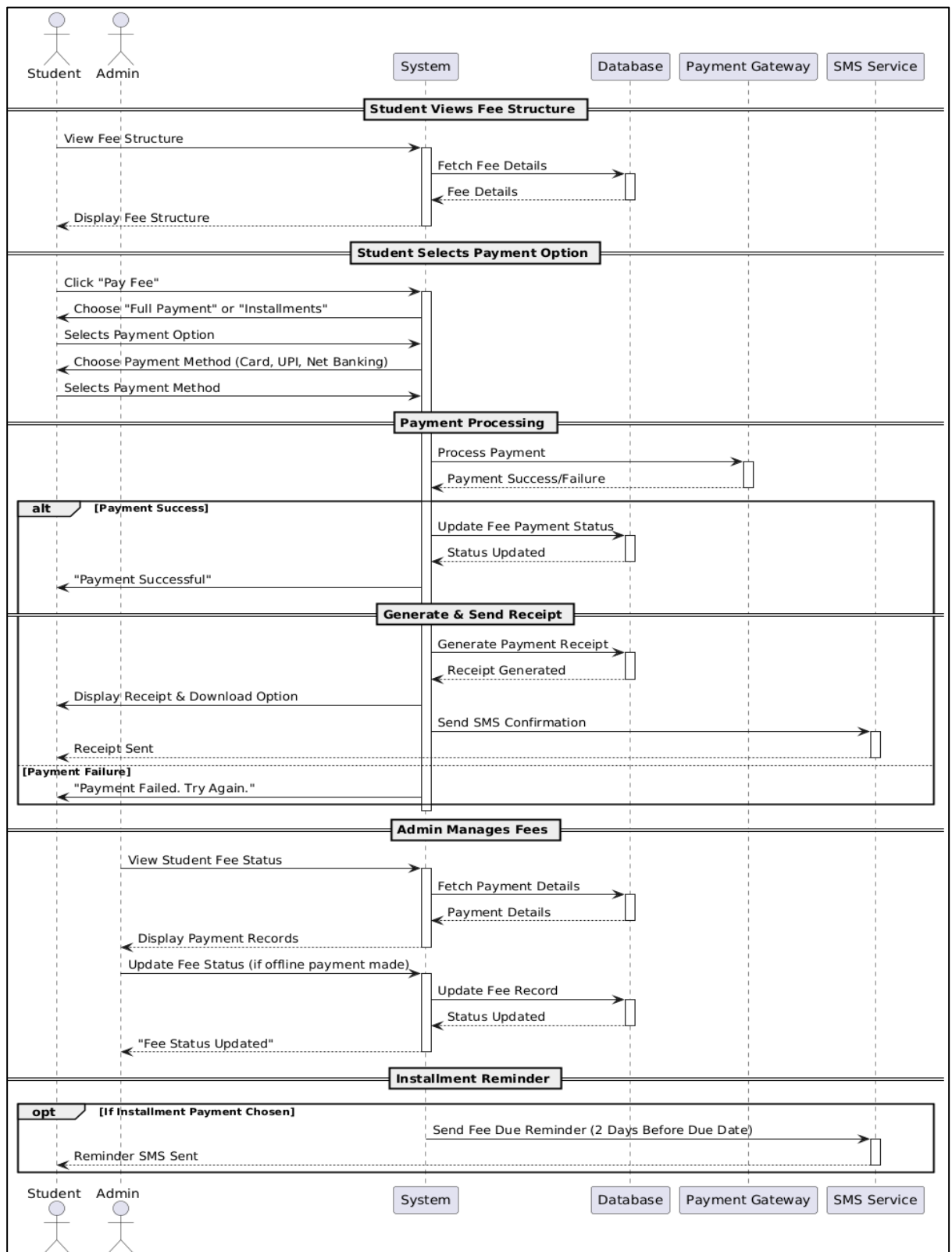
▪ Module – 4 Announcement & Notifications



This sequence diagram outlines the process of sending announcements and notifications. An **Admin** or **Teacher** initiates by entering the title, message, and selecting the recipient category. The **System** then stores the notification data in the **Database**. If the "Send SMS to Parents" option is selected, the system sends the notification message to parents via the **SMS Service**, and confirms the data storage and SMS delivery. The system then displays the notification on the dashboard for the **Recipients**. Finally, the system confirms the successful sending of the notification to the admin or teacher.

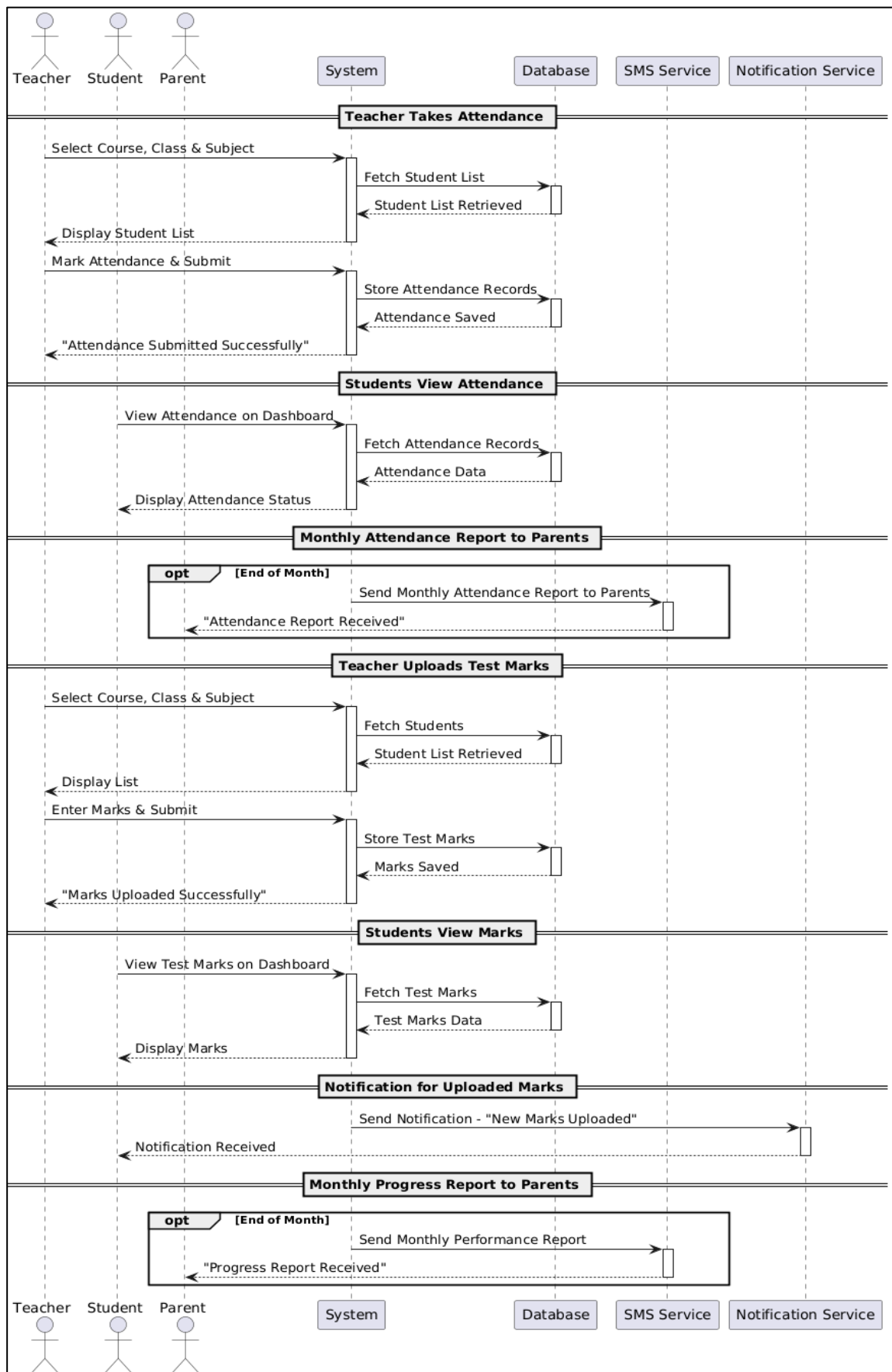
▪ **Module – 5 Fee Management**

This sequence diagram illustrates the fee management process. A **Student** first views the fee structure by requesting it from the **System**. The system fetches the fee details from the **Database** and displays them to the student. The student then selects the payment option, choosing either full payment or installments, and selects a payment method (card, UPI, net banking). The **Payment Gateway** processes the payment and returns a success or failure message to the system. If successful, the system updates the fee payment status in the database, generates a payment receipt, displays it to the student with a download option, and sends an SMS confirmation to the student via the **SMS Service**. If the payment fails, an error message is displayed to the student. An **Admin** can view student fee status by requesting payment details from the system, which fetches them from the database and displays them. The admin can also update the fee status for offline payments, which the system updates in the database and confirms. If the student chose installment payments, the system sends a fee due reminder SMS two days before the due date via the SMS Service. This diagram effectively outlines the interactions for students and admins related to fee management.



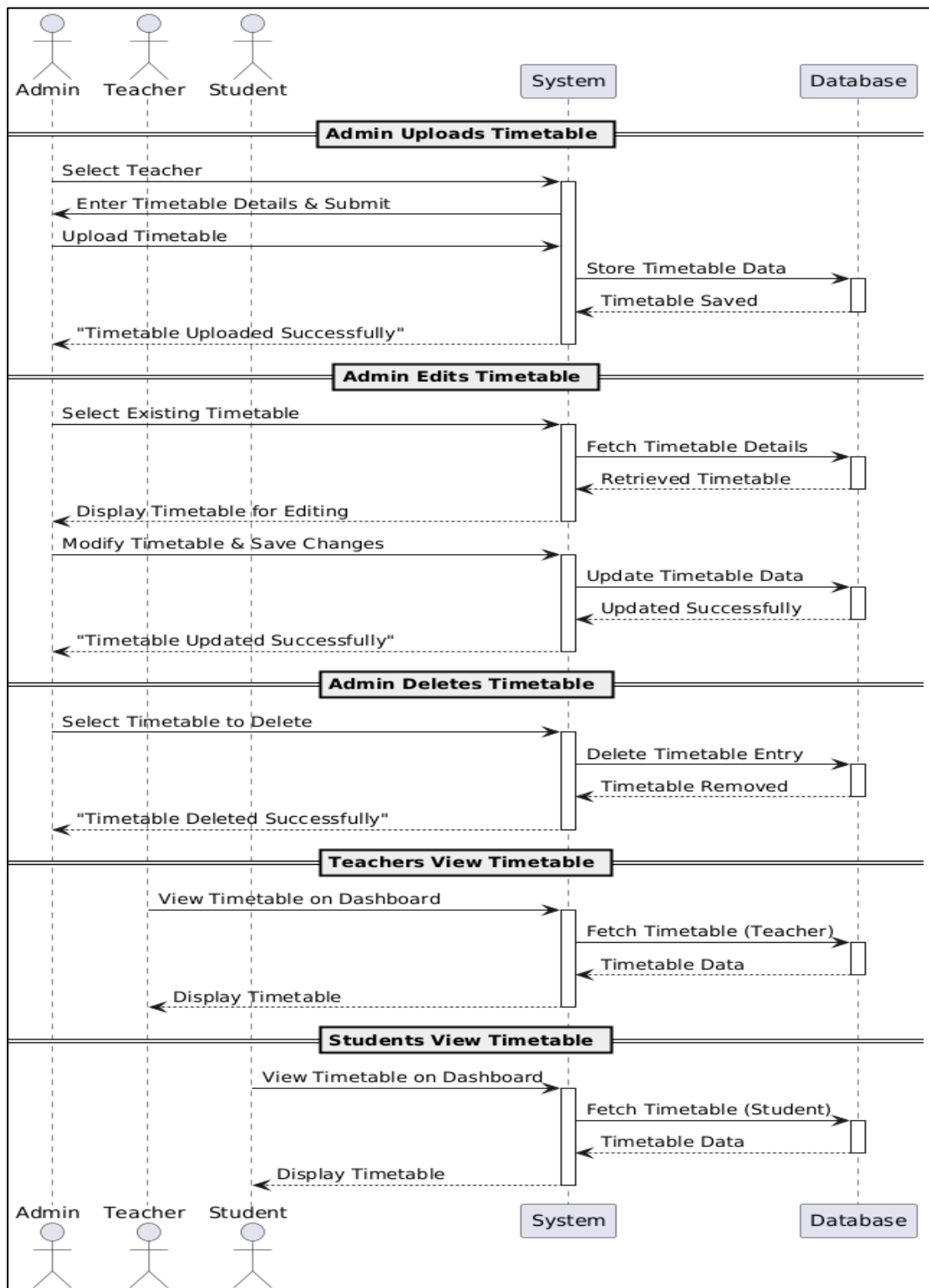
▪ **Module – 6 Uploading Test Marks & Attendance**

This sequence diagram outlines the process of uploading and viewing attendance and test marks. A Teacher selects a course, class, and subject, and the system fetches and displays the student list. The teacher marks attendance or enters test scores, submits them, and the system stores the data, confirms the save, and displays a success message. Students can view their records on the dashboard, fetched from the database. Monthly, the system may send attendance and progress reports to parents via SMS. A notification is also sent when new test marks are uploaded, ensuring seamless tracking of student performance.



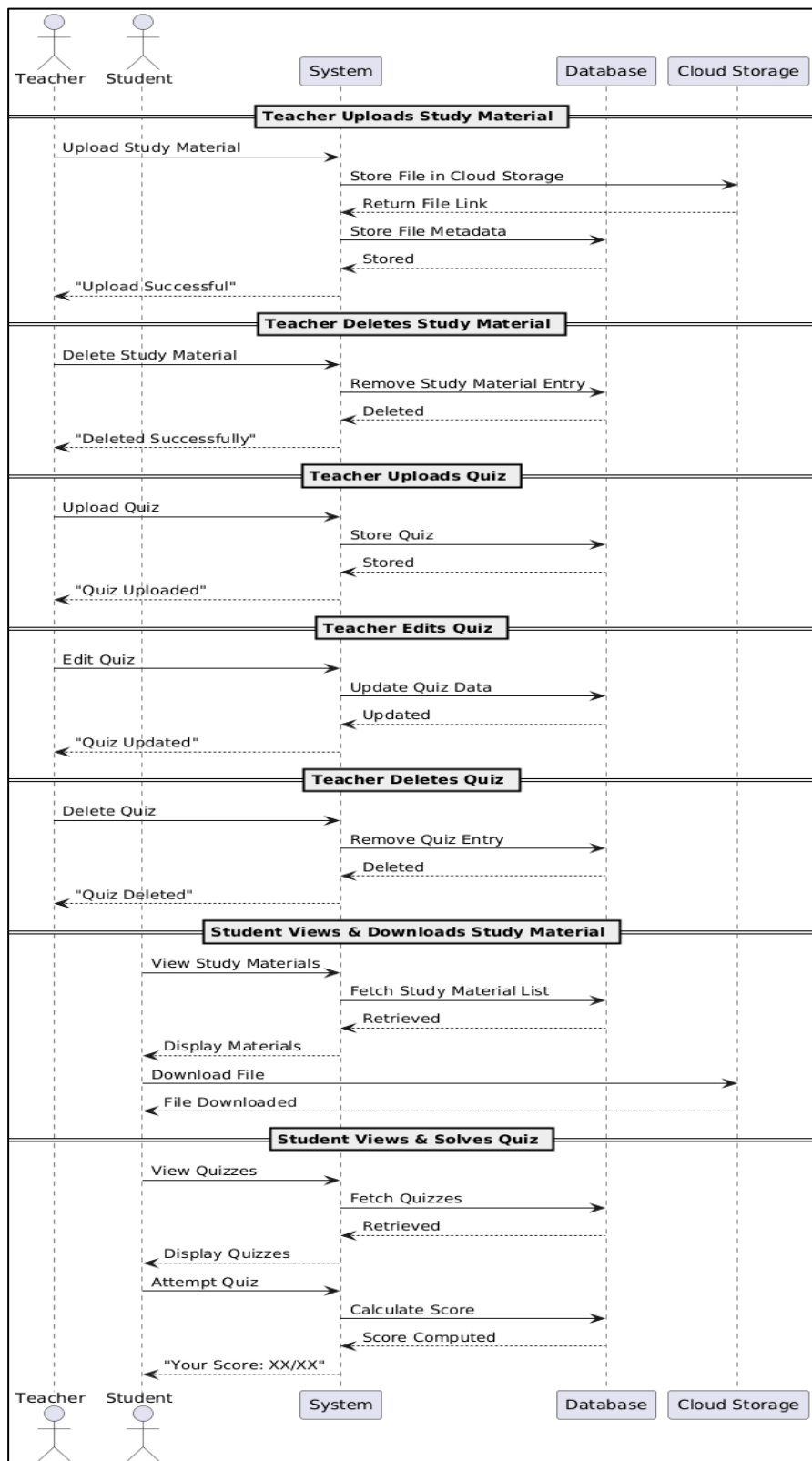
▪ **Module – 7 Timetable Management**

This sequence diagram illustrates the interactions for managing timetables. An **Admin** initiates by selecting whether the timetable is for teachers or students. They then enter the timetable details and submit them. The **System** stores this data in the **Database** and confirms the save. The system then displays a success message to the admin. The admin can also edit an existing timetable. They select the timetable, the system retrieves the details from the database, and displays them for editing. The admin modifies the timetable and saves the changes. The system updates the timetable in the database and confirms the update, displaying a success message. To delete a timetable, the admin selects it, the system removes the entry from the database, and confirms the deletion. **Teachers** can view their timetable on the dashboard. The system fetches the teacher's timetable from the database and displays it. **Students** can also view their timetable on the dashboard. The system fetches the student's timetable from the database and displays it. This diagram outlines the workflow for creating, editing, deleting, and viewing timetables.

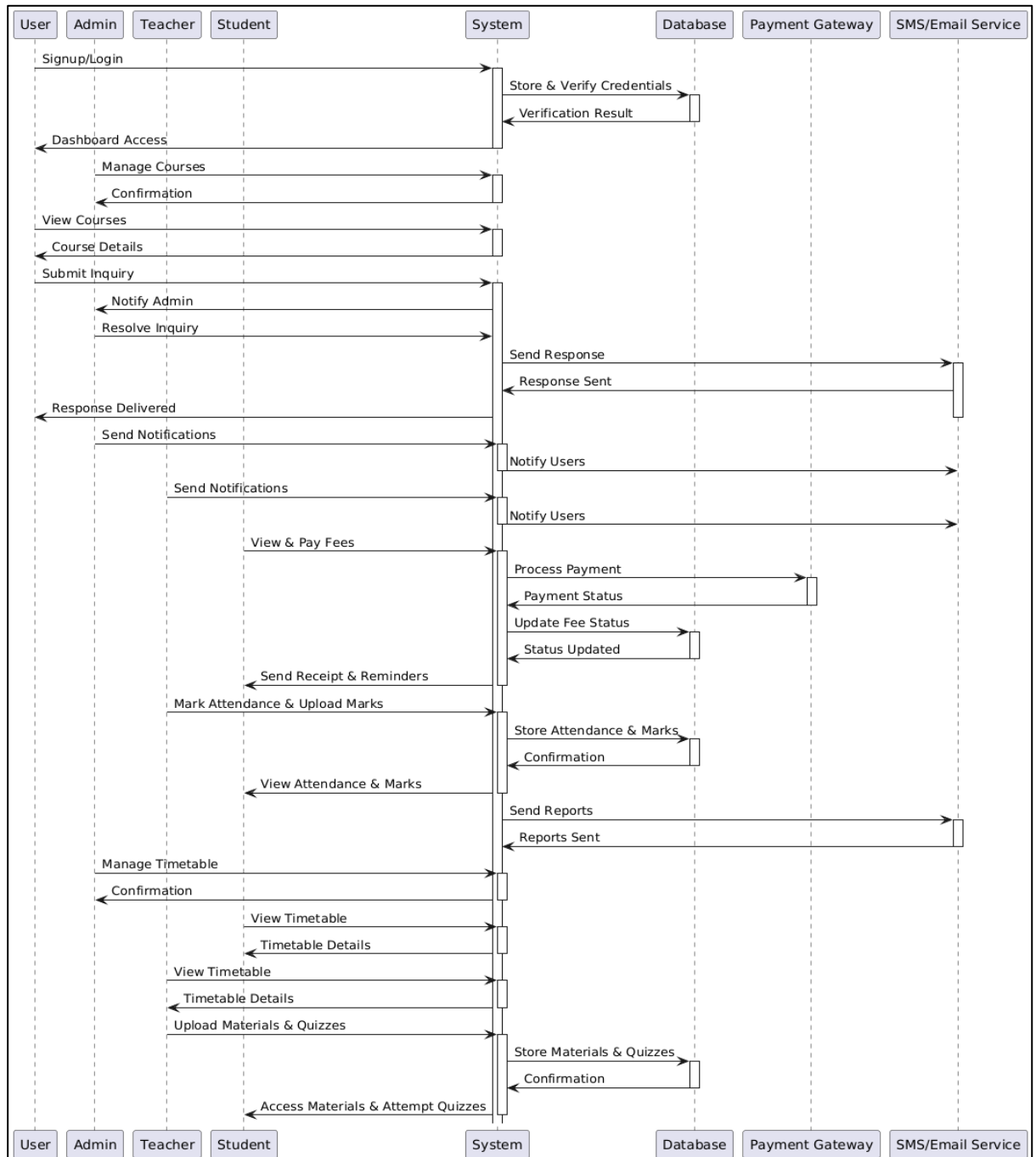


▪ **Module – 8 Uploading Study Material**

This sequence diagram illustrates the workflow for managing study materials and quizzes. A Teacher uploads study materials, which the system stores in Cloud Storage, generating a link saved in the database. Teachers can delete materials, upload, edit, or delete quizzes, with all changes reflected in the database. Students can view, download materials, and attempt quizzes. The system fetches and displays content, calculates quiz scores, and confirms actions. This diagram outlines teacher management and student access efficiently.



Overall Sequence Diagram



This sequence diagram provides an overview of interactions within the Shreeji Classes system. Users (Admins, Teachers, Students) sign up or log in, with the system verifying credentials. Upon successful login, users access the dashboard. Admins manage courses, timetables, notifications, and inquiries, with the system confirming actions and sending responses via SMS/Email. Students view courses, submit inquiries, pay fees, and receive payment confirmations. Teachers mark attendance, upload marks, study materials, and quizzes, while students access these records and attempt quizzes. The system processes payments, updates databases, and sends reports, effectively managing key interactions across all modules.

8.3 Data Modeling

8.3.1 Data Dictionary

The data dictionary provides definitions for each field in the database schema. The following tables outline the fields, data types, constraints, and descriptions for the key entities in the system.

User Collection

Field Name	Data Type	Constraints	Description
name	String	Required	User's full name
email	String	Required, Unique, Validated	User's email ID
password	String	Required, Min Length: 6, Must contain letters and numbers	User's encrypted password
role	String	Required, Enum: [admin, student, teacher]	Role of the user
contactNumber	String	Required, 10 Digits	User's phone number
otp	String	Optional	OTP for authentication
otpExpires	Date	Optional	OTP expiration time
signupToken	String	Optional	Token for signup verification
signupTokenExpires	Date	Optional	Expiry for signup token

parentContactNumber	String	Required for students, 10 Digits	Parent's phone number (for students)
status	String	Default: 'pending', Enum: [pending, approved]	Account approval status
studentClass	Array of Strings	Required for students	The class student belongs to
selectedCourse	Array of Strings	Required for students, Enum: [Grade 6-10, JEE/NEET, etc.]	Course(s) selected by the student
subjects	Array of Strings	Required if selectedCourse is set	Subjects based on selected course
teachesClass	Array of Strings	Required for teachers	Classes a teacher teaches
taughtCourses	Array of Objects	Required for teachers	Courses and subjects a teacher teaches
teacherSubjects	Array of Objects	Required for teachers	Subjects taught by the teacher

Attendance Collection

Field Name	Data Type	Constraints	Description
studentId	ObjectId	Required, References User	Student who attended
teacherId	ObjectId	Required, References User	Teacher marking attendance
date	Date	Required, Default: Date.now	Date of attendance

status	String	Required, Enum: [Present, Absent]	Attendance status
selectedCourse	String	Required	Course for which attendance is marked
studentClass	String	Required	Class student belongs to
subjects	String	Required	Subject of the attendance

Course Collection

Field Name	Data Type	Constraints	Description
category	String	Required	Course category (e.g., JEE, NEET)
board	Array of Strings	Optional	Boards applicable to this course
subjects	Array of Strings	Required	Subjects covered in the course
duration	String	Required	Course duration
description	String	Required	Detailed course description
highlights	Array of Strings	Required	Key highlights of the course
weeklyTests	Boolean	Default: true	Whether the course has weekly tests
imageUrl	String	Required	Image URL for the course
additionalDetails	String	Required	Additional course-related info

Fee Details Collection

Field Name	Data Type	Constraints	Description
selectedCourse	String	Required	Course for which fee details apply

studentClass	Number	Required	Class of the student
subjects	Array of Objects	Required	Subjects with their respective monthly fees
fullPackageMonthlyFee	Number	Required	Monthly fee for the full package
installments	Array of Strings	Optional	Available installment plans

File Collection

Field Name	Data Type	Constraints	Description
fileName	String	Required	Name of the uploaded file
fileUrl	String	Required	URL for accessing the file
fileType	String	Required	Type of file (PDF, DOC, etc.)
uploadedBy	ObjectId	Required, References User	User who uploaded the file
selectedCourse	String	Required, Enum: [Various Course Categories]	Course the file belongs to
studentClass	String	Required, Enum: [Various Classes]	Class level for the file
subjects	String	Required, Enum: [Subject List]	Subject the file is related to
materialType	String	Required, Enum: [Notes, Assignment]	Type of study material
createdAt	Date	Default: Date.now	Upload timestamp

Inquiry Collection

Field Name	Data Type	Constraints	Description
contactNumber	String	Required	Contact number of the inquirer
email	String	Required	Email of the inquirer
enquiryMode	String	Required, Enum: [call, email, in-person]	Mode of inquiry
status	String	Default: "pending", Enum: [pending, resolved]	Status of the inquiry

Marks Collection

Field Name	Data Type	Constraints	Description
studentId	ObjectId	Required, References User	Student who took the test
teacherId	ObjectId	Required, References User	Teacher who assigned the test
studentClass	String	Required, Enum: [Various Classes]	Class level of the student
selectedCourse	String	Required, Enum: [Various Courses]	Course the student is enrolled in
subjects	String	Required, Enum: [Subjects List]	Subject of the test
testTitle	String	Required, Max Length: 100	Title of the test
testDate	Date	Required	Date of the test
marksObtained	Number	Required, Min: 0	Marks scored by the student
totalMarks	Number	Required, Min: 1	Total marks for the test

percentage	Number	Default: Auto-calculated (marksObtained/totalMarks * 100)	Percentage of the student
remarks	String	Max Length: 200	Remarks from the teacher
status	String	Default: "pending", Enum: [pending, completed]	Status of the test

Notification Collection

Field Name	Data Type	Constraints	Description
title	String	Required	Notification title
message	String	Required	Message content
sender	ObjectId	Required, References User	Sender of the notification
sendSMSFlag	Boolean	Default: false	Whether SMS should be sent
recipientType	String	Required, Enum: [Various Recipient Types]	Target audience of the notification
className	String	Optional	Class level the notification is related to
courseName	String	Optional	Course the notification is related to
subjectName	String	Optional	Subject of the notification
sendToParents	Boolean	Default: false	Whether parents should receive the notification
createdAt	Date	Default: Date.now	Notification timestamp

Quiz Collection

Field Name	Data Type	Constraints	Description
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title	String	Required	Title of the quiz
selectedCourse	String	Required, Enum: [Various Courses]	Course the quiz belongs to
studentClass	String	Required, Enum: [Various Classes]	Class level the quiz is intended for
subjects	String	Required, Enum: [Subjects List]	Subject of the quiz
createdBy	ObjectId	Required, References User	Teacher who created the quiz
questions	Array of Objects	Required	List of questions, options, and correct answers
createdAt	Date	Default: Date.now	Quiz creation timestamp

Study Material Collection

Field Name	Data Type	Constraints	Description
title	String	Required	Title of the study material
category	String	Required	Category of the study material
selectedCourse	String	Required, Enum: [Various Courses]	Course the material belongs to
studentClass	String	Required, Enum: [Various Classes]	Class level the material is intended for
subjects	String	Required, Enum: [Subjects List]	Subject of the material
fileUrl	String	Required	URL to access the study material file
uploadedBy	ObjectId	Required, References User	User who uploaded the material

createdAt	Date	Default: Date.now	Material upload timestamp
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TimeTable Collection

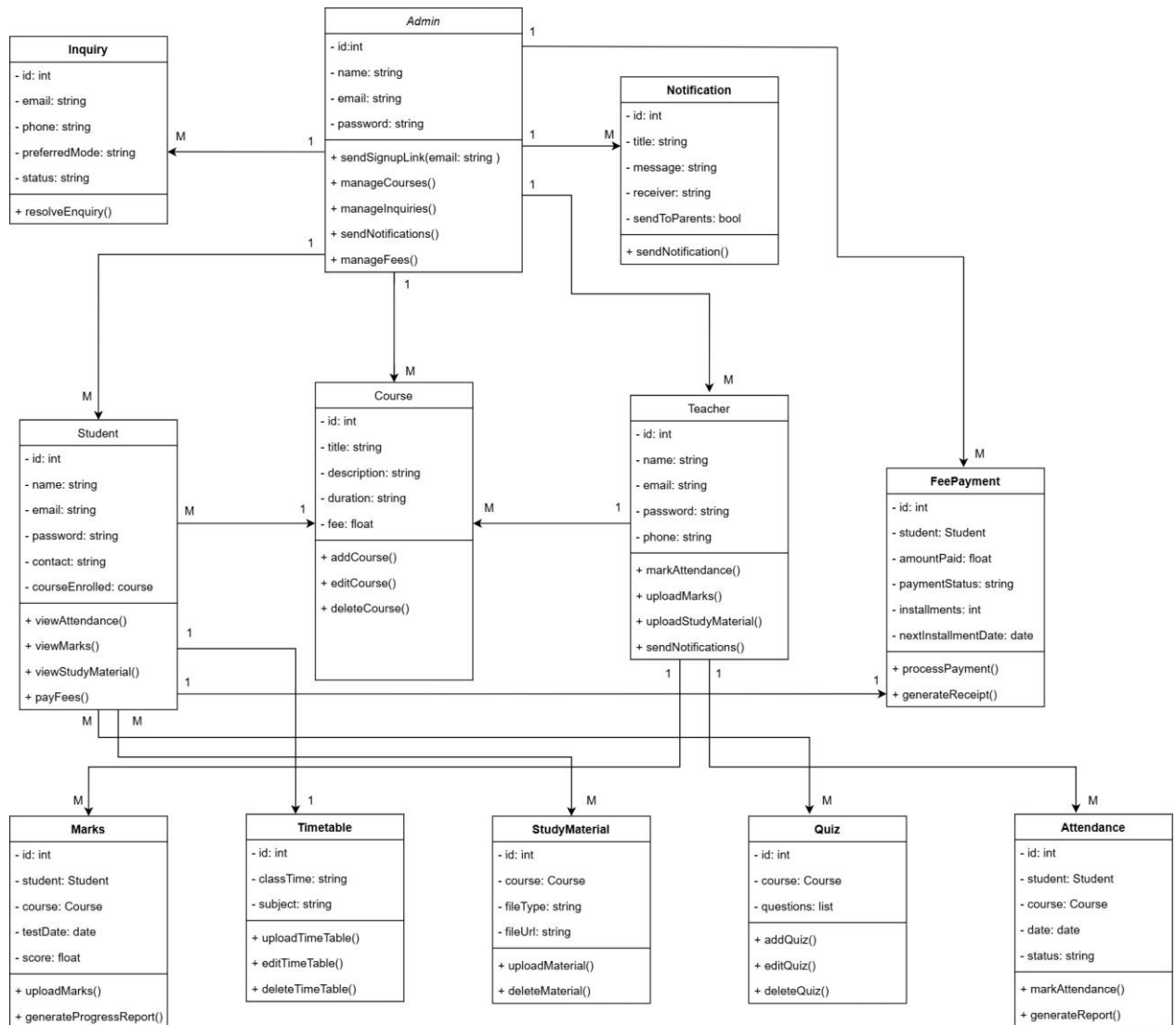
Field Name	Data Type	Constraints	Description
userType	String	Required, Enum: [teacher, student]	Type of user for whom timetable is created
name	String	Required if userType is "teacher"	Teacher's name
teacherId	ObjectId	References User	ID of the teacher (if applicable)
studentId	ObjectId	References User	ID of the student (if applicable)
selectedCourse	String	Required, Enum: [Various Courses]	Course for the timetable
subjects	Array of Strings	Required, Enum: [Subjects List]	Subjects covered in the timetable
timeSlots	Array of Objects	Required	List of time slots for the classes
createdAt	Date	Default: Date.now	Timetable creation timestamp

Payment Collection

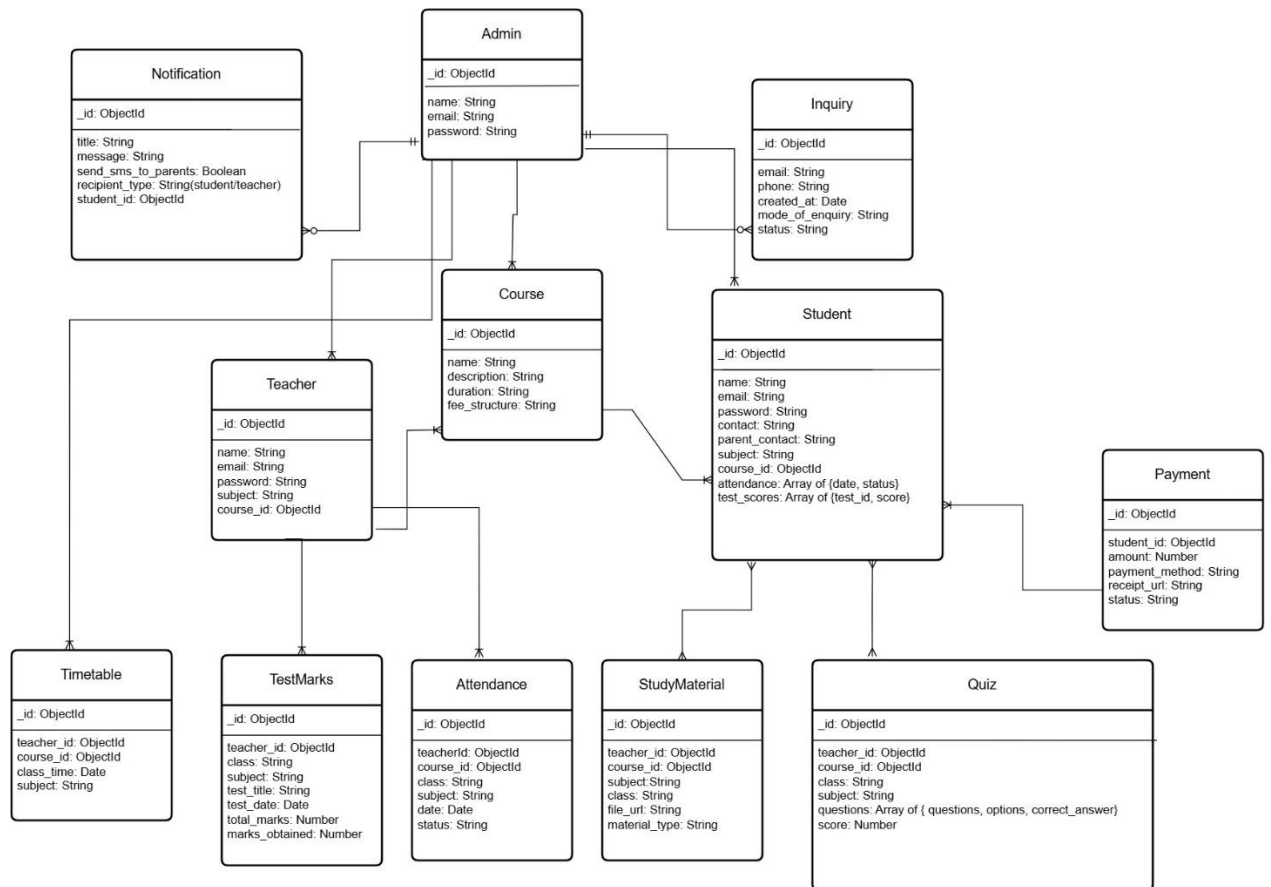
Field Name	Data Type	Constraints	Description
userId	ObjectId	Required, References User	User making the payment
totalFee	Number	Required	Total fee for the course

paidAmount	Number	Default: 0	Total amount paid so far
remainingAmount	Number	Auto-calculated	Remaining amount to be paid
amount	Number	Required	Last paid installment amount
paymentType	String	Required, Enum: [full, installment]	Type of payment
installmentPlan	String	Optional, Enum: [3 Months, 6 Months]	Installment plan chosen
remainingInstallments	Number	Default: 0	Remaining installment count
nextDueDate	Date	Auto-calculated	Due date for next installment
status	String	Default: "pending", Enum: [pending, paid, installments_pending, completed]	Payment status
razorpayOrderId	String	Optional	Razorpay Order ID
razorpayPaymentId	String	Optional	Razorpay Payment ID
createdAt	Date	Default: Date.now	Payment timestamp

8.3.2 Class Diagram

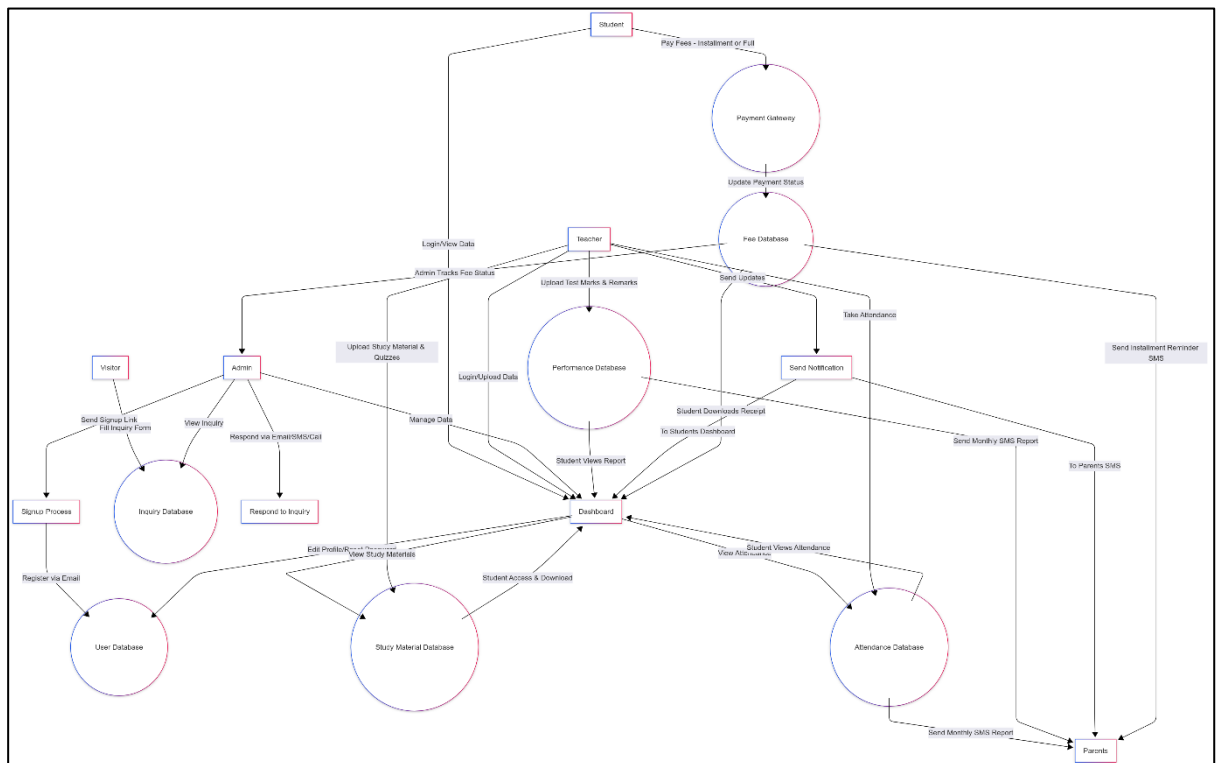


8.3.3 ER Diagram



8.4 Functional And Behaviorial Modeling

8.4.1 Data flow diagram



This Data Flow Diagram (DFD) visually represents the functional aspects of Shreeji Classes, outlining the flow of data between various actors and processes. The system caters to students, administrators, and visitors, each interacting with the platform in distinct ways. Students can log in, view data, pay fees through a payment gateway, and access their attendance records. The administrator plays a crucial role in managing the platform by uploading study materials, marking attendance, creating student news reports, and sending notifications via SMS or email. Visitors can register their email and submit inquiries. The system processes include login/data viewing, task management, content uploading, attendance tracking, and report generation, all centered around a "Dashboard" which acts as a central access point. Data is stored and managed within databases such as User, Enquiry, Study Material, Attendance, Payment, and Fee databases.

9. SYSTEM DESIGN

9.1 Database Design

The Shreeji Classes Web App follows a MongoDB-based database design using Mongoose as the ODM (Object Data Modeling) library. The following database schemas define different aspects of the system:

9.1.1 User Schema

- Role-based user management (Admin, Student, Teacher)
- Secure authentication using password encryption (bcrypt)
- OTP-based verification for account activation and login
- Class and course selection for students
- Subjects taught for teachers

9.1.2 Attendance Schema

- Tracks student attendance status (Present/Absent)
- Stores student and teacher references
- Course and class association for attendance tracking

9.1.3 Course Schema

- Contains course details (category, subjects, board, duration)
- Weekly test flag for test-based courses
- Image URL for course thumbnails

9.1.4 Fee Details Schema

- Stores fee structure based on selected courses and subjects
- Supports monthly fee breakdown and installment plans

9.1.5 File Upload Schema

- Manages study materials (Notes, Assignments)
- Categorized by class, subject, and course
- Stores file URL for easy access

9.1.6 Inquiry Schema

- Handles student/parent inquiries
- Categorizes inquiries by mode (call, email, in-person)
- Tracks inquiry status (Pending/Resolved)

9.1.7 Marks Schema

- Stores student test scores
- Auto-calculates percentage from obtained marks
- Keeps test history and remarks

9.1.8 Notification Schema

- Supports bulk messaging (SMS, in-app notifications)
- Targets specific users (students, teachers, parents)
- Includes class/course-specific announcements

9.1.9 Payment Schema

- Supports installment-based or full payment
- Integrated with Razorpay
- Auto-updates remaining balance and due dates

9.2 Interface Design

The Shreeji Classes Web App UI is designed for intuitive navigation and role-based access, ensuring a smooth experience for students, teachers, and admins.

9.2.1 Admin Interface

- User Management: Approve users and send signup links via Gmail.
- Course Management: Add/update/delete courses
- Fee & Payment Management: Track student payments
- Inquiry Management: View and respond to user inquiries
- Notifications: Send updates to students and teachers

9.2.2 Teacher Interface

- Manage Attendance: Mark students Present/Absent
- Upload Study Materials: Upload notes and assignments
- Enter Student Marks: Record test results
- Notifications: Receive updates from admin

9.2.3 Student Interface

- Attendance Tracking: View attendance history
- Access Study Materials: Download uploaded files
- Test Scores: View test results and progress
- Payments: Make fee payments and track dues

9.3 System Architecture

The system follows a MERN (MongoDB, Express.js, React, Node.js) stack, ensuring scalability and performance.

1. Backend (Node.js + Express.js)

- Handles API requests for authentication, payments, attendance, etc.
- Interacts with MongoDB using Mongoose models
- Implements security (JWT authentication, bcrypt password hashing)
- Manages Razorpay Integration for payments

2. Frontend (React + Tailwind CSS)

- Component-based UI for seamless interactions
- Responsive Design for accessibility across devices
- Role-Based Navigation to restrict unauthorized access

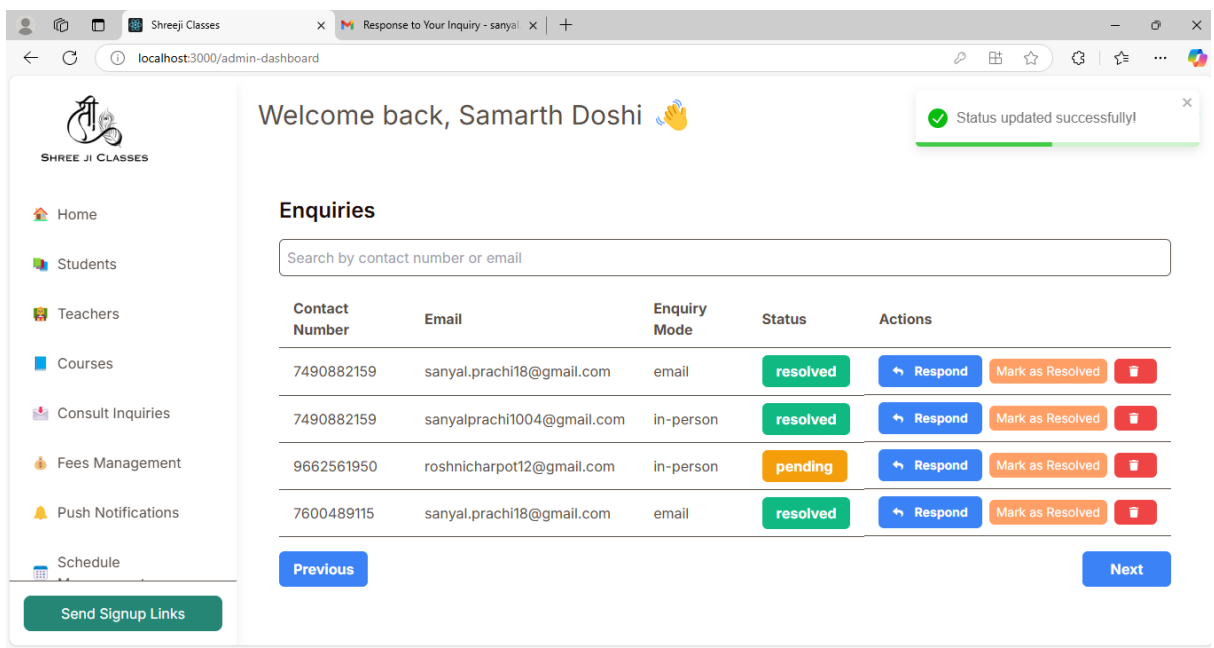
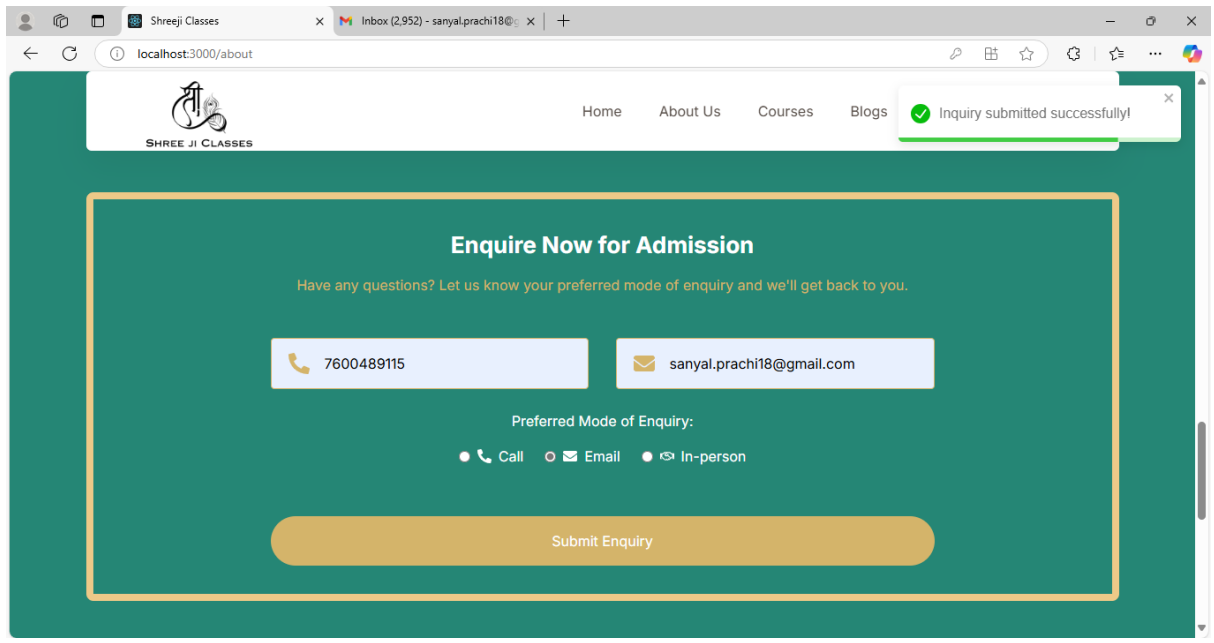
3 Database (MongoDB)

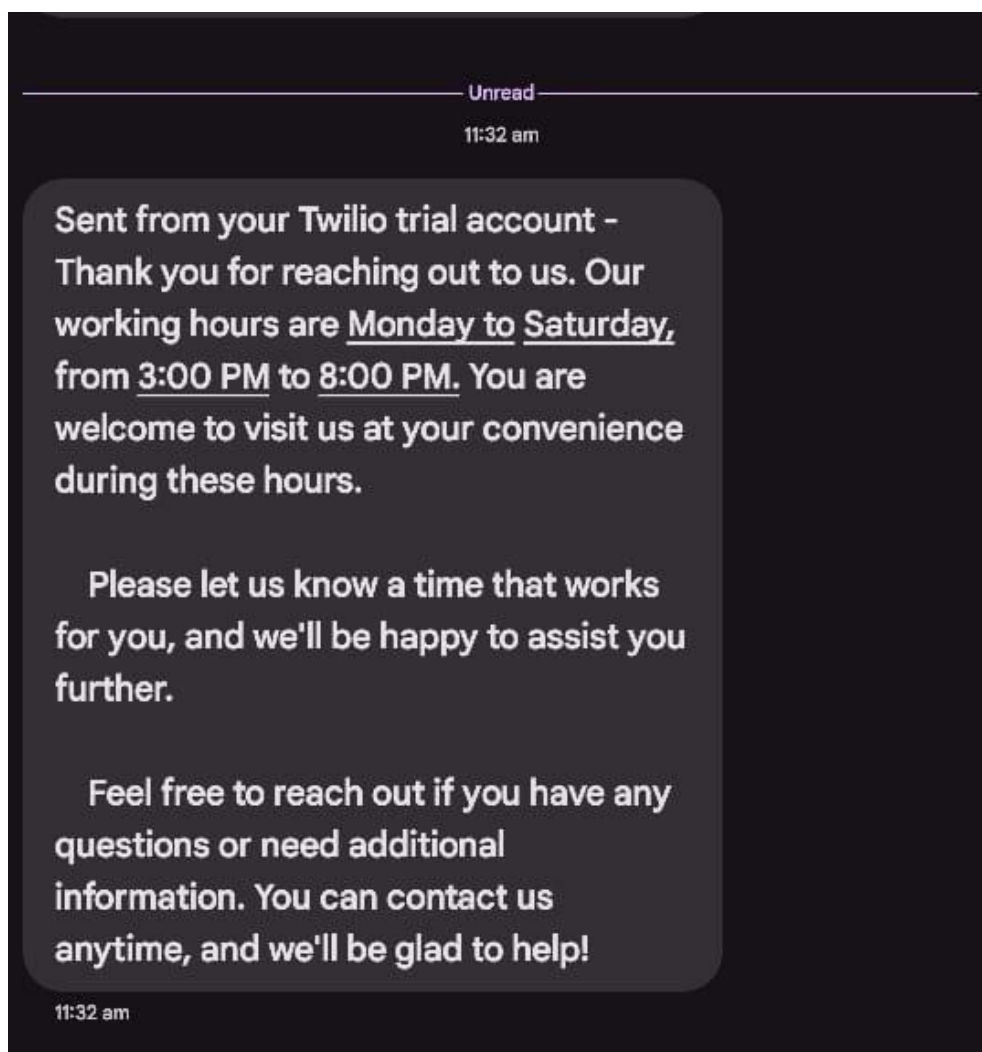
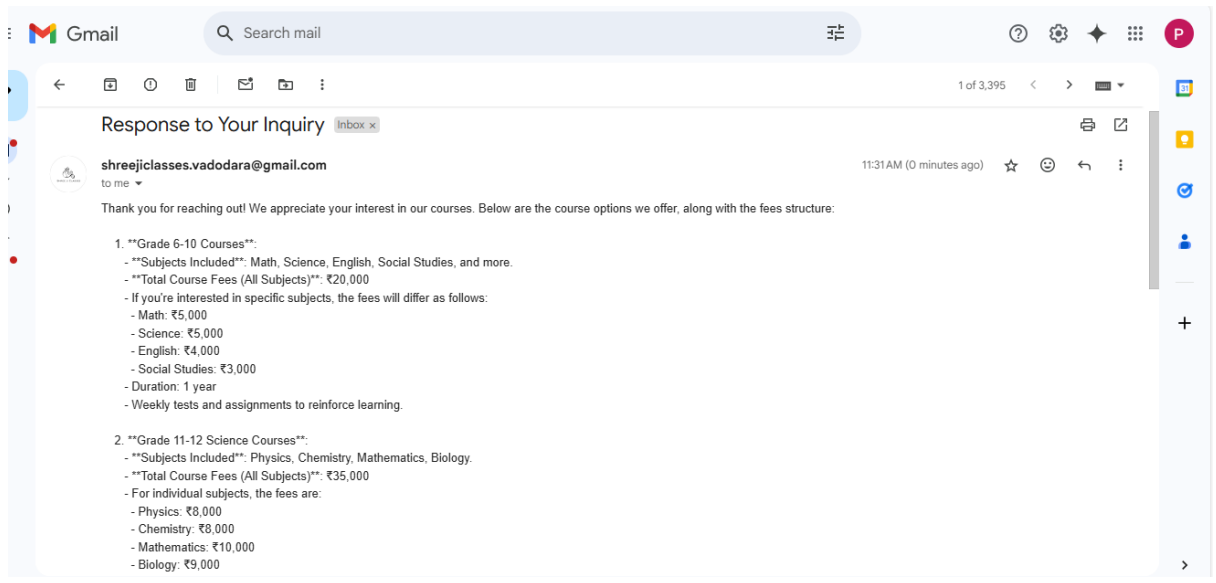
- Schema-driven structure with well-defined relationships
- Data Validation at schema level for consistency

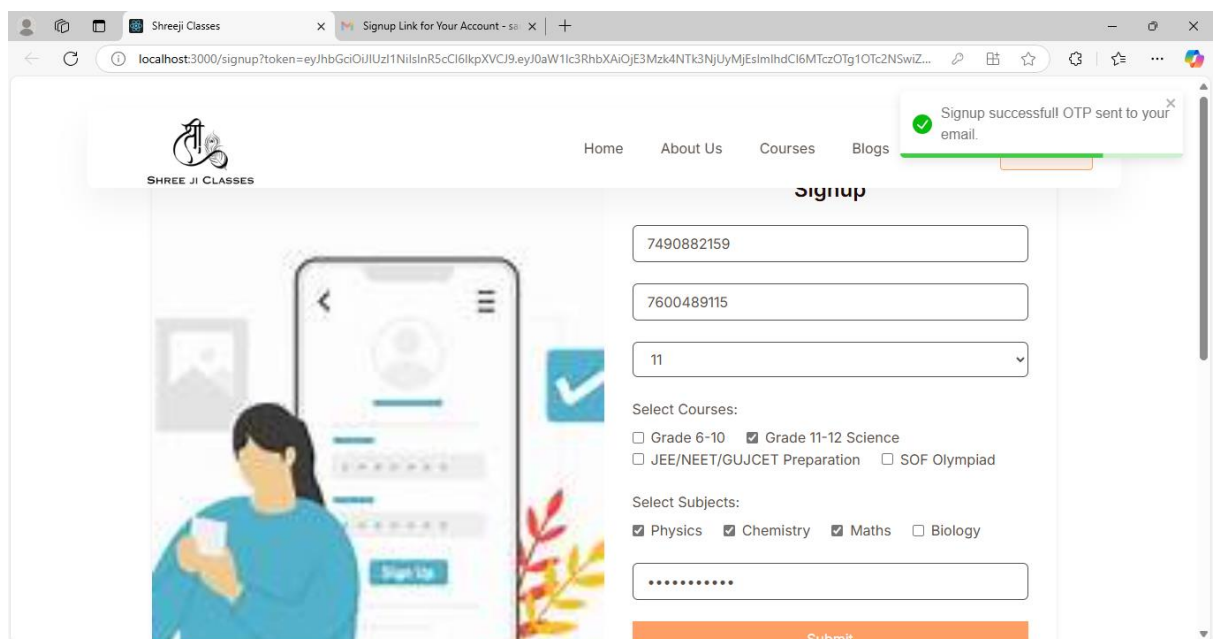
4 Third-Party Integrations

- Razorpay for online fee payments
- Nodemailer for sending emails (e.g., signup links, inquiries)
- Twilio for sending OTPs and monthly performance SMS to parents
- Cloudinary for uploading and managing study materials
- Google Drive for storing performance reports and sharing report links via SMS

9.4 SCREENSHOTS







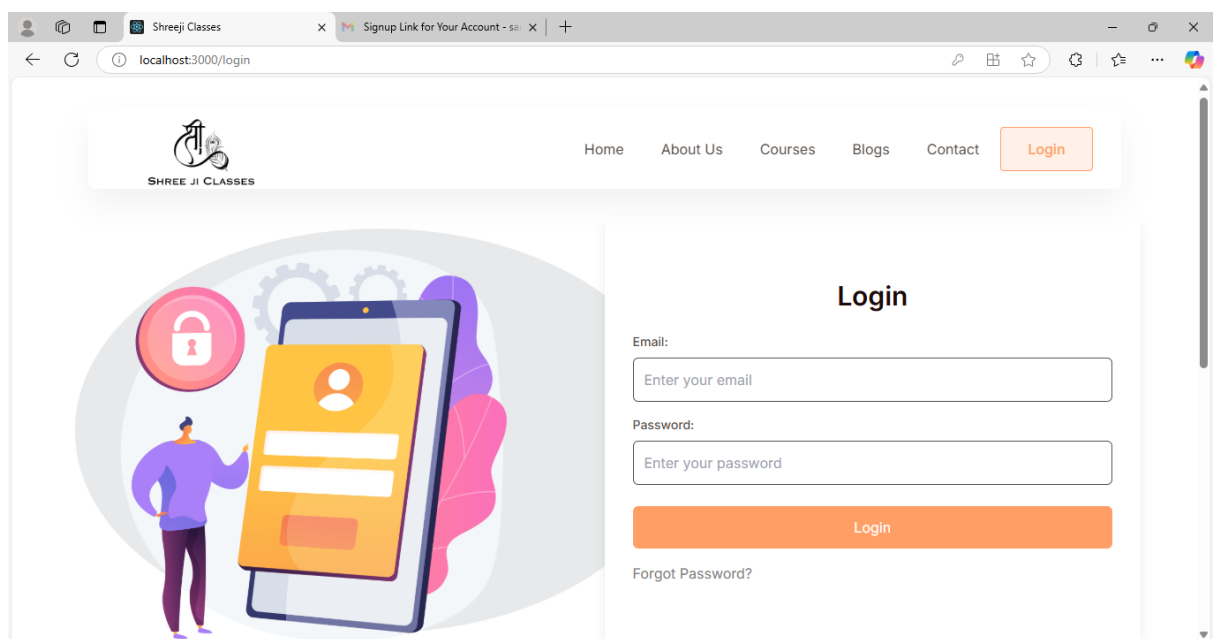
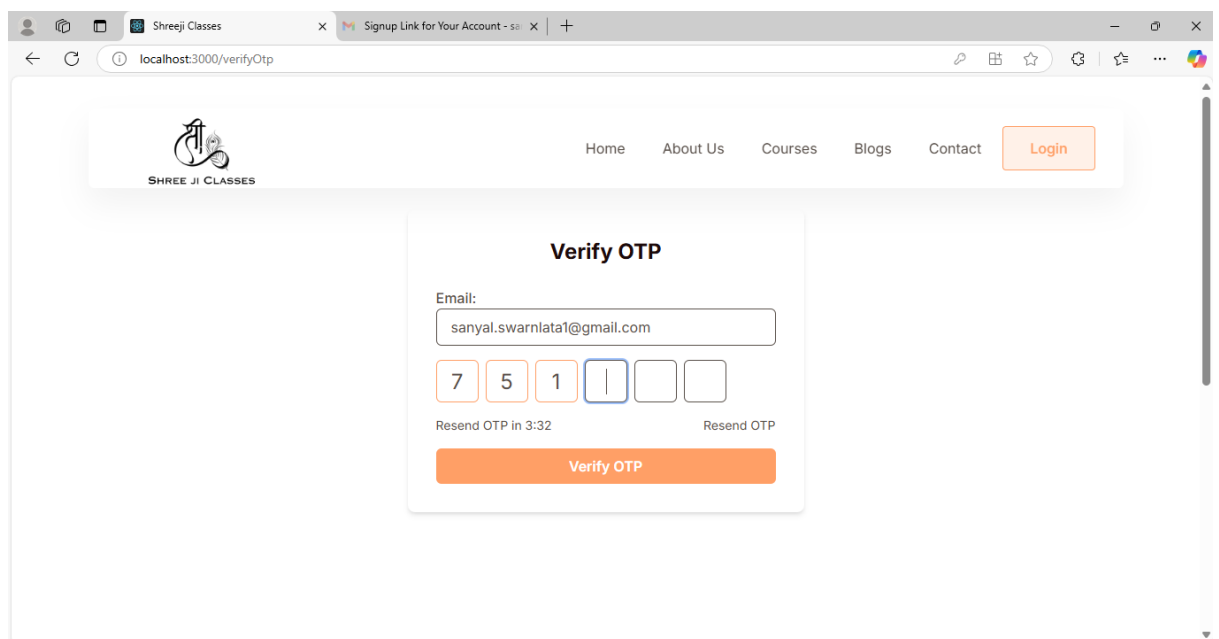
shreejiclasses.va...

12:02



to me ▾

Your OTP code is 751605. It is valid for 5 minutes.



Admin Dashboard

The screenshot shows the Admin Dashboard for 'Shreeji Classes'. The sidebar on the left contains navigation links: Home, Students, Teachers, Courses, Consult Inquiries, Fees Management, Push Notifications, and Schedule. A 'Send Signup Links' button is at the bottom of the sidebar. The main content area displays a welcome message to 'Samarth Doshi', a title 'Welcome to the Admin Dashboard', and a description of the panel's purpose. Below this, a 'Statistics' section shows three cards: 'Total Students' (120), 'Total Teachers' (15), and 'Total Courses' (18). A 'Charts' section features a line chart with four data series: Total Students (teal), Total Teachers (purple), Total Courses (orange), and Total Fees Collected (pink). The chart shows a steady upward trend for all metrics over time.

The screenshot displays the 'Admin TimeTable Management' interface. It includes a sidebar with navigation links: Home, Students, Teachers, Courses, Consult Inquiries, Fees Management, Push Notifications, and Schedule Management. A 'Send Signup Links' button is at the bottom of the sidebar. The main content area is titled 'Admin TimeTable Management' and contains an 'Update Timetable' form. The form has a 'Teacher' dropdown menu set to 'Arun Sanyal'. Below the dropdown is a table with columns for 'Days' and time slots: '3:00 PM - 4:00 PM', '4:00 PM - 5:00 PM', '5:00 PM - 6:00 PM', '6:00 PM - 7:00 PM', and '7:00 PM - 8:00 PM'. The rows represent the days of the week from Monday to Saturday. The table content is as follows:

Days	3:00 PM - 4:00 PM	4:00 PM - 5:00 PM	5:00 PM - 6:00 PM	6:00 PM - 7:00 PM	7:00 PM - 8:00 PM
Monday	11th - Maths	11th - Maths	12th - Maths	Class	Class
Tuesday	12th - Maths	Class	11th - Maths	Class	Class
Wednesday	Class	12th - Maths	12th - Maths	11th - Maths	Class
Thursday	11th - Maths	Class	Class	12th - Maths	Class
Friday	Class	11th - Maths	12th - Maths	Class	Class
Saturday	Class	Class	Class	Class	Class

At the bottom of the form, there are two buttons: 'Update Timetable' (green) and 'Cancel' (orange).



SHREEJI CLASSES

- Teachers
- Courses
- Consult Inquiries
- Fees Management
- Push Notifications
- Schedule Management
- Feedbacks

Send Signup Links

Manage Courses

[Add Course](#)



Grade 11-12 Science
 Focused coaching for Science streams, PCB and PCM.
Weekly Tests: Yes
Additional Details: For students pursuing science in Grades 11-12, this course provides an in-depth understanding of key subjects. Sessions are tailored for board exams and advanced concepts.

[Edit](#) [Delete](#)



GUJCET/NEET/JEE Preparation
 Comprehensive preparation for competitive exams.
Weekly Tests: Yes
Additional Details: This course prepares students for competitive exams with a focus on problem-solving, conceptual clarity, and exam strategies. Students can opt for specific exams or a combined preparation package.

[Edit](#) [Delete](#)



SOF Olympiad
 Specialized coaching for SOF Olympiad exams.
Weekly Tests: Yes
Additional Details: This course equips students for the Science and Math Olympiads conducted by SOF. With a curriculum focused on advanced problem-solving, students gain confidence to tackle challenging questions.

[Edit](#) [Delete](#)



SHREEJI CLASSES

- Teachers
- Courses
- Consult Inquiries
- Fees Management
- Push Notifications
- Schedule Management
- Feedbacks

Send Signup Links

Manage Courses

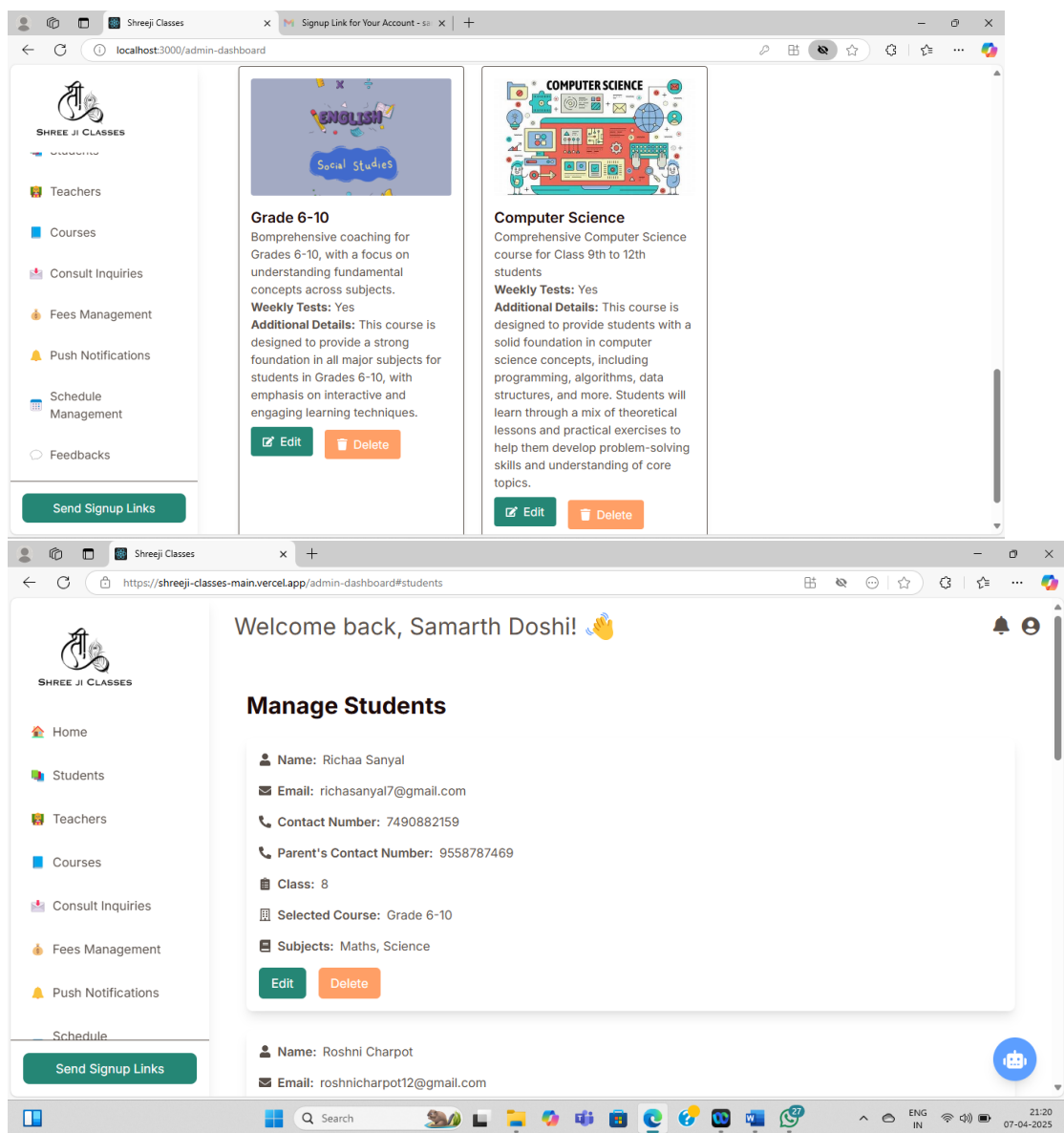
[Add Course](#)

Weekly Tests: ☒

[Choose File](#)

This course is designed to provide students with a solid foundation in computer science concepts, including programming, algorithms, data structures, and more. Students will learn through a mix of theoretical lessons and practical exercises to help them develop problem-solving skills and understanding of core topics.

[Add Course](#)



Shreeji Classes

https://shreeji-classes-main.vercel.app/admin-dashboard#teachers

Welcome back, Samarth Doshi! 🙌

Manage Teachers

Name: Prachil Sanyal
Email: prachisanyal1005@gmail.com
Contact Number: 7490882159
Teaches Class: 6, 7
Teaches Course: Grade 6-10, JEE/NEET/GUJCET Preparation
Subjects: Maths, Science, Physics, Chemistry

[Edit](#) [Delete](#)

Name: Arun Sanyal
Email: sanyal.arun1@gmail.com
Contact Number: 7490882159

SHREEJI CLASSES

- Home
- Students
- Teachers
- Courses
- Consult Inquiries
- Fees Management
- Push Notifications
- Schedule

[Send Signup Links](#)

SHREEJI CLASSES

https://shreeji-classes-main.vercel.app/admin-dashboard#fees

Welcome back, Samarth Doshi! 🙌

Payment Status Details

Filter by Payment Status:

Richaa Sanyal
Course: Grade 6-10
Class: 8
Email: richasanyal7@gmail.com
Subjects: Maths, Science

Paid: 19200
Remaining: 0
Status: paid
Remaining Installments: N/A
Next Due Date: N/A

[Edit](#)

Roshni Charpot
Course: Grade 11-12 Science
Class: 11
Email: roshnicharpot12@gmail.com
Subjects: Physics, Chemistry

Paid: 38400
Remaining: 0
Status: paid
Remaining Installments: N/A
Next Due Date: N/A

[Edit](#)

SHREEJI CLASSES

- Home
- Students
- Teachers
- Courses
- Consult Inquiries
- Fees Management
- Push Notifications
- Schedule

[Send Signup Links](#)

Shreeji Classes

https://shreeji-classes-main.vercel.app/admin-dashboard#fees

Welcome back, Samarth Doshi! 🙌

Payment Status Details

Filter by Payment Status: Pending

Prachi Sanyal

Course: Grade 11-12 Science

Class: 11

Email: sanyalprachi1004@gmail.com

Subjects: Physics, Chemistry, Maths

Paid: 0

Remaining: N/A

Status: pending

Remaining Installments: N/A

Next Due Date: N/A

Edit

Prachi Sanyal

Course: Grade 11-12 Science

Class: 11

Email: sanyal.yana18@gmail.com

Subjects: Physics, Chemistry, Maths

Paid: 0

Remaining: N/A

Status: pending

Remaining Installments: N/A

Next Due Date: N/A

Edit

SHREEJI CLASSES

- Home
- Students
- Teachers
- Courses
- Consult Inquiries
- Fees Management
- Push Notifications
- Schedule

Send Signup Links

Shreeji Classes

Mahavir Jayanti Holiday Notice

https://shreeji-classes-main.vercel.app/admin-dashboard#notifications

Welcome back, Samarth Doshi! 🙌

Notifications

Send a Notification

Holiday

Classes will remain closed on 10th April due to Mahavir Jayanti.

All Students

☐ Send to Parents ☐ Send SMS Notification

Send Notification

testing

testing

By: Samarth Doshi

Delete

SHREEJI CLASSES

- Students
- Teachers
- Courses
- Consult Inquiries
- Fees Management
- Push Notifications
- Schedule Management

Send Signup Links

Shreeji Classes

27

21:20

07-04-2025

21:23

07-04-2025

Teacher Dashboard



SHREEJI CLASSES

- Home
- Schedule
- Take Attendance
- Upload Resources
- Upload Test Marks
- Push Notifications

Send Announcements

Welcome back, Arun Sanyal! 🙌



Quote of the Day
"A teacher affects eternity; he can never tell where his influence stops. - Henry Adams"

Event Calendar

April 2025

MON	TUE	WED	THU	FRI	SAT	SUN
31	1	2	3	4	5	6
7	8	9	10	11	12	13
14	15	16	17	18	19	20
21	22	23	24	25	26	27
28	29	30	1	2	3	4

To-Do List

Add a task...

take attendance

localhost:3000/teacher-dashboard



SHREEJI CLASSES

- Home
- Schedule
- Take Attendance
- Upload Resources
- Upload Test Marks
- Push Notifications

Send Announcements

Welcome back, Arun Sanyal! 🙌

Teacher TimeTable

Day / Time	3:00 PM - 4:00 PM	4:00 PM - 5:00 PM	5:00 PM - 6:00 PM	6:00 PM - 7:00 PM	7:00 PM - 8:00 PM
Monday	11th - Maths	11th - Maths	12th - Maths	-	-
Tuesday	12th - Maths	-	11th - Maths	-	-
Wednesday	-	12th - Maths	12th - Maths	11th - Maths	-
Thursday	11th - Maths	-	-	12th - Maths	-
Friday	-	11th - Maths	12th - Maths	-	-
Saturday	-	-	-	-	-
Sunday	-	-	-	-	-


SHREE JI CLASSES

Home

Schedule

Take Attendance

Upload Resources

Upload Test Marks

Push Notifications

Send Announcements

Welcome back, Prachi Sanya! 🙌

Attendance marked successfully!

Mark Attendance

Grade 6-10

8

Maths

Load Students

Student Name	Attendance
Richa Sanyal	☒ Present ☐ Absent

Submit Attendance

Sent from your Twilio trial account - Dear Parent, your child attended 16/18 days this month.






SHREE JI CLASSES

Home

Schedule

Take Attendance

Upload Resources

Upload Test Marks

Push Notifications

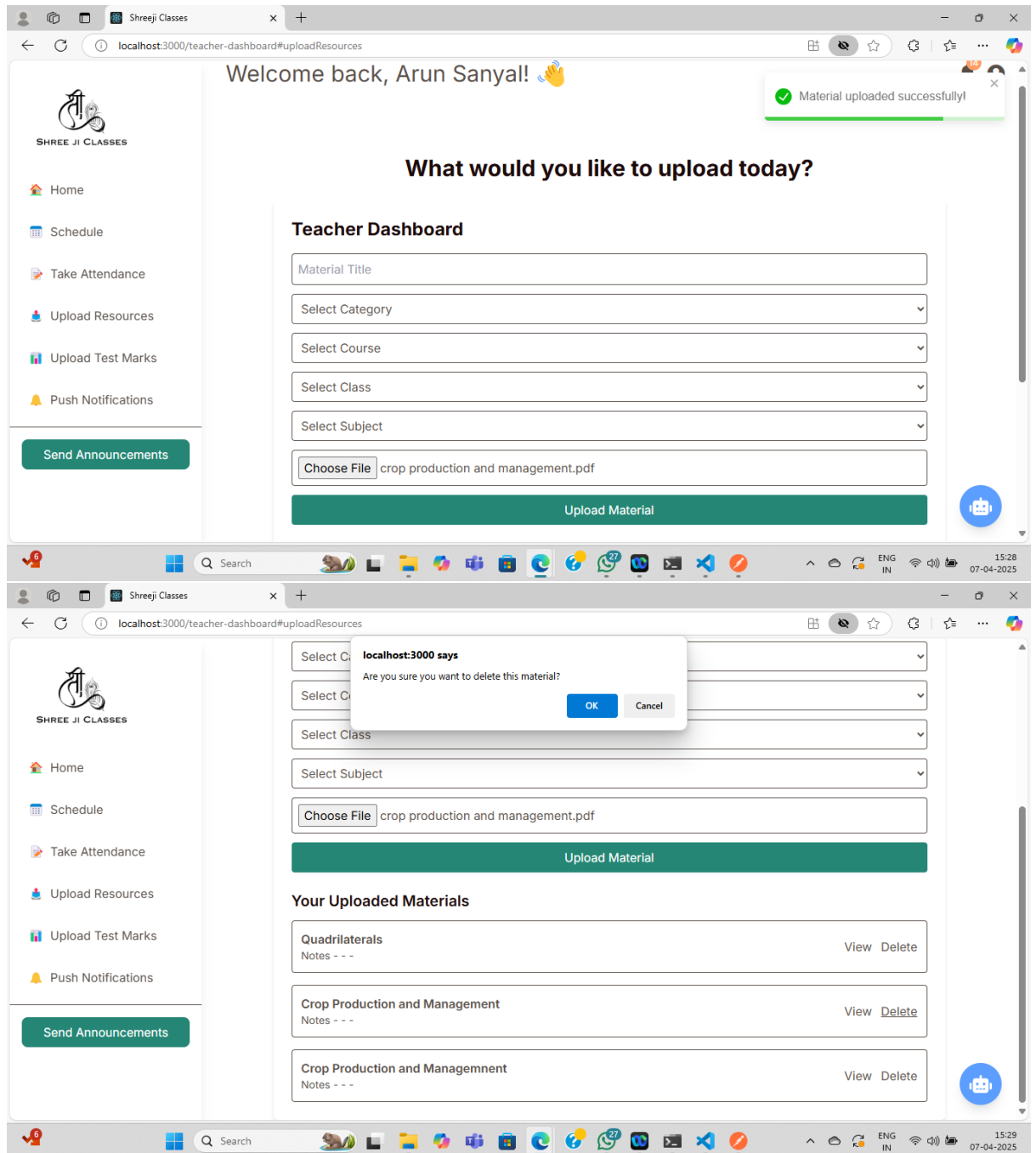
Send Announcements

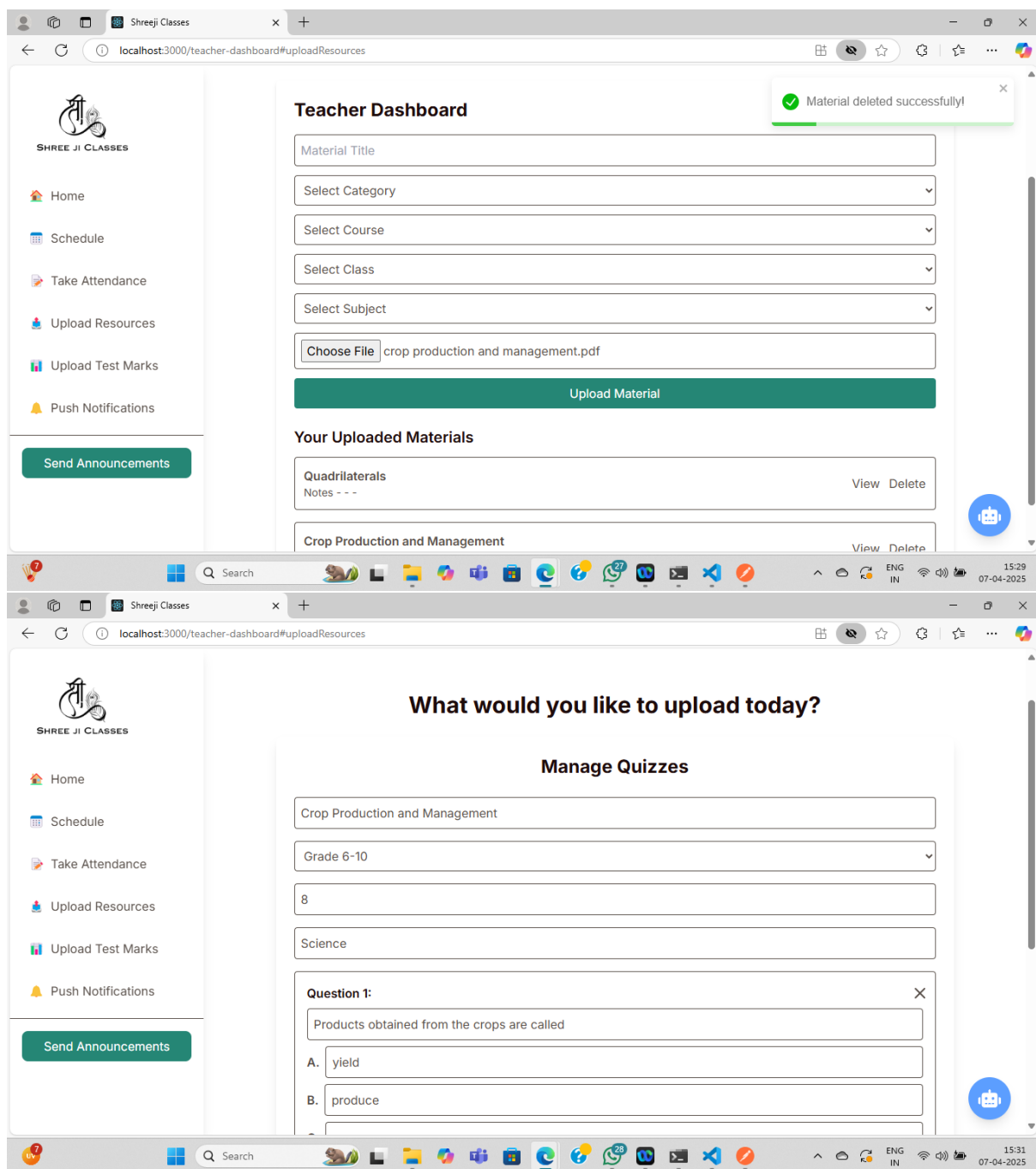
Welcome back, Arun Sanya! 🙌

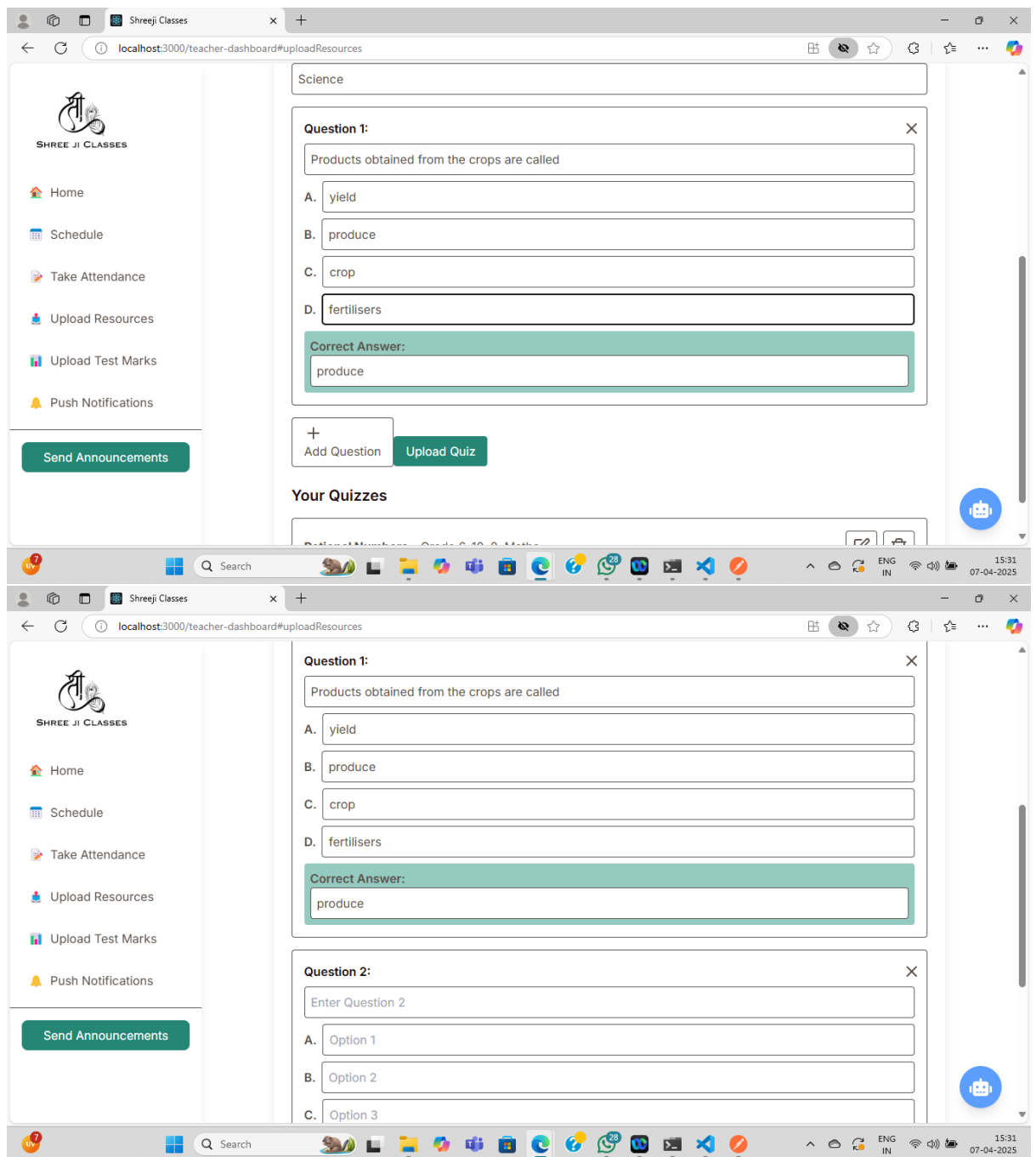
What would you like to upload today?


Study Material


Quiz









SHREEJI CLASSES

- Home
- Schedule
- Take Attendance
- Upload Resources
- Upload Test Marks
- Push Notifications

Send Announcements

A.

B.

C.

D.

Correct Answer:

+ Add Question **Upload Quiz**

Your Quizzes

Rational Numbers - Grade 6-10, 8, Maths

Crop Production and Management - Grade 6-10, 8, Science

Shreeji Classes

localhost:3000/teacher-dashboard#uploadResources



SHREEJI CLASSES

- Home
- Schedule
- Take Attendance
- Upload Resources
- Upload Test Marks
- Push Notifications

Send Announcements

Welcome back, Arun Sanyal! 🙌

Upload Marks

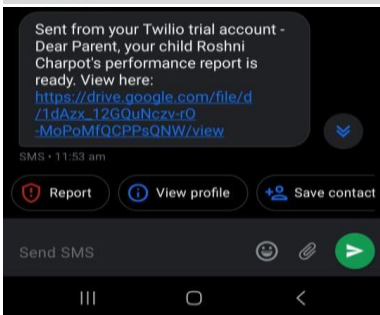
Name	Marks	Remarks
Roshni Charpot	21	Need Improvement
Prachi Sanyal	28	Good
Prachi Sanyal	18	Need Improvement
Radha Sharma	30	Excellent

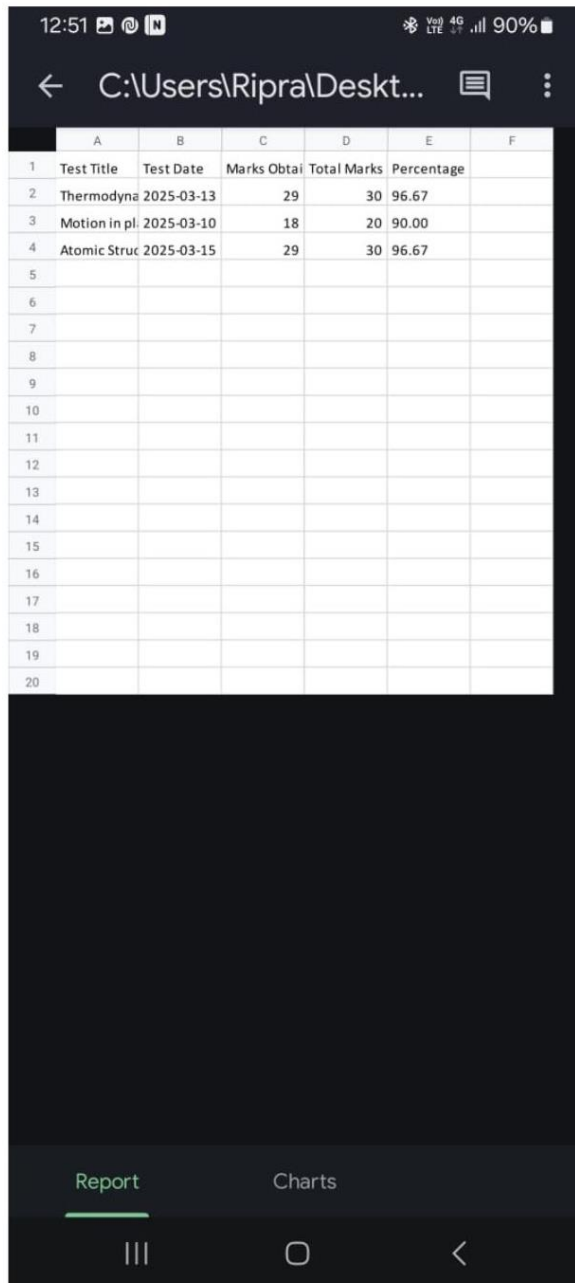
Upload Marks

Uploaded Records

Shreeji Classes

https://shreeji-classes-main.vercel.app/teacher-dashboard#uploadMarks





Welcome back, Prachi Sanyal! 🙌

Teacher Notifications

Send a Notification

Extra Lecture

This is to inform you that on Tuesday, there will be an additional mathematics lecture scheduled at 6:00 PM. Please make a note of this change.

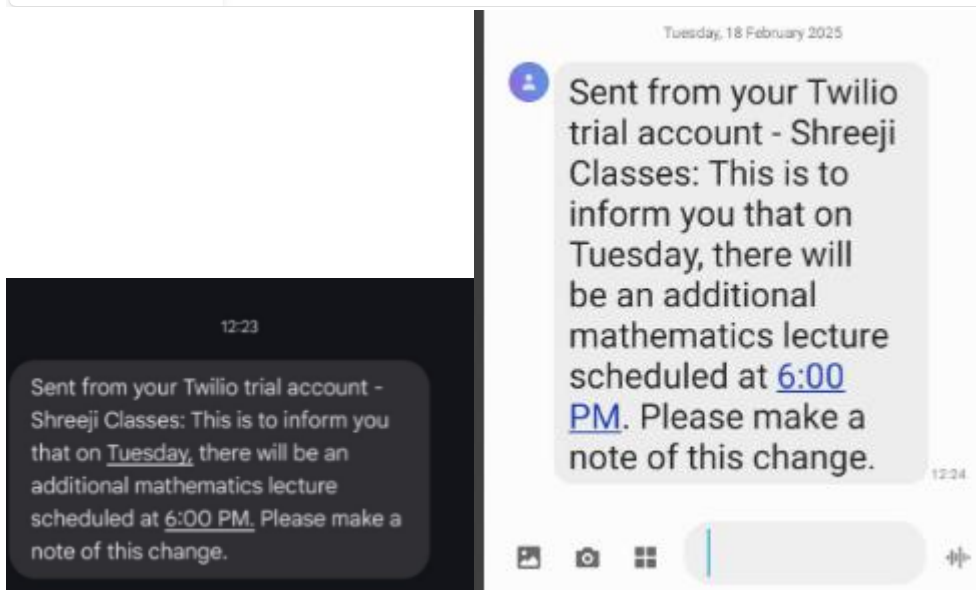
Specific Subject

8

Maths

☒ Send to Parents ☒ Send SMS Notification

Send Announcements **Send Notification**



Student Dashboard



SHREEJI CLASSES

Home

Timetable

Attendance Record

Progress Report

Study Resources

Notifications

Pay Fees

View Fee Details

Welcome back, Roshni Charpot! 🙌

Welcome to Your Student Dashboard

Keep track of your studies, attendance, and progress. Use the sections below to monitor your performance and schedule.

Attendance
50.00%

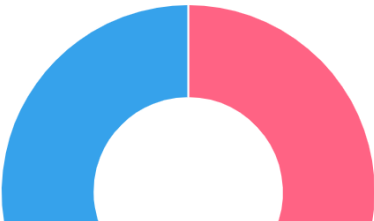
Study Progress
78.33%

Overall Performance
85%

Subject-Wise Performance

Physics

Chemistry



Monthly Attendance

Attendance





SHREEJI CLASSES

Home

Timetable

Attendance Record

Progress Report

Study Resources

Notifications

Pay Fees

View Fee Details

Welcome back, Richa Sanyal 🙌

My TimeTable

Day / Time	3:00 PM - 4:00 PM	4:00 PM - 5:00 PM	5:00 PM - 6:00 PM	6:00 PM - 7:00 PM	7:00 PM - 8:00 PM
Monday	Maths	Science	English	-	-
Tuesday	-	Social Science	Maths	-	-
Wednesday	-	Science	Maths	English	-
Thursday	-	English	Maths	Social Science	-
Friday	-	Science	Social Science	-	-
Saturday	-	English	Maths	-	-
Sunday	-	-	-	-	-



SHREEJI CLASSES

- Home
- Timetable
- Attendance Record
- Progress Report
- Study Resources
- Notifications
- Pay Fees

View Fee Details

Welcome back, Richa Sanyal 🙌

Attendance Records

Attendance Summary

Total Classes: 14
Present Days: 12
Absent Days: 2



< February 2025 >

Date	Status
01 Feb 2025	Present
02 Feb 2025	N/A



SHREEJI CLASSES

- Home
- Timetable
- Attendance Record
- Progress Report
- Study Resources
- Notifications
- Pay Fees

View Fee Details

Welcome back, Richa Sanyal 🙌

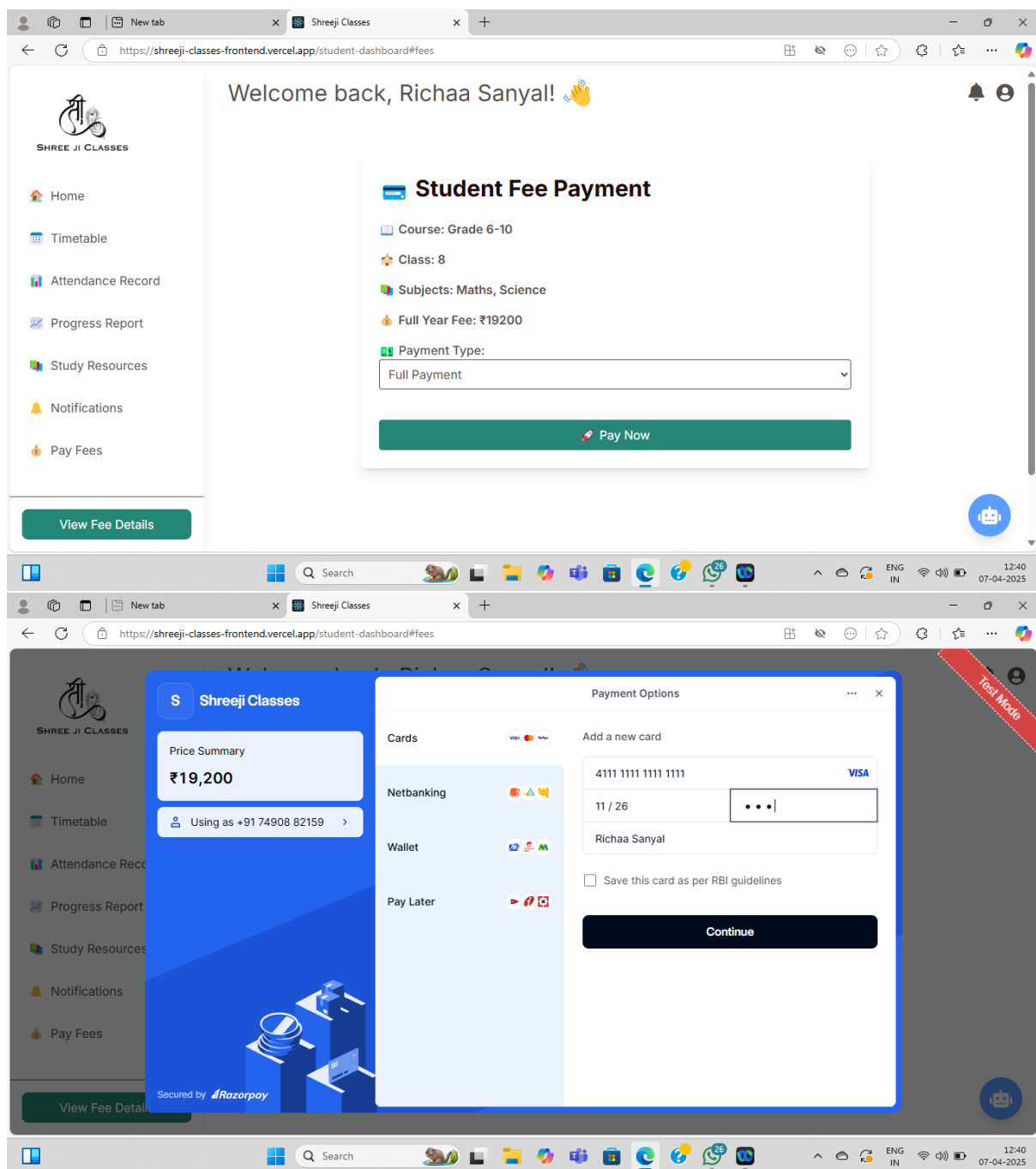
Student Notifications

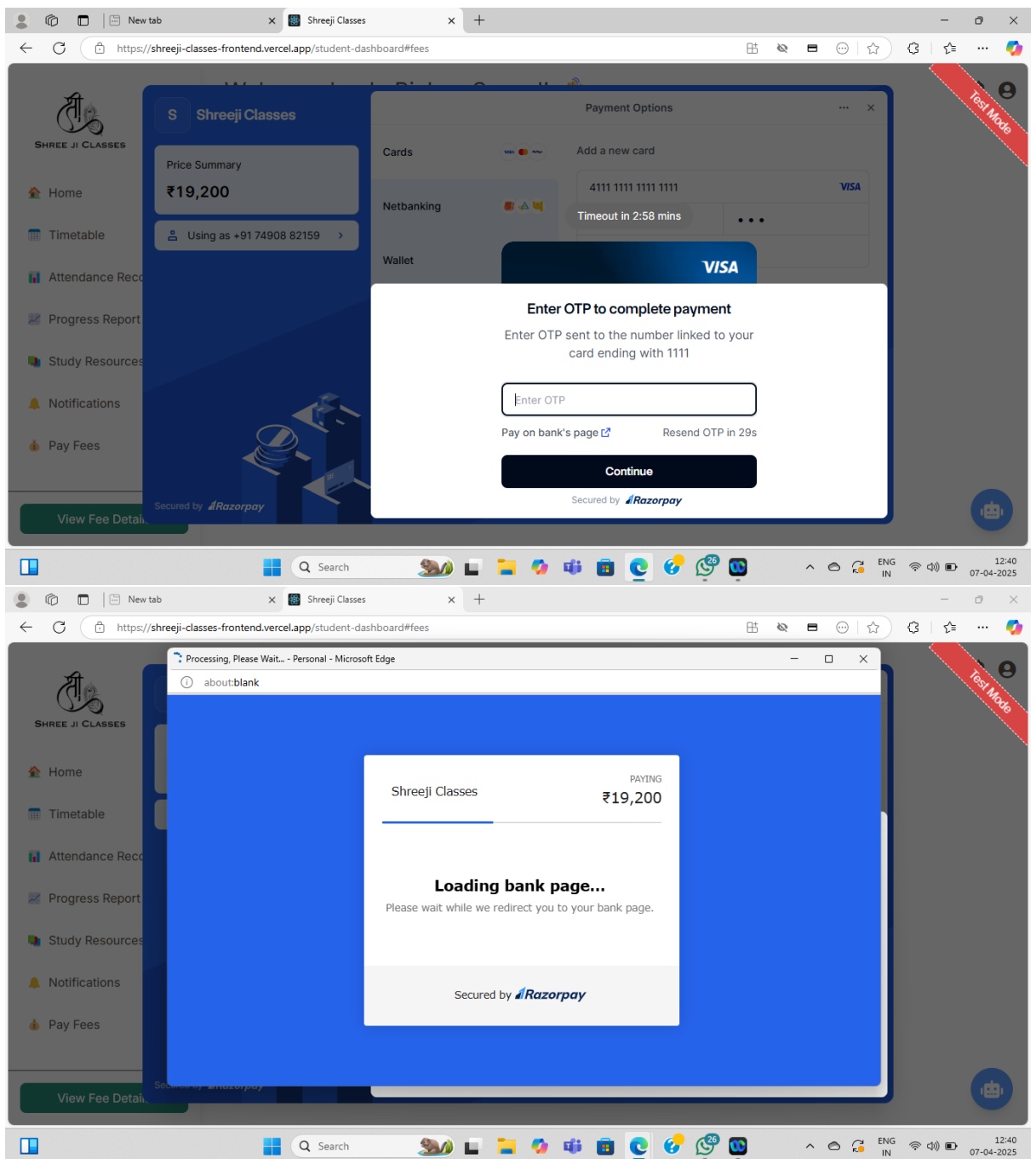
testing
testing
By: Samarth Doshi

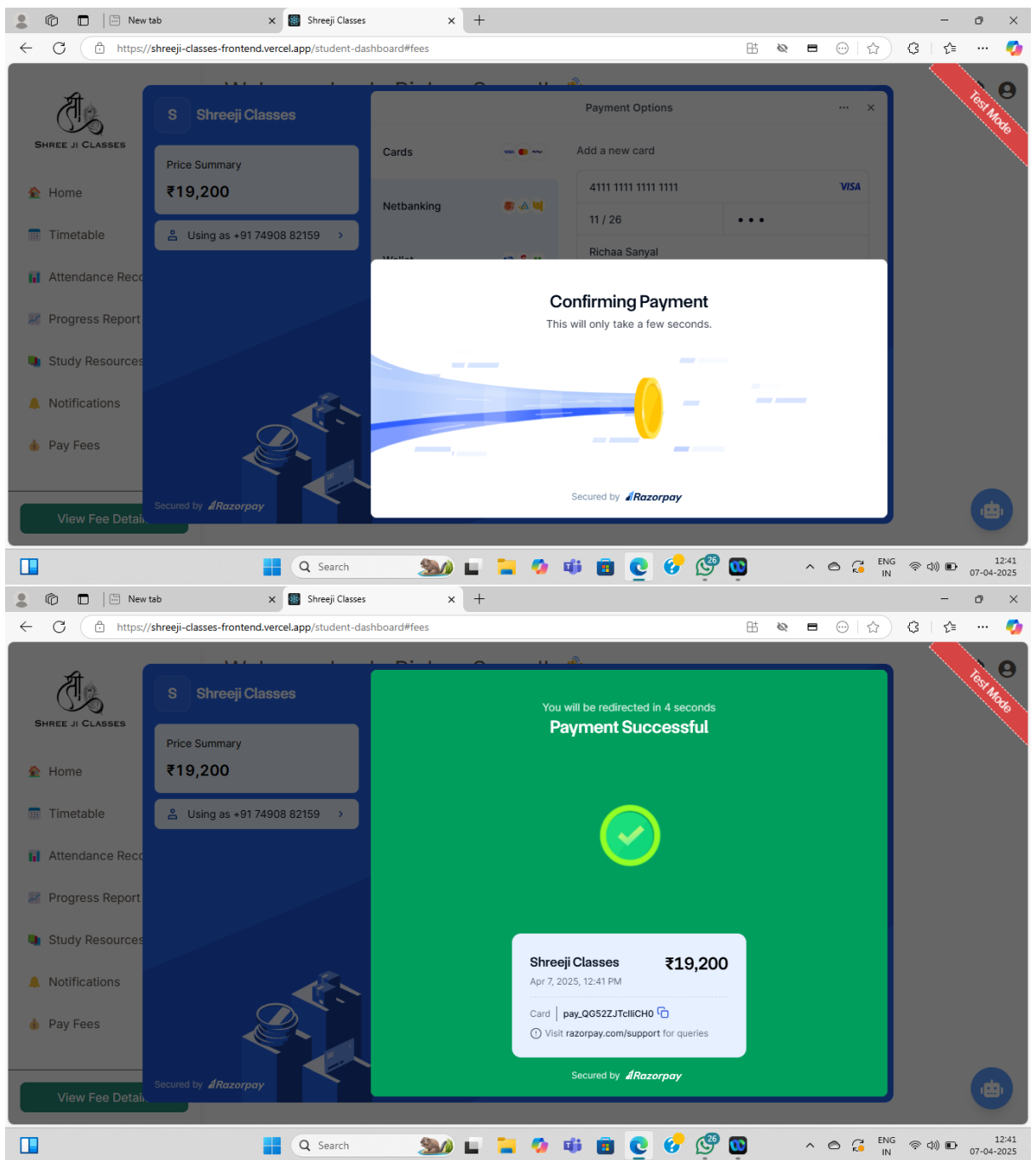
Weekly Test
This week on Saturday, you will have your Maths Test of Chapter 8.
By: Samarth Doshi

summer vaction
your summer vaction will start from 25th
By: Prachi Sanyal

lecture cancel
maths lecture canceled
By: Samarth Doshi







Shreeji Classes

Home

Timetable

Attendance Record

Progress Report

Study Resources

Notifications

Pay Fees

View Fee Details

Welcome back, Richaa Sanyal! 🙌

Payment Successful

- Name: Richaa Sanyal
- Email: richasanyal7@gmail.com
- Contact: 7490882159
- Course: Grade 6-10
- Class: 8
- Subjects: Maths, Science
- Amount Paid: ₹19200
- Remaining Amount: ₹
- Next Due Date:
- Remaining Installments:
- Installment Plan: Full Payment
- Payment ID: pay_QG52ZJTcliCHO
- Order ID: order_QG51zbHIRFJZAM
- Date: 7/4/2025, 12:41:11 pm

Download Receipt

Back

Fee Receipt_Richaa Sanyal.pdf

Welcome back, Richaa Sanyal! 🙌

Payment History

- Amount Paid: ₹19200
- Last Payment Date: 7/4/2025, 12:40:13 pm
- Payment Type: full
- Installment Plan: N/A
- Remaining Amount: ₹0
- Amount to pay now: ₹19200
- Remaining Installments: 0
- Status: COMPLETED

Fee Paid Completely

View Fee Details



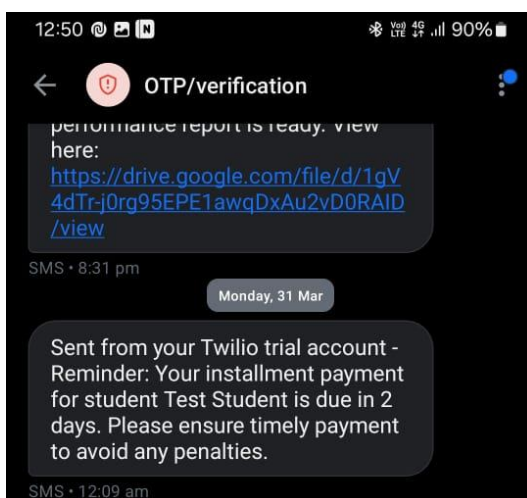
Shreeji Classes

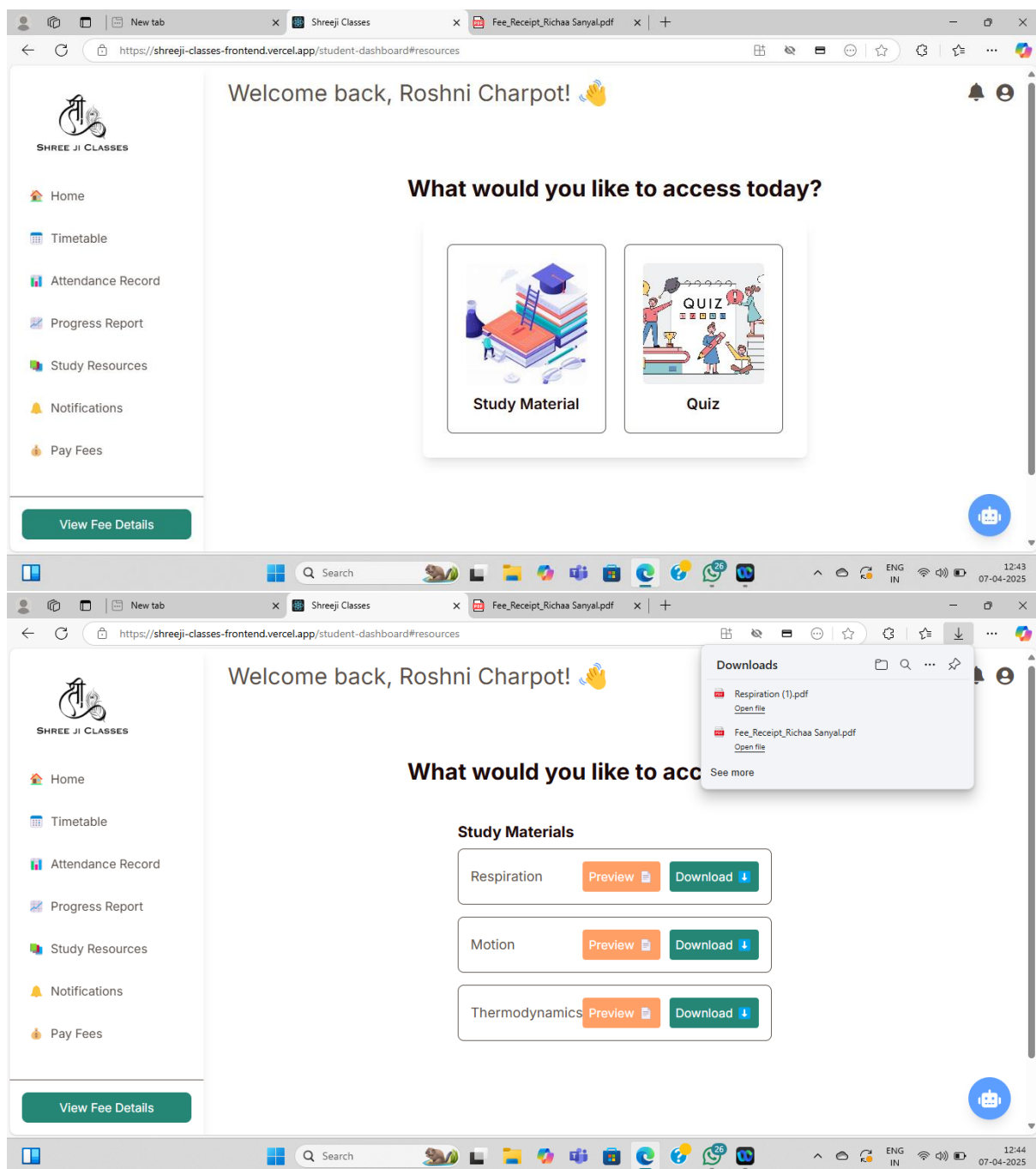
Payment Receipt

name: Richaa Sanyal
email: richasanyal7@gmail.com
contact: 7490882159
course: Grade 6-10
studentClass: 8
subjects: Maths, Science
amountPaid: 19200
installmentPlan: Full Payment
paymentId: pay_QG52ZJTclliCH0
orderId: order_QG51zbHIRFJZAM
date: 7/4/2025, 12:41:20 pm

Authorized Signatory

S. K. Doshi







SHREE JI CLASSES

[Home](#)
[Timetable](#)
[Attendance Record](#)
[Progress Report](#)
[Study Resources](#)
[Notifications](#)
[Pay Fees](#)

[View Fee Details](#)

What would you like to access today?

Physics Quiz

What is the S.I. unit of power?

☐ kilojoule

☐ joules

☒ Watt

☐ kilowatt

One watt is equal to

☐ one kilowatt per second

☒ one kilo joule per second

☐ one joule per second

☐ one kilo joule per minute

[Submit Quiz](#)

Your Score: 1 / 2



SHREE JI CLASSES

[Home](#)
[Timetable](#)
[Attendance Record](#)
[Progress Report](#)
[Study Resources](#)
[Notifications](#)
[Pay Fees](#)

[View Fee Details](#)

Welcome back, Roshni Charpot! 🙌

← April 2025 →

Test Records

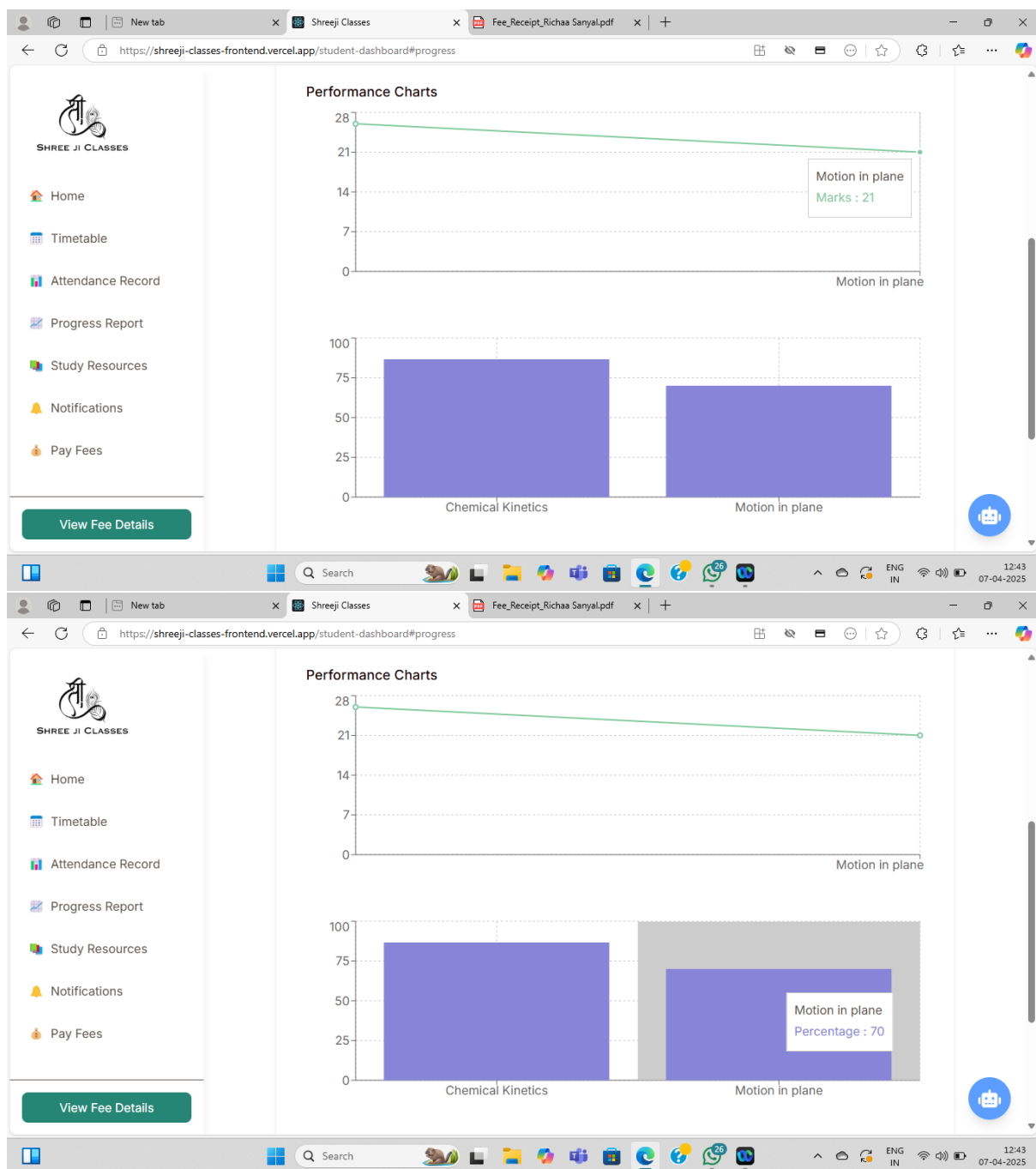
Chemical Kinetics - Chemistry
Date: 2/4/2025
Marks: 26 / 30
Percentage: 86.67%

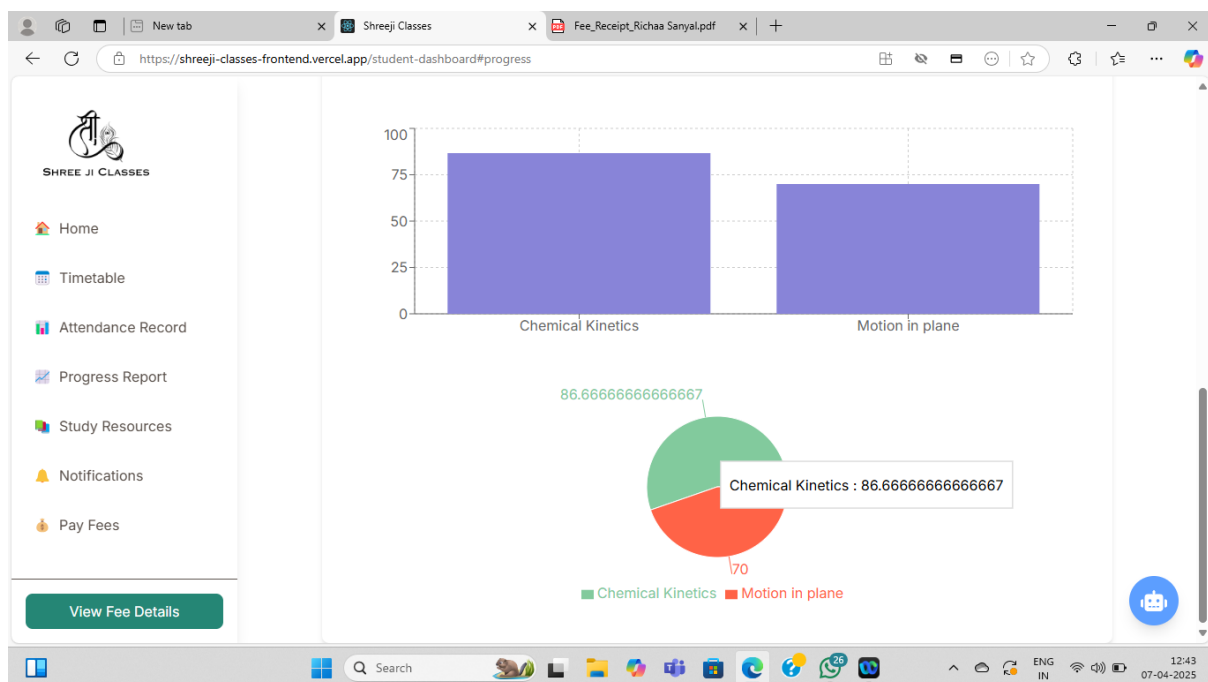
Motion in plane - Physics
Date: 1/4/2025
Marks: 21 / 30
Percentage: 70.00%
Remarks: Need Improvement

Performance Charts



Test	Marks	Percentage
Chemical Kinetics - Chemistry	26 / 30	86.67%
Motion in plane - Physics	21 / 30	70.00%





10. TESTING

10.1 Test Plan

10.1.1 Introduction

The test plan outlines the strategy, objectives, scope, and methodology for testing the Shreeji Classes Web App to ensure functionality, usability, security, and performance.

10.1.2 Objectives

- Verify that all features of the web application work as expected.
- Ensure the UI/UX is user-friendly and responsive.
- Identify and fix any bugs or inconsistencies.
- Test the security of user authentication and data protection.

10.1.3 Scope

- **Functional Testing:** Testing all core functionalities, including student registration, course enrollment, payments, and admin features.
- **UI/UX Testing:** Ensuring proper layout, navigation, and responsiveness.
- **Performance Testing:** Checking load time and app behavior under heavy traffic.
- **Security Testing:** Verifying authentication, authorization, and protection of sensitive data.

10.1.4 Test Approach

- Manual testing was primarily used throughout the development process.
- Core modules were tested individually and in flow to ensure proper integration.
- End-to-end testing was done for user flows like authentication, course enrollment, inquiry, and payment.

10.1.5 Testing Environment

- **Browser Compatibility:** Chrome, Firefox, Edge
- **Devices:** Desktop, Mobile, Tablet
- **Database:** MongoDB
- **Technology Stack:** MERN (MongoDB, Express, React, Node.js)

10.2 Test Methods

10.2.1 Functional Testing

- **Login & Authentication:** Verify login/signup, password reset, and role-based access.
- **Student Management:** Test student registration, profile updates, and course enrollments.
- **Course Management:** Ensure courses are displayed correctly and have proper content visibility.
- **Payment System:** Test online payments, failed transactions, and payment records.

10.2.2 UI/UX Testing

- Verify responsiveness on different screen sizes.
- Ensure proper navigation, button actions, and form validations.

10.2.3 Performance Testing

- Check website speed and response time.
- Test with multiple users to check server response.

10.2.4 Security Testing

- Validate encryption of passwords.

10.3 Test Cases

Test Case ID	Test Case Description	Preconditions	Steps	Expected Result
UA-01	Verify that only students and teachers can sign up via invite link	Admin has sent invite link to student/teacher	1. Click on invite link 2. Fill in details 3. Submit form	User should receive OTP for verification
UA-02	Verify OTP authentication	User has received OTP in email	1. Enter OTP 2. Click Submit	If OTP is correct, user should be successfully signed up
UA-03	Verify login functionality	User is registered and has valid credentials	1. Enter email and password 2. Click Login	User should be redirected to their dashboard
UA-04	Verify login with invalid credentials	User account exists	1. Enter wrong email/password 2. Click Login	Error message should appear
UA-05	Verify forgot password functionality	User is registered	1. Click on 'Forgot Password' 2. Enter email 3. Click Submit	Password reset email should be sent
UA-06	Verify logout functionality	User is logged in	1. Click on Logout button	User should be logged out and redirected to login page

Test Case ID	Test Case Description	Preconditions	Steps	Expected Result
CM-01	Verify admin can add new course	Admin is logged in	1. Go to Course Management 2. Click "Add Course" 3. Enter details 4. Click "Save"	Course should be added successfully
CM-02	Verify admin can edit a course	A course exists	1. Select a course 2. Click "Edit" 3. Modify details 4. Click "Save"	Course details should be updated
CM-03	Verify admin can delete a course	A course exists	1. Select a course 2. Click "Delete" 3. Confirm deletion	Course should be deleted
CM-04	Verify users can view courses	Courses are added in system	1. Visit homepage 2. Click on "Courses" section	List of available courses should be displayed

Test Case ID	Test Case Description	Preconditions	Steps	Expected Result
IC-01	Verify inquiry form submission	Inquiry form is available	1. Fill in Name, Email, Contact Number 2. Select mode of inquiry 3. Click Submit	Inquiry should be saved in DB

IC-02	Verify admin can view inquiries	Inquiry exists in DB	1. Login as Admin 2. Navigate to "Inquiries" section	All inquiries should be listed
IC-03	Verify inquiry auto-responses	Inquiry mode is email	1. Submit inquiry 2. Click "Resolve"	Email should be sent with course details
IC-04	Verify SMS notification for in-person visit	Inquiry mode is "in-person visit"	1. Submit inquiry 2. Click "Resolve"	SMS should be sent with visit timing

Test Case ID	Test Case Description	Preconditions	Steps	Expected Result
NT-01	Verify admin can send notifications	Admin is logged in	1. Navigate to Notifications 2. Enter Title, Message 3. Select Recipients 4. Click Send	Notification should be sent successfully
NT-02	Verify SMS notification to parents	Admin selects "Send SMS to Parents"	1. Send a notification with SMS option enabled	Parents should receive SMS

Test Case ID	Test Case Description	Preconditions	Steps	Expected Result
FP-01	Verify student can view fee structure	Student is logged in	1. Go to "Fees" section	Fee details should be displayed
FP-02	Verify student can pay fees fully	Student is logged in	1. Click "Pay Fees" 2. Select "Full Payment" 3. Choose payment method 4. Click Pay	Payment should be successful
FP-03	Verify installment payments	Student is logged in	1. Click "Pay Fees" 2. Select "Installments" 3. Choose payment method 4. Click Pay	Partial payment should be processed
FP-04	Verify receipt download	Student has paid fees	1. Navigate to Fees 2. Click on "Download Receipt"	PDF receipt should be downloaded

Test Case ID	Test Case Description	Preconditions	Steps	Expected Result
AT-01	Verify teacher can take attendance	Teacher is logged in	1. Select Course, Class, Subject 2. Mark Present/Absent 3. Click Submit	Attendance should be recorded
AT-02	Verify student can view attendance	Student is logged in	1. Go to Attendance section	Attendance report should be displayed
AT-03	Verify marks upload	Teacher is logged in	1. Select Course, Class, Subject 2. Enter Marks 3. Click Submit	Marks should be stored in DB
AT-04	Verify student can view marks	Marks are uploaded	1. Go to "Test Marks" section	Marks should be visible

Test Case ID	Test Case Description	Preconditions	Steps	Expected Result
TT-01	Verify admin can upload timetable	Admin is logged in	1. Go to Timetable 2. Upload timetable 3. Click Save	Timetable should be saved
TT-02	Verify student/teacher can view timetable	Timetable is uploaded	1. Login as Student/Teacher 2. Go to "Timetable" section	Respective timetable should be displayed

Test Case ID	Test Case Description	Preconditions	Steps	Expected Result
SM-01	Verify teacher can upload study material	Teacher is logged in	1. Select Course, Class, Subject 2. Upload File 3. Click Submit	File should be saved
SM-02	Verify student can download study material	Study material is uploaded	1. Navigate to "Study Material" 2. Click Download	File should be downloaded
SM-03	Verify student can attempt quiz	Quiz is uploaded	1. Go to "Quizzes" 2. Attempt questions 3. Submit Quiz	Quiz should be scored instantly

10.4 Test Report

During the development of the Shreeji Classes Web App, rigorous testing was conducted at every stage to ensure the system's functionality, reliability, and performance. Instead of preparing a separate formal test report, testing was integrated into the development process itself.

Key aspects such as functionality, responsiveness, data integrity, and API performance were continuously verified through manual testing, debugging, and iterative improvements. Any identified issues were promptly addressed and resolved to maintain a smooth user experience.

The system underwent multiple rounds of checks, including:

- **Component-Level Testing:** Ensuring that each individual module, including frontend components and backend APIs, functioned as expected.
- **Integration Testing:** Verifying seamless interaction between different system components, such as database operations, API requests, and UI interactions.
- **UI & Responsiveness Testing:** Checking the application's layout and usability across different screen sizes and devices.
- **Database & API Validation:** Ensuring proper data handling, validation, and secure communication between the frontend and backend.

Since testing was an ongoing and iterative process, any bugs, inconsistencies, or optimization needs were promptly identified and resolved. As a result, the final version of the web application is stable, functional, and ready for deployment without the need for a separate formal test report.

11. SECURITY FEATURES

To ensure data security and protect user information, the following security measures have been implemented in the **Shreeji Classes Web App**:

- **User Authentication & OTP Verification:** Only authorized users can sign up using invite links, and OTP authentication adds an extra layer of security.
- **Password Encryption:** User passwords are stored in encrypted format using **bcrypt hashing** to prevent unauthorized access.
- **Role-Based Access Control (RBAC):** Different user roles (Admin, Teacher, Student) have restricted access based on their permissions.
- **Secure Payment Integration:** Payments are processed through **secure gateways (e.g., Razorpay)** with SSL encryption to prevent fraud.
- **Session Management & Auto Logout:** User sessions expire after inactivity to prevent unauthorized access.

12. CODING STANDARDS FOLLOWED

To maintain code quality and readability, the following coding standards were followed:

- **Modular Code Structure:** Followed MVC (Model-View-Controller) architecture for better maintainability.
- **Consistent Naming Conventions:** Used camelCase for JavaScript & backend variables, PascalCase for class names.
- **Proper Indentation & Formatting:** Code formatting maintained manually using VS Code.
- **Code Reusability & DRY Principle:** Avoided redundant code by creating reusable components and functions.
- **Comments & Documentation:** Added meaningful comments for better understanding and future modifications.
- **Security Best Practices:** Avoided hardcoding sensitive information, used environment variables for API keys.

13. LIMITATIONS & FUTURE ENHANCEMENTS

13.1 Limitations

Despite its robust features, the **Shreeji Classes Web App** has some limitations:

- **No Mobile App:** Currently, the platform is only web-based; mobile application support is not available.
- **Limited Vendor Integration:** Only specific payment gateways are integrated, restricting payment flexibility.
- **Basic Communication Features:** There is no in-app chat or direct messaging between students and teachers.
- **Manual Course & Fee Updates:** Admin needs to manually update courses and fee structures.
- **Limited Reporting & Analytics:** The system lacks advanced insights and graphical analytics for performance tracking.

13.2 Future Enhancements

To improve the platform, the following features can be added in future updates:

- **Mobile App Development** – A dedicated Android/iOS app for better accessibility.
- **Advanced AI-Based Chatbot** – For student queries and automated support.
- **Multiple Payment Gateways** – Support for UPI, Net Banking, and other digital wallets.
- **Enhanced Attendance System** – Biometric or QR-based attendance marking.

14. EXPERIENCE & LEARNING

Working on the **Shreeji Classes Web App** provided a deep learning experience in:

- **Full-Stack Development:** Hands-on experience with **MERN stack**.
- **Payment Gateway Integration:** Learned how to integrate **Razorpay** securely.
- **Bug Fixing & Testing:** Understood the importance of **test cases, debugging, and performance optimization**.
- **Security Best Practices:** Gained knowledge of **encryption, access control, and secure coding**.
- **User Experience (UX) Design:** Focused on **UI improvements** for better usability.
- **Project Management:** Learned how to manage **tasks, deadlines, and handling project iterations** effectively.

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